ECON6103 Econometrics for Honours Course Notes

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I use course materials from semester 1, 2025 to make most of the notes and then use 2026's materials to refine the notes.
Many of the derivations in Eric's slides are incomplete. I filled in the missing steps and also wrote my own answers to some of the questions. There were also a few minor typos in Eric's slides, and I corrected all of them (thanks to GPT and Teresa).

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1 Math Review

1.1 Probability Theory

Random Variable:

Informally, a **Random Variable (RV)** takes on numerical values determined by an experiment. In fact, it is neither random nor a variable, but rather a function: a mapping from abstract outcomes of a hypothetical experiment to (typically real, but could be complex) numbers.

Example 1.1.1. Consider an experiment in which a fair coin is tossed. Then, the possible outcomes are Head and Tail, i.e., $\{H, T\}$, and we define a random variable X as follows:

$$X = \begin{cases} 0 & \text{if the outcome is } T \\ 1 & \text{if the outcome is } H \end{cases}$$

- X could be either 0 or 1 whenever the coin is tossed.
- Each time, the realisation of X would be different.
- The uncertainty can be summarised as $\Pr(X = 0) = 0.5$.

Probability Functions:

More generally, the uncertainty associated with a random variable (X) is characterised by a probability distribution.

Definition 1.1.2. A probability distribution is represented by its cumulative distribution function (CDF), defined as

$$F_X(x) := \Pr(X \leq x).$$

Definition 1.1.3. When $F_X(x)$ is a continuous function, X is said to be a continuous random variable.

Definition 1.1.4. For a continuous random variable, the probability density function (PDF), denoted by $f_X(x)$, is a function that satisfies

$$\Pr(a < X \leq b) = \int_a^b f_X(x)\, dx = F_X(b) - F_X(a).$$

If $f_X(x)$ exists and if $F_X(x)$ is differentiable, we have

$$f_X(x) = \frac{d}{dx} F_X(x).$$

If $F_X(x)$ is a step function, X is discrete, which we do not cover today.

Expectation:

A function $g(\cdot)$ of a random variable X is also a random variable.

Example 1.1.5. $g(X) = \exp(X)$, $g(X) = \ln(X)$, etc.

Definition 1.1.6. Suppose X has the PDF $f_X(x)$. The expectation of $g(X)$ is

$$E[g(X)] := \int_{-\infty}^{\infty} g(x) \cdot f_X(x)\, dx.$$

which represents the central tendency of the random variable, $g(X)$.

Definition 1.1.7. As a special case with $g(X) = X$,

$$E[X] := \int_{-\infty}^{\infty} x \cdot f_X(x)\, dx.$$

which is the expectation of X, or the expected value or the mean.

Definition 1.1.8. Consider another case with $g(X) = (X - E[X])^2$.

$$V(X) := E[(X - E[X])^2] = E[X^2] - E[X]^2,$$

which is called the variance that represents the variability of X. Note that $\sqrt{V(X)}$ is the standard deviation of X.

Property 1.1.9. *1. $E[c] = c$, for any constant c.*

2. $E[aX + b] = aE[X] + b$ for any constants a and b.

3. If $a_1, \ldots, a_n$ are constants, and $X_1, \ldots, X_n$ are RVs, then

$$E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i].$$

As a special case, we have

$$E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i].$$

4. $V(c) = 0$, for any constant c.

5. $V(X) = 0$ if and only if $\Pr(X = c) = 1$ for some constant c.

6. $V(aX + b) = a^2 Var(X)$ for any constants a and c.

Joint Distribution:

Suppose X and Y are jointly distributed with joint CDF

$$F_{XY}(x, y) = \Pr(X \leq x,\, Y \leq y).$$

The joint PDF is

$$f_{XY}(x, y) = \frac{d^2}{dx\, dy} F_{XY}(x, y),$$

if the derivative exists.
The marginal PDF of X can be obtained by integrating out Y,

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dy.$$

X and Y are independent if and only if for all x and y

$$f_{XY}(x, y) = f_X(x) f_Y(y).$$

The expectation of $g(X, Y)$ for some function $g(X, Y)$ is:

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{XY}(x, y)\, dx\, dy.$$

Definition 1.1.10. If $g(X, Y) = (X - E[X])(Y - E[Y])$, then we have the covariance,

$$C(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y].$$

If $C(X, Y) > 0$, then X and Y "likely move in the same direction". If $C(X, Y) < 0$, they "likely move in opposite directions".

$|C(X, Y)| \leq \sqrt{V(X)}\sqrt{V(Y)}$. So, the correlation coefficient

$$\rho_{XY} := \frac{C(X, Y)}{\sqrt{V(X)}\sqrt{V(Y)}}$$

is always between -1 and 1.

If X and Y are independent, $C(X, Y) = 0$, but the converse is generally not true.

Example 1.1.11. $C(X, Y) = 0$ but $C(X^2, Y^2) \neq 0$.

Conditional Distribution:

Suppose X and Y are jointly distributed with the joint PDF $f_{XY}(x, y)$. Then, the conditional PDF of Y given $X = x$ is

$$f_{Y|X}(y|x) := \frac{f_{XY}(x, y)}{f_X(x)},$$

which summarises the distribution of Y when x is the realisation of X.

The conditional CDF is now derived from the conditional PDF as:

$$F_{Y|X}(y|x) = \int_{-\infty}^{y} f_{Y|X}(y|x)\, dy.$$

If X and Y are independent, $f_{Y|X}(y|x) = f_Y(y)$ and $f_{X|Y}(x|y) = f_X(x)$.

The conditional expectation of Y given $X = x$ is

$$E[Y|X = x] = \int_{-\infty}^{\infty} y f_{Y|X}(y|x)\, dy.$$

The conditional variance of Y given $X = x$ is

$$V(Y|X = x) = E\Big[(Y - E[Y|X = x])^2 \,\Big|\, X = x\Big].$$

$E[Y|X = x]$ and $V(Y|X = x)$ are functions of x. When x is not specified, $E[Y|X]$ and $V(Y|X)$ are random variables since they are functions of X: i.e., $g(X) \equiv E[Y|X]$ or $g(X) \equiv V[Y|X]$.

Property 1.1.12. *Basic properties of conditional expectations:*

1. $E[c(X)|X] = c(X)$ *for any function* $c(\cdot)$.
2. $E[a(X)Y + b(X)|X] = a(X)E[Y|X] + b(X)$ *for any functions* $a(\cdot)$, $b(\cdot)$.
3. *If* X *and* Y *are independent,* $E[Y|X] = E[Y]$.
4. $E[E[Y|X]] = E[Y]$. *This is called the law of iterated expectations.*
   - *More generally,* $E[g(X)E[Y|X]] = E[g(X)Y]$.
5. *If* X *and* Y *are independent,* $V(Y|X) = V(Y)$.
6. $E[V(Y|X)] + V(E[Y|X]) = V(Y)$. *This is called the law of total variance.*

Other useful properties can often be easily derived from the basic properties on the previous slide.

Example 1.1.13. If $E[Y|X] = a_0 + a_1 X$ then

$$a_1 = \frac{C(X,Y)}{V(X)} \quad \text{and} \quad a_0 = E[Y] - a_1 E[X].$$

Proof. If $E[Y|X] = a_0 + a_1 X$ then $E[Y] = a_0 + a_1 E[X]$ and $E[XY] = a_0 E[X] + a_1 E[X^2]$. Now, solve for a_1 to get:

$$a_1 = \frac{E[XY] - E[X]E[Y]}{E[X^2] - E[X]^2} = \frac{C(X,Y)}{V(X)}.$$

□

Normal Distribution:

Suppose a random variable Y follows a normal distribution with mean μ and variance σ^2. We denote this by:

$$Y \sim \mathcal{N}(\mu, \sigma^2).$$

Y takes on values from $\mathbb{R} = (-\infty, \infty)$.

Every independent draw from $\mathcal{N}(\mu, \sigma^2)$ is a new realisation of Y. The probability that any two draws have exactly the same value is zero!

The uncertainty is summarised by the normal PDF:

$$f_Y(y|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y-\mu)^2\right).$$

The shape of $f_Y(y|\mu, \sigma^2)$ is given as follows when $\mu = 0$ and $\sigma^2 = 1$ (see Figure 1). It is symmetric about μ and the dispersion is determined by $\sigma := \sqrt{\sigma^2}$. The whole shape is completely determined by (μ, σ^2).

If $Y \sim \mathcal{N}(\mu, \sigma^2)$ then $E[Y] = \mu$ and $V(Y) = \sigma^2$. All higher-order expectations $E[(Y - E[Y])^k]$, $k = 3, 4, \ldots$, exist and are completely determined by $V(Y)$: the normal distribution is the only possible distribution with this feature!

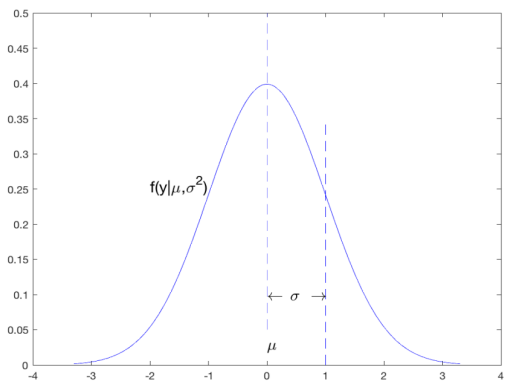

Figure 1: The shape of $f_Y(y|\mu, \sigma^2)$ is given as follows when $\mu = 0$ and $\sigma^2 = 1$.

If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = a_0 + a_1 X$, then

$$Y \sim \mathcal{N}(a_0 + a_1\mu, a_1^2\sigma^2).$$

A normal random variable $Y \sim \mathcal{N}(\mu, \sigma^2)$ can always be standardized:

$$X := \frac{Y - \mu}{\sqrt{\sigma^2}}$$

implies $X \sim \mathcal{N}(0, 1)$. $\mathcal{N}(0, 1)$ is called the standard normal distribution, and it plays a crucial role in statistical inference.

PDF and CDF values are easily computed for the standard normal: e.g., if $Y \sim \mathcal{N}(0, 1)$, then

$$F_Y(1.65) = F_Y(1.96) - F_Y(-1.96) = 0.95.$$

1.2 Matrix Algebra

We rarely work with one or two RVs (e.g. X and Y) at a time in econometrics! We use vector and matrix notation to keep things manageable when working with many (often unspecified number of) RVs.

Vectors and matrices are collections of numbers in a certain form. Let

$$\boldsymbol{a} := \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \quad \text{and} \quad \boldsymbol{b} := \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \text{ be } n \text{ dimensional column vectors.}$$

Then,

$$\boldsymbol{a}' = (a_1 \ \cdots \ a_n),$$

the transpose of $\boldsymbol{a}$, is an n-dim row vector.

Addition is defined when two vectors are in the same form.

$$\boldsymbol{a} + \boldsymbol{b} = \begin{pmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{pmatrix} \quad \text{and} \quad \boldsymbol{a}' + \boldsymbol{b}' = (\boldsymbol{a} + \boldsymbol{b})'. \text{ But } \boldsymbol{a}' + \boldsymbol{b} \text{ is not defined.}$$

Inner product

$$(\boldsymbol{a}, \boldsymbol{b}) := \boldsymbol{a}' \cdot \boldsymbol{b} = (a_1 \ \cdots \ a_n) \cdot \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = \sum_{i=1}^{n} a_i b_i.$$

Consider an $n \times k$ dimensional matrix;

$$\boldsymbol{A} := \begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nk} \end{pmatrix} = (\boldsymbol{a}_1 \ \cdots \ \boldsymbol{a}_k),$$

where

$$\boldsymbol{a}_j = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{pmatrix} \quad \text{for } j = 1, \ldots, k.$$

a vector is a special case of a matrix (e.g., $k = 1$).

Transpose:

$$\boldsymbol{A}' := \begin{pmatrix} a_{11} & \cdots & a_{n1} \\ \vdots & \ddots & \vdots \\ a_{1k} & \cdots & a_{nk} \end{pmatrix} = \begin{pmatrix} \boldsymbol{a}_1' \\ \vdots \\ \boldsymbol{a}_k' \end{pmatrix}.$$

Useful property: $(\boldsymbol{AB})' = \boldsymbol{B}'\boldsymbol{A}'$.

Addition is allowed between two matrices with the same dimensions. Let

$$\boldsymbol{B} := \begin{pmatrix} b_{11} & \cdots & b_{1k} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nk} \end{pmatrix} = (\boldsymbol{b}_1 \ \cdots \ \boldsymbol{b}_k),$$

where $\boldsymbol{b}_j$ denotes the jth column of $\boldsymbol{B}$. Then,

$$\boldsymbol{A} + \boldsymbol{B} = \begin{pmatrix} a_{11} + b_{11} & \cdots & a_{1k} + b_{1k} \\ \vdots & \ddots & \vdots \\ a_{n1} + b_{n1} & \cdots & a_{nk} + b_{nk} \end{pmatrix}.$$

But, $\boldsymbol{A}' + \boldsymbol{B}$ is not defined unless $k = n$, i.e., $\boldsymbol{A}$ and $\boldsymbol{B}$ are square.
To define multiplication, we construct a $k \times m$ matrix

$$\boldsymbol{C} := \begin{pmatrix} c_{11} & \cdots & c_{1m} \\ \vdots & \ddots & \vdots \\ c_{k1} & \cdots & c_{km} \end{pmatrix}.$$

Then,

$$\boldsymbol{A} \times \boldsymbol{C} := \begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nk} \end{pmatrix} \begin{pmatrix} c_{11} & \cdots & c_{1m} \\ \vdots & \ddots & \vdots \\ c_{k1} & \cdots & c_{km} \end{pmatrix} := \begin{pmatrix} \sum_{\ell=1}^{k} a_{1\ell} c_{\ell 1} & \sum_{\ell=1}^{k} a_{1\ell} c_{\ell 2} & \cdots & \sum_{\ell=1}^{k} a_{1\ell} c_{\ell m} \\ \sum_{\ell=1}^{k} a_{2\ell} c_{\ell 1} & \sum_{\ell=1}^{k} a_{2\ell} c_{\ell 2} & \cdots & \sum_{\ell=1}^{k} a_{2\ell} c_{\ell m} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{\ell=1}^{k} a_{n\ell} c_{\ell 1} & \sum_{\ell=1}^{k} a_{n\ell} c_{\ell 2} & \cdots & \sum_{\ell=1}^{k} a_{n\ell} c_{\ell m} \end{pmatrix}.$$

Another way to view matrix multiplication is to denote $\boldsymbol{A}$ in terms of its rows and $\boldsymbol{C}$ in terms of its columns:

$$\boldsymbol{A} = \begin{pmatrix} \boldsymbol{a}_1' \\ \vdots \\ \boldsymbol{a}_n' \end{pmatrix}, \qquad \boldsymbol{C} = (\boldsymbol{c}_1 \ \cdots \ \boldsymbol{c}_m),$$

where $\boldsymbol{a}_i$ is $k \times 1$ and $\boldsymbol{c}_j$ is $k \times 1$.

Then,

$$\boldsymbol{A} \times \boldsymbol{C} = \begin{pmatrix} \boldsymbol{a}_1'\boldsymbol{c}_1 & \cdots & \boldsymbol{a}_1'\boldsymbol{c}_m \\ \vdots & \ddots & \vdots \\ \boldsymbol{a}_n'\boldsymbol{c}_1 & \cdots & \boldsymbol{a}_n'\boldsymbol{c}_m \end{pmatrix}.$$

The $(i,j)^{\text{th}}$ entry of $\boldsymbol{A} \times \boldsymbol{C}$ is

$$\boldsymbol{a}_i'\boldsymbol{c}_j = \sum_{\ell=1}^{k} a_{i\ell} c_{\ell j}.$$

Multiplication is defined only when the number of columns of the first matrix is equal to the number of rows of the second matrix.

Example 1.2.1. For example, $\boldsymbol{A} \times \boldsymbol{C}$ is well defined and the resulting matrix is $n \times m$ dimensional but $\boldsymbol{C} \times \boldsymbol{A}$ is not defined unless $m = n$.

Suppose $\boldsymbol{A}$ is a square matrix, i.e., $n = k$.

- a_{ij} is called a diagonal element if $i = j$; otherwise, it is off-diagonal.
- If off-diagonal elements are all zero, $\boldsymbol{A}$ is said to be a diagonal matrix.
- If $\boldsymbol{A}' = \boldsymbol{A}$, it is symmetric ($a_{ij} = a_{ji}$ for all (i,j)).

If $\boldsymbol{A}$ is diagonal and all diagonal elements are ones, $\boldsymbol{A}$ is a n-dimensional identity matrix, which we denote by $\boldsymbol{I}_n$. So,

$$\boldsymbol{I}_n := \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Multiplication with the identity does not change the other matrix. That is,

$$\boldsymbol{A} \times \boldsymbol{I}_n = \boldsymbol{A} \quad \text{and} \quad \boldsymbol{I}_n \times \boldsymbol{B} = \boldsymbol{B}.$$

Essentially, $\boldsymbol{I}_n$ is the matrix version of the number 1.
For a square $n \times n$ matrix $\boldsymbol{A}$, if there is another $n \times n$ matrix $\boldsymbol{B}$ such that

$$\boldsymbol{I}_n = \boldsymbol{A}\boldsymbol{B} = \boldsymbol{B}\boldsymbol{A},$$

then $\boldsymbol{B}$ is called the inverse of $\boldsymbol{A}$ and denoted by $\boldsymbol{A}^{-1}$. In this case, $\boldsymbol{A}$ is said to be invertible or nonsingular.

1. $(c\boldsymbol{A})^{-1} = \frac{1}{c}\boldsymbol{A}^{-1}$ for a scalar c,
2. $(\boldsymbol{A}\boldsymbol{B})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$,

3. $(\boldsymbol{A}')^{-1} = (\boldsymbol{A}^{-1})'$.

If $\boldsymbol{A}^{-1} = \boldsymbol{A}'$ then $\boldsymbol{A}$ is called a unitary matrix.
Every $n \times n$ unitary matrix satisfies

$$\boldsymbol{A}'\boldsymbol{A} = \boldsymbol{A}\boldsymbol{A}' = \boldsymbol{I}_n.$$

Essentially, a unitary matrix is the matrix version of ± 1.

Example 1.2.2.

$$\boldsymbol{A} = \begin{pmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{pmatrix},$$

where $\boldsymbol{A} = \boldsymbol{I}_n$ iff $a = 0$, $\boldsymbol{A} = -\boldsymbol{I}_n$ iff $a = \pi$ and

$$\boldsymbol{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

iff $a = \pi/2$.

For any $n \times n$ matrix $\boldsymbol{A}$, the trace of $\boldsymbol{A}$, $\mathrm{tr}(\boldsymbol{A})$, is the sum of diagonal elements

$$\mathrm{tr}(\boldsymbol{A}) = \sum_{i=1}^{n} a_{ii}.$$

Property 1.2.3. *Properties of the trace:*

1. $\mathrm{tr}(\boldsymbol{I}_n) = n$,
2. $\mathrm{tr}(\boldsymbol{A}') = \mathrm{tr}(\boldsymbol{A})$,
3. $\mathrm{tr}(\boldsymbol{A} + \boldsymbol{B}) = \mathrm{tr}(\boldsymbol{A}) + \mathrm{tr}(\boldsymbol{B})$,
4. $\mathrm{tr}(c\boldsymbol{A}) = c \cdot \mathrm{tr}(\boldsymbol{A})$ *for a scalar* c,
5. $\mathrm{tr}(\boldsymbol{A}\boldsymbol{B}) = \mathrm{tr}(\boldsymbol{B}\boldsymbol{A})$ *for any dimension-conformable* $\boldsymbol{A}$ *and* $\boldsymbol{B}$.
6. $\mathrm{tr}(\boldsymbol{A}\boldsymbol{B}\boldsymbol{A}') = \mathrm{tr}(\boldsymbol{A}'\boldsymbol{B}\boldsymbol{A}) = \mathrm{tr}(\boldsymbol{B})$ *if* $\boldsymbol{A}$ *is unitary.*

For any $n \times n$ matrix $\boldsymbol{A}$, we can also compute the determinant of $\boldsymbol{A}$, $\det(\boldsymbol{A}) \equiv |\boldsymbol{A}|$. For $n = 2$,

$$|\boldsymbol{A}| = a_{11}a_{22} - a_{12}a_{21}.$$

The formula for general n is complicated, but analytically tractable: it can be expressed directly in terms of the elements of $\boldsymbol{A}$ or recursively using the $n = 2$ formula above.

Property 1.2.4. *Properties of the determinant:*

1. $|\boldsymbol{A}| = a_{11} \cdots a_{nn}$ *if* $a_{ij} = 0$ *for either all* $j > i$ *or all* $i < j$ *(or both),*
2. $|\boldsymbol{A}'| = |\boldsymbol{A}|$,
3. $|\boldsymbol{A}|^d = |\boldsymbol{A}^d|$ *for any integer* d *(incl. negative, so* $|\boldsymbol{A}^{-1}| = |\boldsymbol{A}|^{-1}$*),*
4. $|\boldsymbol{A}| = \pm 1$ *if* $\boldsymbol{A}$ *is unitary,*

5. $|c\boldsymbol{A}| = c^n \cdot |\boldsymbol{A}|$ *for a scalar* c,

6. $|\boldsymbol{AB}| = |\boldsymbol{A}||\boldsymbol{B}|$ *if* $\boldsymbol{A}$ *and* $\boldsymbol{B}$ *are both square,*

7. $|\boldsymbol{A}+\boldsymbol{B}| \neq |\boldsymbol{A}| + |\boldsymbol{B}|$ *in general, but*

8. $|\boldsymbol{A}+\boldsymbol{BDC}'| = |\boldsymbol{A}||\boldsymbol{D}||\boldsymbol{D}^{-1}+\boldsymbol{C}'\boldsymbol{A}^{-1}\boldsymbol{B}|$ *if* $\boldsymbol{A}$ *and* $\boldsymbol{D}$ *are square, but not necessarily the same dimension.*

Definition 1.2.5. Let $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ be a set of $n \times 1$ vectors and $c_1, \ldots, c_n$ scalars. These are linearly independent if and only if

$$c_1\boldsymbol{a}_1 + c_2\boldsymbol{a}_2 + \cdots + c_n\boldsymbol{a}_n = \boldsymbol{0}$$

implies $c_1 = c_2 = \cdots = c_n = 0$.

When $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ are linearly independent, $\boldsymbol{a}_j$ cannot be represented as a linear combination of other $\boldsymbol{a}_i$, $i \neq j$.

Definition 1.2.6. Let $\boldsymbol{A}$ be a $n \times k$ matrix. The rank of $\boldsymbol{A}$, denoted by $\text{rank}(\boldsymbol{A})$, is the maximum number of linearly independent columns in $\boldsymbol{A}$.

In general, $\text{rank}(\boldsymbol{A}) \leq \min\{n, k\}$. If $\text{rank}(\boldsymbol{A}) = k$, $\boldsymbol{X}$ has full column rank (which requires $n \geq k$).

Definition 1.2.7. Let $\boldsymbol{A}$ be a $n \times n$ symmetric matrix and $\boldsymbol{x}$ a $n \times 1$ vector. The quadratic form of $\boldsymbol{A}$ is the function $f_{\boldsymbol{A}}(\boldsymbol{x}) = \boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}$.

Definition 1.2.8. $\boldsymbol{A}$ is positive definite (p.d.) if $f_{\boldsymbol{A}}(\boldsymbol{x}) > 0$ for all $\boldsymbol{x} \neq \boldsymbol{0}$. $\boldsymbol{A}$ is positive semi-definite (p.s.d.) if $f_{\boldsymbol{A}}(\boldsymbol{x}) \geq 0$ for all $\boldsymbol{x} \neq \boldsymbol{0}$.

Property 1.2.9. *Properties of p.d. and p.s.d. matrices:*

1. *If* $\boldsymbol{A}$ *is p.d., then* $a_{ii} > 0$ *for all* i.

2. *If* $\boldsymbol{A}$ *is p.s.d., then* $a_{ii} \geq 0$ *for all* i.

3. *If* $\boldsymbol{A}$ *is p.d., then* $\boldsymbol{A}$ *is invertible, i.e.,* $\boldsymbol{A}^{-1}$ *exists.*

4. *If* $\boldsymbol{A}$ *is* $n \times k$*, then* $\boldsymbol{A}'\boldsymbol{A}$ *and* $\boldsymbol{A}\boldsymbol{A}'$ *are p.s.d.*

5. *If* $\boldsymbol{A}$ *is* $n \times k$ *and* $\text{rank}(\boldsymbol{A}) = k$*, then* $\boldsymbol{A}'\boldsymbol{A}$ *is p.d. and invertible.*

For a given $n \times 1$ vector $\boldsymbol{a}$, consider the linear function

$$g_{\boldsymbol{a}}(\boldsymbol{x}) = \boldsymbol{a}'\boldsymbol{x}$$

for all $n \times 1$ vectors $\boldsymbol{x}$. Then,

$$\frac{\partial g_{\boldsymbol{a}}(\boldsymbol{x})}{\partial \boldsymbol{x}} = \boldsymbol{a}.$$

For an $n \times n$ symmetric matrix $\boldsymbol{A}$, recall the quadratic function

$$f_{\boldsymbol{A}}(\boldsymbol{x}) = \boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}.$$

Then,

$$\frac{\partial f_{\boldsymbol{A}}(\boldsymbol{x})}{\partial \boldsymbol{x}} = 2\boldsymbol{A}\boldsymbol{x}.$$

Note that

$$\frac{\partial f_{\boldsymbol{A}}(\boldsymbol{x})}{\partial \boldsymbol{x}'} = \left(\frac{\partial f_{\boldsymbol{A}}(\boldsymbol{x})}{\partial \boldsymbol{x}}\right)' \quad \text{and} \quad \frac{\partial g_{\boldsymbol{a}}(\boldsymbol{x})}{\partial \boldsymbol{x}'} = \left(\frac{\partial g_{\boldsymbol{a}}(\boldsymbol{x})}{\partial \boldsymbol{x}}\right)'.$$

Let $\boldsymbol{y}$ be a $n \times 1$ vector and $\boldsymbol{X}$ be a $n \times k$ matrix. Consider

$$\min_{\beta_1,\ldots,\beta_k} \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{k} x_{ij}\beta_j\right)^2 = \min_{\boldsymbol{\beta}} (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}).$$

Since $(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})' = \boldsymbol{y}' - (\boldsymbol{X}\boldsymbol{\beta})' = \boldsymbol{y}' - \boldsymbol{\beta}'\boldsymbol{X}'$, the objective function is

$$(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) = (\boldsymbol{y}' - \boldsymbol{\beta}'\boldsymbol{X}')(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$$

$$= \boldsymbol{y}'\boldsymbol{y} - \boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{y} + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{y}'\boldsymbol{y} - 2\boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}.$$

Therefore,

$$\min_{\boldsymbol{\beta}} (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) = \min_{\boldsymbol{\beta}} \left(\boldsymbol{y}'\boldsymbol{y} - 2\boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}\right).$$

Then, the first order necessary condition (FOC) is:

$$\frac{\partial}{\partial \boldsymbol{\beta}} \left(\boldsymbol{y}'\boldsymbol{y} - 2\boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}\right) = -2\frac{\partial}{\partial \boldsymbol{\beta}}\boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} + \frac{\partial}{\partial \boldsymbol{\beta}}\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}.$$

Applying the rules above,

$$\frac{\partial}{\partial \boldsymbol{\beta}}\boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}'\boldsymbol{y}, \qquad \frac{\partial}{\partial \boldsymbol{\beta}}\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} = 2\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}.$$

The FOC is therefore:

$$-2\boldsymbol{X}'\boldsymbol{y} + 2\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{0} \iff \boldsymbol{X}'\boldsymbol{y} = \boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta},$$

at the optimal solution for $\boldsymbol{\beta}$, say $\hat{\boldsymbol{\beta}}$.
If $\boldsymbol{X}'\boldsymbol{X}$ is invertible, the unique solution is:

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}.$$

Recall that $\boldsymbol{X}'\boldsymbol{X}$ is invertible if and only if $\text{rank}(\boldsymbol{X}) = k$. What happens if $\text{rank}(\boldsymbol{X}) < k$? What does it mean?
Suppose we have n random variables, $Y_1, \ldots, Y_n$. Then, define the random vector:

$$\boldsymbol{y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \text{such that} \qquad E[\boldsymbol{y}] = \begin{pmatrix} E[Y_1] \\ \vdots \\ E[Y_n] \end{pmatrix}.$$

Similarly,

$$V(\boldsymbol{y}) = E[(\boldsymbol{y} - E[\boldsymbol{y}])(\boldsymbol{y} - E[\boldsymbol{y}])'],$$

where

- The $(i,j)^{\text{th}}$ element of $V(\boldsymbol{y})$ is $C(Y_i, Y_j)$.
- The ith diagonal element of $V(\boldsymbol{y})$ is $C(Y_i, Y_i) = V(Y_i)$.
- $V(\boldsymbol{y})$ is symmetric because $C(Y_i, Y_j) = C(Y_j, Y_i)$.

We can also define a $n \times m$ random matrix consisting of nm random variables $Y_{11}, \ldots, Y_{nm}$:

$$\boldsymbol{Y} = \begin{pmatrix} Y_{11} & \cdots & Y_{1m} \\ \vdots & \ddots & \vdots \\ Y_{n1} & \cdots & Y_{nm} \end{pmatrix},$$

with $E[\boldsymbol{Y}]$ the $n \times m$ matrix whose $(i,j)^{\text{th}}$ element is $E[Y_{ij}]$. However, $V(\boldsymbol{Y})$ is not well defined. Let $\boldsymbol{X}$ be another random matrix (with random vector as a special case). Then, the conditional mean and variance are:

$$E[\boldsymbol{y} \mid \boldsymbol{X}] = \begin{pmatrix} E[Y_1 \mid \boldsymbol{X}] \\ \vdots \\ E[Y_n \mid \boldsymbol{X}] \end{pmatrix}$$

and

$$V(\boldsymbol{y} \mid \boldsymbol{X}) = E[(\boldsymbol{y} - E[\boldsymbol{y} \mid \boldsymbol{X}])(\boldsymbol{y} - E[\boldsymbol{y} \mid \boldsymbol{X}])' \mid \boldsymbol{X}].$$

In this case, conditional on $\boldsymbol{X}$ is short-hand notation for conditional on the RVs $X_{11}, \ldots, X_{nm}$. If we let $\mathbb{X}$ denote the set $\{X_{11}, \ldots, X_{nm}\}$ and $\boldsymbol{x}$ denote the $nm \times 1$ random vector $(X_{11}, \ldots, X_{nm})'$, then

$$E[\boldsymbol{y} \mid \boldsymbol{X}] \equiv E[\boldsymbol{y} \mid \mathbb{X}] \equiv E[\boldsymbol{y} \mid \boldsymbol{x}],$$
$$V(\boldsymbol{y} \mid \boldsymbol{X}) \equiv V(\boldsymbol{y} \mid \mathbb{X}) \equiv V(\boldsymbol{y} \mid \boldsymbol{x}).$$

2 Ordinary Least Squares Estimation

2.1 Foundations: Data, Random Variables, and Probability Models

	$\boldsymbol{y}$	$\boldsymbol{x}_1$	$\boldsymbol{x}_2$	$\boldsymbol{x}_3$	$\boldsymbol{x}_4$	$\boldsymbol{x}_5$	$\boldsymbol{x}_6$
1	-1.1185	1	0	0.8328	1.0798	18	-0.1694
2	0.0385	1	0	0.6080	0.8328	16	-0.2481
3	-1.4322	1	1	0.1969	0.6080	15	-1.5417
4	1.8653	1	0	-0.5335	0.1969	11	0.1757
5	-0.6481	1	1	-1.1494	-0.5335	7	-1.4942
6	-0.2756	1	0	-1.7754	-1.1494	5	0.3935
7	-0.4702	1	1	-0.5716	-1.7754	10	0.3012
8	0.9111	1	0	0.7090	-0.5716	2	-0.3986
9	1.2303	1	0	-1.8803	0.7090	4	-1.0730
10	0.6417	1	1	-0.9231	-1.8803	1	0.4680
11	0.2097	1	0	-0.5524	-0.9231	0	0.0477
12	0.9120	1	0	-0.3975	-0.5524	6	0.1169
13	-3.4325	1	0	1.0823	-0.3975	5	-0.1766
14	-0.0056	1	0	-1.2701	1.0823	8	0.2709
15	0.4931	1	1	0.5890	-1.2701	5	-0.5692

Goal: extract *useful information* from a *table of numbers.*

The data table has n rows (observations), indexed $i = 1, \ldots, n$. In the example, $n = 15$. Each row in the table is an observation, indexed $1, \ldots, n$, with $n = 15$. Each column in the table is a $n \times 1$ **vector**.

Assumption 2.1.1. We view each observed column vector as *one realisation* of a corresponding *random vector.*

We have arbitrarily labelled the random vectors associated to realisations in each column: $\boldsymbol{y}, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$, with $k = 6$.

Assumption 2.1.2. Assume there exists a joint distribution that characterises all uncertainty associated with $\boldsymbol{y}, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$.

Assuming that observed data was generated by random variables with a joint probability distribution is very general. It is possible to have $\Pr(X_{i1} = X_{j1}) = 1$ for all $i = 1, \ldots, n$ and $j = 1, \ldots, n$ or $\Pr(X_{i4} = X_{(i-1)3}) = 1$ for all $i = 2, \ldots, n$. The only substantive implication: observed data is not the result of chaos (i.e. infinite uncertainty).

To derive practical methods, we need further assumptions about the joint distribution of $\boldsymbol{y}, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$. This is equivalent to constructing a probability model.

- **Intuition**: features of the joint distribution characterise the useful information we aim to extract, along with uncertainty.
- **Advantage**: we can use probability / maths to rigorously analyse the useful information and extraction methods.

Definition 2.1.3 (Probability Model)**.** The sole purpose of a probability model is to describe the joint distribution of $\boldsymbol{y}, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$ in a useful way.

Remark. 1. A probability model is not the true joint distribution (which is itself an assumption) or any other real feature of the world. A probability model is a useful tool for extracting useful info from data. Uncertainty regarding which models are most useful is inherent: continuously evaluate, compare and update.

2. A probability model does not need to be complete: it does not need to rigorously describe every detail of the joint distribution. Useful models focus on features of the distribution related to the information of interest.

3. A probability model is not an economic model. Must be explicitly connected to extract economic information.
   Econometric model = a probability model rigorously connected with an economic model.

A useful model of the joint distribution of $\boldsymbol{y}, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$ is often derived from a model of a joint distribution involving additional random variables. There are two ways to do so:

1. Consider the additional random variables $Y_i, X_{i1}, \ldots, X_{ik}$ for $i = n+1, \ldots, N$, for some integer $N \geq n$. Let $\tilde{\boldsymbol{y}}$ denote the $N \times 1$ random vector with elements Y_i, $i = 1, \ldots, N$. For each $j = 1, \ldots, k$, let $\tilde{\boldsymbol{x}}_j$ denote the $N \times 1$ random vector with elements X_{ij}, $i = 1, \ldots, N$. Note that $\tilde{\boldsymbol{y}}$ nests $\boldsymbol{y}$ as a sub-vector and $\tilde{\boldsymbol{x}}_j$ nests $\boldsymbol{x}_j$.

2. Similarly, consider the additional random variables $X_{i(k+1)}, \ldots, X_{iK}$ for $i = 1, \ldots, N$ and $K \geq k$. Using the same notation as above, these define the additional random vectors $\tilde{\boldsymbol{x}}_{k+1}, \ldots, \tilde{\boldsymbol{x}}_K$.

2.2 Conditional Moments

Let $\boldsymbol{X} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k)$ and $\tilde{\boldsymbol{X}} = (\tilde{\boldsymbol{x}}_1, \ldots, \tilde{\boldsymbol{x}}_K)$; hence, $\boldsymbol{X}$ is a $n \times k$ matrix and $\tilde{\boldsymbol{X}}$ is a $N \times K$ matrix, which nests $\boldsymbol{X}$ as its first n rows and first k columns. If $F_{\tilde{\boldsymbol{y}},\tilde{\boldsymbol{X}}}$ is the joint CDF of $\tilde{\boldsymbol{y}}, \tilde{\boldsymbol{x}}_1, \ldots, \tilde{\boldsymbol{x}}_K$, then $F_{\boldsymbol{y},\boldsymbol{X}}$ is the implied marginal CDF of $\boldsymbol{y}, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$.

Suppose the purpose is to forecast Y_{n+i}. For any "prediction" $\hat{Y}_{n+i}$, we accept that there is a good chance the prediction will not be perfect. Hence, we define an expected loss:

$$E\Big[(Y_{n+i} - \hat{Y}_{n+i})^2 \mid \tilde{\boldsymbol{X}}\Big] = E\Big[Y_{n+i}^2 \mid \tilde{\boldsymbol{X}}\Big] - 2E\Big[Y_{n+i} \mid \tilde{\boldsymbol{X}}\Big]\,\hat{Y}_{n+i} + \hat{Y}_{n+i}^2.$$

This expected loss is minimised at $\hat{Y}_{n+i} = E\Big[Y_{n+i} \mid \tilde{\boldsymbol{X}}\Big]$. Hence, the minimum expected loss is $V\Big(Y_{n+i} \mid \tilde{\boldsymbol{X}}\Big)$.

Table 1: Forecast Scenario

	$\boldsymbol{y}$	$\boldsymbol{x}_1$	$\boldsymbol{x}_2$	$\boldsymbol{x}_3$	$\boldsymbol{x}_4$	$\boldsymbol{x}_5$	$\boldsymbol{x}_6$
1	-1.1185	1	0	0.8328	1.0798	18	-0.1694
2	0.0385	1	0	0.6080	0.8328	16	-0.2481
3	-1.4322	1	1	0.1969	0.6080	15	-1.5417
4	1.8653	1	0	-0.5335	0.1969	11	0.1757
5	-0.6481	1	1	-1.1494	-0.5335	7	-1.4942
6	-0.2756	1	0	-1.7754	-1.1494	5	0.3935
7	-0.4702	1	1	-0.5716	-1.7754	10	0.3012
8	0.9111	1	0	0.7090	-0.5716	2	-0.3986
9	1.2303	1	0	-1.8803	0.7090	4	-1.0730
10	0.6417	1	1	-0.9231	-1.8803	1	0.4680
11	0.2097	1	0	-0.5524	-0.9231	0	0.0477
12	0.9120	1	0	-0.3975	-0.5524	6	0.1169
13	-3.4325	1	0	1.0823	-0.3975	5	-0.1766
14	-0.0056	1	0	-1.2701	1.0823	8	0.2709
15	0.4931	1	1	0.5890	-1.2701	5	-0.5692
16	?	1	0	1.1756	0.4775	3	-1.3012

In the preceding formulation, the features of $F_{\tilde{\boldsymbol{y}},\tilde{\boldsymbol{X}}}$ that stand out are the conditional mean and conditional variance.

- $E\Big[Y_{n+i} \mid \tilde{\boldsymbol{X}}\Big]$ is the optimal predictor.
- $V\Big(Y_{n+i} \mid \tilde{\boldsymbol{X}}\Big)$ measures uncertainty associated with the optimal predictor.

It makes sense to choose $N = n + 1$, $K = k$ and focus on modelling $E\Big[Y_i \mid \tilde{\boldsymbol{X}}\Big]$, $V\Big(Y_i \mid \tilde{\boldsymbol{X}}\Big)$, $C\Big(Y_i, Y_j \mid \tilde{\boldsymbol{X}}\Big)$.

Alternatively, suppose the purpose is to assess to what extent X_{ij} helps to predict Y_i for some $i = 1, \ldots, n$ and $j = 1, \ldots, k$. Let $\boldsymbol{X}_{-ij}$ denote the set of all elements in $\boldsymbol{X}$ excluding X_{ij}. We can

evaluate the difference in expected loss:

$$E\Big[(Y_i - E[Y_i \mid \boldsymbol{X}_{-ij}])^2 \mid \boldsymbol{X}\Big] - V(Y_i \mid \boldsymbol{X}) = (E[Y_i \mid \boldsymbol{X}] - E[Y_i \mid \boldsymbol{X}_{-ij}])^2 .$$

In this case it makes sense to set $N = n$, $K = k$ and again focus on modelling $E[Y_i \mid \boldsymbol{X}]$, $V(Y_i \mid \boldsymbol{X})$, $C(Y_i, Y_j \mid \boldsymbol{X})$.

Question: why does "additional random vectors" matter here?

Answer:

1. Forecasting.
2. Asymptotics.
3. Omitted variables.
4. ...

2.3 Linear Conditional Mean and the Linear Regression Model

Recall that $E\Big[Y_i \mid \tilde{\boldsymbol{X}}\Big]$ and $V\Big(Y_i \mid \tilde{\boldsymbol{X}}\Big)$ are functions of the NK random variables $X_{11}, \ldots, X_{NK}$.

Without loss of generality (we can always rearrange), we may set $E\Big[Y_i \mid \tilde{\boldsymbol{X}}\Big] = E[Y_i \mid \tilde{\boldsymbol{x}}_i]$, where $\tilde{\boldsymbol{x}}_i = (X_{i1}, \ldots, X_{iK})$ is the ith row of $\tilde{\boldsymbol{X}}$.

With only a small loss of generality (Taylor Expansion), we may assume that $E[Y_i \mid \tilde{\boldsymbol{x}}_i]$ is a linear function: There exist quantities $\alpha_{i1}, \ldots, \alpha_{iK}$ such that

$$E[Y_i \mid \tilde{\boldsymbol{x}}_i] = \sum_{j=1}^{K} X_{ij}\alpha_{ij} = \tilde{\boldsymbol{x}}_i'\tilde{\boldsymbol{\alpha}}_i,$$

where $\tilde{\boldsymbol{\alpha}}_i = (\alpha_{i1}, \ldots, \alpha_{iK})$ is a $K \times 1$ **vector**. $\tilde{\boldsymbol{\alpha}}_i$ does not depend on any element of $\tilde{\boldsymbol{x}}_i$.

Let $\boldsymbol{x}_i = (X_{i1}, \ldots, X_{ik})'$ denote the observed k regressors (we don't observe extra regressors). Then

$$E[Y_i \mid \boldsymbol{x}_i] = E\left(\boldsymbol{x}_i'\boldsymbol{\alpha}_i + \sum_{j=k+1}^{K} X_{ij}\alpha_{ij} \,\middle|\, \boldsymbol{x}_i\right) = \boldsymbol{x}_i'\boldsymbol{\alpha}_i + \sum_{j=k+1}^{K} E[X_{ij} \mid \boldsymbol{x}_i]\alpha_{ij}.$$

A **stronger assumption** imposes that $E[Y_i \mid \boldsymbol{x}_i]$ is linear:

$$E[Y_i \mid \boldsymbol{x}_i] = \boldsymbol{x}_i'\boldsymbol{\beta}_i,$$

which requires either $E[X_{ij} \mid \boldsymbol{x}_i]$ is linear in $\boldsymbol{x}_i$ or $\alpha_{ij} = 0$ for $j = k+1, \ldots, K$.

Since $E[\boldsymbol{x}_i E[Y_i \mid \boldsymbol{x}_i]] = E[\boldsymbol{x}_i Y_i]$, linearity implies the **moment equation**

$$E[\boldsymbol{x}_i Y_i] = E[\boldsymbol{x}_i\boldsymbol{x}_i']\boldsymbol{\beta}_i.$$

$\boldsymbol{\beta}_i$ is uniquely determined iff rank $E[\boldsymbol{x}_i\boldsymbol{x}_i'] = k$.

- If rank $E[\boldsymbol{x}_i\boldsymbol{x}_i'] = k$, then

  $$E[Y_i \mid \boldsymbol{x}_i] = \boldsymbol{x}_i'\boldsymbol{\beta}_i = \boldsymbol{x}_i' E[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1} E[\boldsymbol{x}_i Y_i].$$

  Note that $E[\boldsymbol{x}_i\boldsymbol{x}_i']$ is a matrix, where elements are all integral.

- If rank $E[\boldsymbol{x}_i\boldsymbol{x}_i'] < k$, then $E[\boldsymbol{x}_i\boldsymbol{x}_i']$ is not positive definite., and there exists $\boldsymbol{c}_i \neq \boldsymbol{0}$ such that

$$\boldsymbol{c}_i' E[\boldsymbol{x}_i\boldsymbol{x}_i']\boldsymbol{c}_i = 0 \;\Rightarrow\; E[(\boldsymbol{c}_i'\boldsymbol{x}_i)^2] = 0 \;\Rightarrow\; \boldsymbol{c}_i'\boldsymbol{x}_i = \sum_{j=1}^{k} X_{ij}c_{ij} = 0 \text{ with probability 1.}$$

  Hence some X_{ij} are redundant: linear functions of other elements of $\boldsymbol{x}_i$. In other words, they are linear dependent!

Assumption 2.3.1 (A1 + A2)**.**

$$\text{(i) } E[Y_i \mid \boldsymbol{x}_i] \text{ is linear,} \qquad \text{(ii) rank } E[\boldsymbol{x}_i\boldsymbol{x}_i'] = k, \qquad i = 1, \ldots, n.$$

Remark. (i) "Strong" does not mean "wrong"; it means the assumption should be reassessed in the context of the application. Often, relaxing it does *not* lead to a better practical method.

(ii) A1 implicitly implies strong exogeneity.

Assumption 2.3.2 (A3)**.** Assume $E[\boldsymbol{x}_i\boldsymbol{x}_i']$ and $E[\boldsymbol{x}_iY_i]$ are the same for $i = 1, \ldots, n$. That is

$$\boldsymbol{M}_x = E[\boldsymbol{x}_i\boldsymbol{x}_i'], \qquad \boldsymbol{m}_{y,x} = E[\boldsymbol{x}_iY_i], \qquad \boldsymbol{\beta}_i = \boldsymbol{M}_x^{-1}\boldsymbol{m}_{y,x} \equiv \boldsymbol{\beta}.$$

Then $E[Y_i \mid \boldsymbol{x}_i]$ is determined by the $k(k+3)/2$ unconditional moments in $E[\boldsymbol{x}_i\boldsymbol{x}_i']$ and $E[\boldsymbol{x}_iY_i]$.

However, we *cannot* compute these moments directly from the observed data: they are integrals involving $F_{y,X}$. We need a link from model objects (expectations) to sample objects.

One way (of many) to establish a link between the probability model we have constructed and the data is to define

Definition 2.3.3 (Ordinary Least Squares (OLS) Estimator)**.**

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} = \left(\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{x}_i\boldsymbol{x}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{x}_iY_i\right),$$

where $\hat{\boldsymbol{\beta}}$ is called the Ordinary Least Squares estimator and is a $k \times 1$ random vector because it is a function the random variables Y_i and X_{ij}, $i = 1, \ldots, n$, $j = 1, \ldots, k$.

Remark. (i) $\hat{\boldsymbol{\beta}}$ is only a function of RVs for which we have observed one realisation: we can plug in the data in place of $\boldsymbol{X}$ and $\boldsymbol{y}$ to obtain one realisation of $\hat{\boldsymbol{\beta}}$. However, $\boldsymbol{\beta}$ has no uncertainty; it is not a RV,

(ii) Note the connections: $\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{x}_i\boldsymbol{x}_i' \approx E[\boldsymbol{x}_i\boldsymbol{x}_i']$, $\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{x}_iY_i \approx E[\boldsymbol{x}_iY_i]$.

OLS solves the least squares problem:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{b}} \frac{1}{n}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}) = \arg\min_{\boldsymbol{b}} \frac{1}{n}\sum_{i=1}^{n}(Y_i - \boldsymbol{x}_i'\boldsymbol{b})^2.$$

This connects to the conditional expected loss viewpoint:

$$\frac{1}{n}\sum_{i=1}^{n}(Y_i - \boldsymbol{x}_i'\boldsymbol{b})^2 \approx E[(Y_i - \boldsymbol{x}_i'\boldsymbol{b})^2 \mid \boldsymbol{X}].$$

Under the linearity (A1), rank (A2), and invariant moments (A3) assumptions:

$$E[\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}] = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'E[\boldsymbol{y} \mid \boldsymbol{X}] = (\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{X})\boldsymbol{\beta} = \boldsymbol{\beta} = E[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}E[\boldsymbol{x}_iY_i] = \boldsymbol{M}_x^{-1}\boldsymbol{m}_{y,x}.$$

2.4 The OLS Estimator and Properties

Thus OLS is **unbiased** (and mean-independent of $\boldsymbol{X}$ in the lecture's phrasing).

Theorem 2.4.1 (Unbiasedness)**.**

$$E[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta} = E[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}E[\boldsymbol{x}_iY_i].$$

Example 2.4.2 (Intercept + one regressor)**.** Let $k = 2$ with $X_{i1} = 1$ for all i. Then

$$\beta_2 = \frac{C(X_{i2}, Y_i)}{V(X_{i2})}.$$

In this case, the reduction in expected loss from using X_{i2} is

$$\Big(E[Y_i \mid X_{i1}] - E[Y_i \mid X_{i1}, X_{i2}]\Big)^2 = (E[Y_i] - E[Y_i \mid X_{i2}])^2 = \beta_2^2\,(X_{i2} - E[X_{i2}])^2.$$

So β_2^2 measures how much X_{i2} reduces expected loss *per squared deviation from its mean.* The OLS slope is

$$\hat{\beta}_2 = \frac{\sum_{i=1}^n (X_{i2} - \bar{X}_2)(Y_i - \bar{Y})}{\sum_{i=1}^n (X_{i2} - \bar{X}_2)^2}.$$

Moreover,

$$E[\hat{\beta}_2^2 \mid \boldsymbol{X}] = V(\hat{\beta}_2 \mid \boldsymbol{X}) + \beta_2^2 = \frac{\sum_{i=1}^n \sum_{j=1}^n (X_{i2} - \bar{X}_2)(X_{j2} - \bar{X}_2)\, C(Y_i, Y_j \mid \boldsymbol{X})}{\Big(\sum_{i=1}^n (X_{i2} - \bar{X}_2)^2\Big)^2} + \beta_2^2.$$

More generally, $V(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) = E[\hat{\boldsymbol{\beta}}\hat{\boldsymbol{\beta}}' \mid \boldsymbol{X}] - \boldsymbol{\beta}\boldsymbol{\beta}'$, and

$$\Big(E[Y_i] - E[Y_i \mid \boldsymbol{x}_i]\Big)^2 = \mathrm{tr}\Big(\boldsymbol{\beta}\boldsymbol{\beta}'(\boldsymbol{x}_i - E[\boldsymbol{x}_i])(\boldsymbol{x}_i - E[\boldsymbol{x}_i])'\Big).$$

$V(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X})$ measures the conditional bias of $\hat{\boldsymbol{\beta}}\hat{\boldsymbol{\beta}}'$ as an estimator of $\boldsymbol{\beta}\boldsymbol{\beta}'$.

$V(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X})$ depends on $V(Y_i \mid \boldsymbol{X})$ and $C(Y_i, Y_j \mid \boldsymbol{X})$:

$$V(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\,V(\boldsymbol{y} \mid \boldsymbol{X})\,\boldsymbol{X}\,(\boldsymbol{X}'\boldsymbol{X})^{-1}.$$

Recall from ECON3330: $V(\boldsymbol{y} \mid \boldsymbol{X})$ is called covariance matrix.

Assumption 2.4.3 (A4 + A5)**.** $C(Y_i, Y_j \mid \boldsymbol{X}) = 0$ and $V(Y_i \mid \boldsymbol{X}) = \sigma_y^2$ for all $i = 1, \ldots, n$, $j \neq i$; equivalently $V(\boldsymbol{y} \mid \boldsymbol{X}) = \sigma_y^2\boldsymbol{I}_n$.

$C(Y_i, Y_j \mid \boldsymbol{X}) = E[Y_iY_j \mid \boldsymbol{X}] - \boldsymbol{\beta}'\boldsymbol{x}_i\boldsymbol{x}_j'\boldsymbol{\beta} = 0$ implies

$$E[Y_iY_j] = E[\boldsymbol{x}_iY_i]'E[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}E[\boldsymbol{x}_i\boldsymbol{x}_j']E[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}E[\boldsymbol{x}_iY_i].$$

$V(Y_i \mid \boldsymbol{X}) = E[Y_i^2 \mid \boldsymbol{X}] - \boldsymbol{\beta}'\boldsymbol{x}_i\boldsymbol{x}_i'\boldsymbol{\beta} = \sigma_y^2$ implies

$$\sigma_y^2 = E[Y_i^2] - E[\boldsymbol{x}_iY_i]'E[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}E[\boldsymbol{x}_iY_i].$$

Under the invariant moments ($E[\boldsymbol{x}_i Y_i]$ and $E[\boldsymbol{x}_i \boldsymbol{x}_i']$) assumption, this implies $E[Y_i^2]$ and $E[Y_i Y_j]$ are also invariant.

$V(\boldsymbol{y} \mid \boldsymbol{X}) = \sigma_y^2 \boldsymbol{I}_n$ implies

$$V(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) = \sigma_y^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}.$$

Theorem 2.4.4 (Gauss–Markov Theorem)**.** *Let $\tilde{\boldsymbol{\beta}} = \boldsymbol{W}\boldsymbol{y}$ be an alternative linear, unbiased estimator of $\boldsymbol{\beta}$. Assume $V(\boldsymbol{y} \mid \boldsymbol{X}) = \sigma_y^2 \boldsymbol{I}_n$. If the linearity, rank, and invariance assumptions also hold, then*

$$V(\tilde{\boldsymbol{\beta}} \mid \boldsymbol{X}) - V(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) \text{ is positive semi-definite.}$$

This is just for fun...

Proof. Let $\boldsymbol{A} = \boldsymbol{W} - (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$ (a $k \times n$ matrix). Then $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}} + \boldsymbol{A}\boldsymbol{y}$. Moreover,

$$E[\tilde{\boldsymbol{\beta}} \mid \boldsymbol{X}] = E[\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}] + \boldsymbol{A}E[\boldsymbol{y} \mid \boldsymbol{X}] = \boldsymbol{\beta} + \boldsymbol{A}\boldsymbol{X}\boldsymbol{\beta}.$$

Thus $\tilde{\boldsymbol{\beta}}$ is unbiased iff $\boldsymbol{A}\boldsymbol{X} = \boldsymbol{0}$. Then

$$\begin{aligned} V(\tilde{\boldsymbol{\beta}} \mid \boldsymbol{X}) &= \Big((\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' + \boldsymbol{A}\Big) V(\boldsymbol{y} \mid \boldsymbol{X}) \Big((\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' + \boldsymbol{A}\Big)' \\ &= \sigma_y^2 \Big[(\boldsymbol{X}'\boldsymbol{X})^{-1} + \boldsymbol{A}\boldsymbol{A}' + \boldsymbol{A}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{A}'\Big] \\ &= V(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) + \sigma_y^2 \boldsymbol{A}\boldsymbol{A}', \end{aligned}$$

where the cross terms vanish because $\boldsymbol{A}\boldsymbol{X} = \boldsymbol{0}$. Since $\boldsymbol{A}\boldsymbol{A}'$ is p.s.d., the claim follows. □

2.5 Confidence Intervals in Two Ways

Gauss–Markov Theorem is a neat result, but how does $\hat{\beta}_j$ help us measure how much X_{ij} reduces expected loss? Given the preceding assumptions, and setting $\boldsymbol{S} = (\boldsymbol{X}'\boldsymbol{X})^{-1}$, we have $V(\hat{\beta}_j \mid \boldsymbol{X}) = \sigma_y^2 S_{jj}$. The standard error is: $se(\hat{\beta}_j) = \sqrt{\sigma_y^2 S_{jj}}$.

We may now ask: How many standard errors below β_j would constitute a very unlikely realisation of $\hat{\beta}_j$? Formally, we:

1. quantify "very unlikely" with a small probability α (e.g. $\alpha = 0.025$ or $\alpha = 0.05$);
2. formulate the probability statement:

$$\Pr\Big(\hat{\beta}_j \leq \beta_j - z_l \, se(\hat{\beta}_j)\Big) = \alpha;$$

3. invert the equation to solve for z_l as a function of α.

Similarly, we may ask: How many standard errors above β_j would constitute a very unlikely realisation of $\hat{\beta}_j$? compute z_u from $\Pr\Big(\hat{\beta}_j \geq \beta_j + z_u \, se(\hat{\beta}_j)\Big) = \alpha$.

Combining very unlikely small with very unlikely large yields:

$$\Pr\Big(\hat{\beta}_j \leq \beta_j - z_l \, se(\hat{\beta}_j) \text{ or } \hat{\beta}_j \geq \beta_j + z_u \, se(\hat{\beta}_j)\Big) = 2\alpha,$$

since the two events are mutually exclusive.

This implies that realisations of $\hat{\beta}_j$ will most likely be within z_l std errors below β_j and within z_u std errors above β_j:

$$\Pr\Big(\beta_j - z_l\, se(\hat{\beta}_j) \leq \hat{\beta}_j \leq \beta_j + z_u\, se(\hat{\beta}_j)\Big) = 1 - 2\alpha.$$

It is often interpreted as follows.

Definition 2.5.1. Consider the interval CI $\equiv \Big[\hat{\beta}_j - z_u\, se(\hat{\beta}_j),\ \hat{\beta}_j + z_l\, se(\hat{\beta}_j)\Big]$. The probability that the lower bound $\hat{\beta}_j - z_u\, se(\hat{\beta}_j)$ is below β_j and simultaneously the upper bound $\hat{\beta}_j + z_l\, se(\hat{\beta}_j)$ is above β_j, is precisely $1 - 2\alpha$. CI is called the $(1-2\alpha) \times 100\%$ **confidence interval** for β_j.

If $\alpha = 0.025$, we may conclude the random interval CI (since the lower and upper bounds are random variables) contains the deterministic quantity β_j with 95% probability.

Confidence intervals can often be easily transformed. Let $\hat{l}_j \equiv \hat{\beta}_j - z_u\, se(\hat{\beta}_j)$ and $\hat{u}_j \equiv \hat{\beta}_j + z_l\, se(\hat{\beta}_j)$; $\hat{l}_j \leq \beta_j \leq \hat{u}_j$ implies:

$$\begin{array}{ll} \hat{l}_j^2 \leq \beta_j^2 \leq \hat{u}_j^2 & \text{if} \quad \hat{l}_j \geq 0, \\ \hat{u}_j^2 \leq \beta_j^2 \leq \hat{l}_j^2 & \text{if} \quad \hat{u}_j \leq 0, \\ 0 \leq \beta_j^2 \leq \max\{|\hat{l}_j|, |\hat{u}_j|\}^2 & \text{if} \quad \hat{l}_j < 0 < \hat{u}_j. \end{array}$$

Consequently, an interval formed using the bounds defined above will have confidence level at least 95%.

Question: How much does X_{ij} reduce expected loss?

Answer:

1. $\hat{\boldsymbol{\beta}}$ can be labeled "point estimate" but that doesn't give it any useful interpretation in itself.
2. The answer comes entirely from the CI. Say the 95% CI is $[0.1, 1.5]$. Then the answer is, "the reduction in expected loss is between 0.1 and 1.5 with 95% confidence"

Remark. All values outside the interval are equally unreasonable; all values inside the interval are equally reasonable. $\hat{\boldsymbol{\beta}}$ can only help us construct the CI; nothing else. It has no other interpretation.

To actually compute CIs in practice, we need $se(\hat{\beta}_j)$, z_l and z_u. $se(\hat{\beta}_j)$ depends on σ_y^2, which we have so far left unspecified.

Recall that $\sigma_y^2 = E[Y_i^2] - E[\boldsymbol{x}_i Y_i]' E[\boldsymbol{x}_i \boldsymbol{x}_i']^{-1} E[\boldsymbol{x}_i Y_i]$. As with $\hat{\boldsymbol{\beta}}$, we can try "plugging in" sample moments for all the expectations in the expression above:

$$\hat{\sigma}_y^2 = \frac{1}{n}\boldsymbol{y}'\Big(\boldsymbol{I}_k - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\Big)\boldsymbol{y} = \frac{1}{n}\sum_i Y_i^2 - \Big(\frac{1}{n}\boldsymbol{x}_i Y_i\Big)'\Big(\frac{1}{n}\boldsymbol{x}_i\boldsymbol{x}_i'\Big)^{-1}\Big(\frac{1}{n}\boldsymbol{x}_i Y_i\Big).$$

Theorem 2.5.2 (Mean-Independence of $\hat{\sigma}_y^2$ and $\boldsymbol{X}$)**.** *Set* $\boldsymbol{M}_x = \boldsymbol{I}_n - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$ *and* $\hat{\boldsymbol{u}} = \boldsymbol{M}_x\boldsymbol{y}$. *Then,* $\hat{\sigma}_y^2 = \frac{1}{n}\hat{\boldsymbol{u}}'\hat{\boldsymbol{u}}$, $E[\hat{\sigma}_y^2 \mid \boldsymbol{X}] = \frac{n-k}{n}\sigma_y^2$, $C(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}, \hat{\boldsymbol{u}} \mid \boldsymbol{X}) = 0$.

Proof. It can be easily verified that $\boldsymbol{M}_x^2 = \boldsymbol{M}_x$, $\operatorname{tr}\boldsymbol{M}_x = n - k$ and $\boldsymbol{M}_x\boldsymbol{X} = 0$. It follows immediately that $\hat{\sigma}_y^2 = \frac{1}{n}\hat{\boldsymbol{u}}'\hat{\boldsymbol{u}}$,

$$E[\hat{\sigma}_y^2 \mid \boldsymbol{X}] = \frac{1}{n}E[\boldsymbol{y}'\boldsymbol{M}_x\boldsymbol{y} \mid \boldsymbol{X}] = \frac{1}{n}E[\operatorname{tr}(\boldsymbol{y}'\boldsymbol{M}_x\boldsymbol{y}) \mid \boldsymbol{X}] = \frac{1}{n}E[\operatorname{tr}(\boldsymbol{M}_x\boldsymbol{y}\boldsymbol{y}') \mid \boldsymbol{X}] = \frac{1}{n}\operatorname{tr}\Big(\boldsymbol{M}_x E[\boldsymbol{y}\boldsymbol{y}' \mid \boldsymbol{X}]\Big)$$

$$= \frac{1}{n}\operatorname{tr}\Big(\boldsymbol{M}_x\Big(V(\boldsymbol{y} \mid \boldsymbol{X}) + E[\boldsymbol{y} \mid \boldsymbol{X}]E[\boldsymbol{y} \mid \boldsymbol{X}]'\Big)\Big) = \frac{\operatorname{tr}\boldsymbol{M}_x}{n}\sigma_y^2 + \frac{1}{n}\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{M}_x\boldsymbol{X}\boldsymbol{\beta} = \frac{n-k}{n}\sigma_y^2,$$

and

$$\begin{aligned} C(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}, \hat{\boldsymbol{u}} \mid \boldsymbol{X}) &= E[(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\hat{\boldsymbol{u}}' \mid \boldsymbol{X}] - E[\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \mid \boldsymbol{X}]\, E[\hat{\boldsymbol{u}}' \mid \boldsymbol{X}] \\ &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'E[\boldsymbol{y}\boldsymbol{y}' \mid \boldsymbol{X}]\, \boldsymbol{M}_x - \boldsymbol{\beta}\, E[\boldsymbol{y} \mid \boldsymbol{X}]'\boldsymbol{M}_x = \sigma_y^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{M}_x. \end{aligned}$$

□

The important conclusion is that $\hat{\sigma}_y^2$ is mean-independent of $\boldsymbol{X}$. It can be easily made unbiased by re-scaling:

$$E\left[\frac{n}{n-k}\hat{\sigma}_y^2 \,\middle|\, \boldsymbol{X}\right] = \sigma_y^2.$$

We may use $\hat{\sigma}_y^2$ instead of σ_y^2 in $\widehat{se}(\hat{\beta}_j) = \sqrt{\hat{\sigma}_y^2 S_{jj}}$ and:

$$\Pr\Big(\hat{\beta}_j \le \beta_j - z_l\, \widehat{se}(\hat{\beta}_j)\Big) = \alpha, \qquad \Pr\Big(\hat{\beta}_j \ge \beta_j + z_u\, \widehat{se}(\hat{\beta}_j)\Big) = \alpha.$$

However, this is still not enough: to obtain z_l and z_u from α.

- We need the CDF for the RV $\hat{z} \equiv (\hat{\beta}_j - \beta_j)/\widehat{se}(\hat{\beta}_j)$.
- Hence, we need to at least specify the conditional distribution $(\boldsymbol{y} \mid \boldsymbol{X})$.

Assumption 2.5.3 (A6 (i))**.** $(\boldsymbol{y} \mid \boldsymbol{X}) \sim \mathcal{N}(\boldsymbol{X}\boldsymbol{\beta}, \sigma_y^2\boldsymbol{I}_n)$.

This assumption implies $E(Y_i \mid \boldsymbol{x}_i) = \boldsymbol{x}_i'\boldsymbol{\beta}$ and $V(\boldsymbol{y} \mid \boldsymbol{X}) = \sigma_y^2\boldsymbol{I}_n$. In fact, Y_i and Y_j are conditionally independent. This turns out to be the only practically useful choice to obtain an exact distribution (and, therefore, exact CDF) for $\hat{z}$.

From conditional normality, we obtain:

- $(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) \sim \mathcal{N}\Big(\boldsymbol{\beta}, \sigma_y^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\Big)$, which implies $(\hat{\beta}_j - \beta_j)/se(\hat{\beta}_j) \sim \mathcal{N}(0,1)$.
- $\hat{\boldsymbol{u}}'\hat{\boldsymbol{u}}/\sigma_y^2 \sim \chi^2_{n-k}$ (Chi-squared distribution).
- $C(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}, \hat{\boldsymbol{u}} \mid \boldsymbol{X}) = 0$ implies $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ and $\hat{\boldsymbol{u}}$ are independent.
  - $(\hat{\beta}_j - \beta_j)/se(\hat{\beta}_j)$ and $\hat{\sigma}_y^2$ are independent.
  - $\sqrt{\frac{n-k}{n}}\,(\hat{\beta}_j - \beta_j)/\widehat{se}(\hat{\beta}_j) \sim t_{n-k}$ (Student-t distribution).

Therefore, for any z,

$$\Pr\Big(\hat{\beta}_j \ge \beta_j + z\, \widehat{se}(\hat{\beta}_j)\Big) = \Pr\Big(\hat{\beta}_j \le \beta_j - z\, \widehat{se}(\hat{\beta}_j)\Big) = 1 - \Pr\left(\sqrt{\frac{n-k}{n}}\,\frac{\hat{\beta}_j - \beta_j}{\widehat{se}(\hat{\beta}_j)} \le z\sqrt{\frac{n-k}{n}}\right),$$

which yields $z = z_l = z_u$ and $t = z\sqrt{(n-k)/n}$ is computed as the $(1-\alpha)$th percentile of the Student-t distribution; the $(1-2\alpha) \times 100\%$ CI is simply

$$\hat{\beta}_j \pm t\sqrt{\frac{n}{n-k}}\,\widehat{se}(\hat{\beta}_j).$$

If we do not like the assumption $(\boldsymbol{y} \mid \boldsymbol{X}) \sim \mathcal{N}(\boldsymbol{X}\boldsymbol{\beta}, \sigma_y^2 \boldsymbol{I}_n)$, we can replace it with alternative assumptions.

Let us keep k fixed but consider additional RVs, Y_i and $\boldsymbol{x}_i$ such that the OLS estimator and the variance estimator are:

$$\tilde{\boldsymbol{\beta}} = (\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{X}})^{-1}\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{y}}, \qquad \tilde{\sigma}_y^2 = \frac{1}{N}\tilde{\boldsymbol{y}}'\Big(\boldsymbol{I}_k - \tilde{\boldsymbol{X}}(\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{X}})^{-1}\tilde{\boldsymbol{X}}'\Big)\tilde{\boldsymbol{y}}.$$

All assumptions except the Normal distribution assumption are extended to the additional random variables.

To replace the Normal distribution assumption, instead let $U_i \equiv Y_i - E[Y_i \mid \boldsymbol{x}_i] = Y_i - \boldsymbol{x}_i'\boldsymbol{\beta}$; $\tilde{\boldsymbol{u}} = (U_1, \ldots, U_N)'$.

Assumption 2.5.4 (A6 (ii))**.** Assume $(\boldsymbol{x}_i, U_i)$ and $(\boldsymbol{x}_j, U_j)$ are independent for all $i \neq j$.

The independence assumption (along with the other assumptions) implies:

- $\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{X}}/N \xrightarrow{p} E[\boldsymbol{x}_i\boldsymbol{x}_i'] \equiv \boldsymbol{Q}$ as $N \to \infty$ (WLLN);
- $\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{u}}/N \xrightarrow{p} E[\boldsymbol{x}_i U_i] = \boldsymbol{0}$ as $N \to \infty$ (WLLN);
- $\tilde{\boldsymbol{u}}'\tilde{\boldsymbol{u}}/N \xrightarrow{p} E[U_i^2] = \sigma^2$ as $N \to \infty$ (WLLN);
- Consequently,

$$\tilde{\sigma}_y^2 = \frac{\tilde{\boldsymbol{u}}'\tilde{\boldsymbol{M}}_x\tilde{\boldsymbol{u}}}{N} = \frac{\tilde{\boldsymbol{u}}'\tilde{\boldsymbol{u}}}{N} - \left(\frac{\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{u}}}{N}\right)'\left(\frac{\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{X}}}{N}\right)^{-1}\left(\frac{\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{u}}}{N}\right) \xrightarrow{p} \sigma_y^2.$$

- $\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{u}}/\sqrt{N} \xrightarrow{d} \mathcal{N}(\boldsymbol{0}, \sigma_y^2\boldsymbol{Q})$ as $N \to \infty$ (Lindberg-Feller CLT);
- $\sqrt{N}(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \left(\frac{\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{X}}}{N}\right)^{-1}\left(\frac{\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{u}}}{\sqrt{N}}\right) \xrightarrow{d} \mathcal{N}(0, \sigma_y^2\boldsymbol{Q}^{-1})$ as $N \to \infty$ (Slutsky Lemma).

Result: Letting $\tilde{\boldsymbol{S}} = (\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{X}})^{-1}$ and $\widetilde{se}(\tilde{\beta}_j) = \sqrt{\tilde{\sigma}_y^2\tilde{S}_{jj}}$, we have $N\tilde{\boldsymbol{S}} \xrightarrow{p} \boldsymbol{Q}^{-1}$ and

$$\frac{\tilde{\beta}_j - \beta_j}{\widetilde{se}(\tilde{\beta}_j)} = \frac{\sqrt{N}(\tilde{\beta}_j - \beta_j)}{\sqrt{\tilde{\sigma}_y^2(N\tilde{S}_{jj})}} \xrightarrow{d} \mathcal{N}(0,1).$$

Putting everything all together

- If $(\boldsymbol{y} \mid \boldsymbol{X}) \sim N(\boldsymbol{X}\boldsymbol{\beta}, \sigma_y^2\boldsymbol{I}_n)$ (or $(\boldsymbol{u} \mid \boldsymbol{X}) \sim N(\boldsymbol{0}, \sigma_y^2\boldsymbol{I}_n)$) and the other assumptions hold for $i = 1, \ldots, n$, then we can construct **exact** $(1-2\alpha) \times 100\%$ CIs using $\hat{\boldsymbol{\beta}}$, $\hat{\sigma}_y^2$ and the Student-t CDF.
- If $(\boldsymbol{x}_i, U_i)$ are independent and satisfy the other assumptions for $i = 1, \ldots, N$, we can *hypothetically* construct CIs using $\tilde{\boldsymbol{\beta}}$, $\tilde{\sigma}_y^2$ and the standard Normal CDF:
  - CI significance level is approximately $(1-2\alpha) \times 100\%$,
  - approximation error goes to 0 as $N \to \infty$,
  - Student-t with $N-k$ d.f. converges to standard Normal as $N \to \infty$ (with k fixed),

– in practice we never observe $N > n$; we use $\hat{\boldsymbol{\beta}}, \hat{\sigma}_y^2$ to approximate the asymptotic CI. These are **asymptotic CIs**.

We compute

$$CI: \quad \hat{\beta}_j \pm z\, \widehat{se}(\hat{\beta}_j),$$

regardless of whether or not normality is assumed. The only difference is how we set z:

- **Normality assumption:** set $z = t\sqrt{\frac{n}{n-k}}$, where t is the $(1-\alpha)$th percentile of the Student-t CDF.
- **Alternative assumption:** set z as the $(1-\alpha)$th percentile of the standard Normal CDF.

Question: When is the difference more important or less important?

Answer:

1. Numerical difference: driven largely by degrees of freedom $n-k$.
2. The normality assumption only helps if the *other* assumptions also hold: if $V(Y_i \mid \boldsymbol{x}_i) \neq V(Y_j \mid \boldsymbol{x}_j)$ or $C(Y_i, Y_j \mid \boldsymbol{X}) \neq 0$ for some $i \neq j$, then normality does not rescue classical CI formulas.

2.6 Summary

To extract information from a given set of data using statistics tools, we always begin by assuming the data is a realisation of some RVs. We assumed the relevant information is contained the conditional mean; we focused on inference from $E[Y_i \mid \boldsymbol{x}_i]$ based on confidence intervals.

We developed inference from $E[Y_i \mid \boldsymbol{x}_i]$ based on the following assumptions:

- **A1 (linearity):** $E[Y_i \mid \boldsymbol{x}_i]$ is a linear function for $i = 1, \ldots, n$.
- **A2 (full rank):** $\operatorname{rank} E[\boldsymbol{x}_i \boldsymbol{x}_i'] = k$ for $i = 1, \ldots, n$.
- **A3 (invariant moments):** $E[\boldsymbol{x}_i \boldsymbol{x}_i']$ and $E[\boldsymbol{x}_i Y_i]$ are constant for $i = 1, \ldots, n$.
- **A4 (no correlation):** $C(Y_i, Y_j \mid \boldsymbol{X}) = 0$ for all $i \neq j$, $i = 1, \ldots, n$, $j = 1, \ldots, n$.
- **A5 (homoscedasticity):** $V(Y_i \mid \boldsymbol{X})$ is constant for $i = 1, \ldots, n$.
- **A6(i) (normality):** $(\boldsymbol{y} \mid \boldsymbol{X}) \sim N(\boldsymbol{X}\boldsymbol{\beta}, \sigma_y^2 \boldsymbol{I}_n)$, or
- **A6(ii) (independence):** A1–A5 are satisfied for $i = 1, \ldots, N$ RVs ($N \geq n$), and $(\boldsymbol{x}_i, U_i)$ is independent of $(\boldsymbol{x}_j, U_j)$ for all $i \neq j$, where $U_i = Y_i - E[Y_i \mid \boldsymbol{x}_i]$.

Assumptions A1–A5 are ordered from least to most restrictive; A6(i) is the most restrictive but yields an *exact* CI.

Assumptions A1–A6 constitute a linear regression model:

$$Y_i = \boldsymbol{x}_i' \boldsymbol{\beta} + U_i.$$

Remark. Everything so far is probability and statistics; there is no explicit connection to economics. To apply these techniques to economic phenomena, we must connect the probability model to economic theory. Key takeaway: except in a very special case (A1–A5 plus A6(i)), any CI is an approximation; the quality of approximation depends on how well finite-sample information approximates information in a hypothetical infinite sample.

3 Regression Modeling and M-Estimation

Recall the linear regression (LR) model:

$$Y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + U_i,$$

which (under assumptions A1–A6) provides a method to obtain inference on the conditional mean $E[Y \mid X]$, a feature of the conditional distribution $(Y \mid X)$.

A key question: *why* should we care about $E[Y \mid X]$?

Answer. Under the simple counting model, of course we can do some things, such as forecasting. But if we stop there, the conditional expectation we get from regression is essentially just a statistical object. (**Recall**: even the idea that $(Y \mid X)$ exists is an assumption we impose.)

Common side-questions / terminology: What do people mean by *true distribution* or *true model*? What do people mean by *true parameters*? What do people mean by *mis-specification*? What do people mean by *large sample*...

To endow $(Y \mid X)$ with *economic meaning* and to put the side-questions into a practically useful context, we need *economic models*. Useful economic models:

1. Characterise economic phenomena in a useful way:
   - They are rigorous quantitative representations.
   - They incorporate uncertainty through meaningful random variables.
   - They guarantee a unique solution for variables that can be meaningfully linked to observed data.
2. Often describe an *economic equilibrium* (micro/macro theory).
3. Can describe the outcome of a *controlled experiment* (experimental economics).

Economic models have the following elements:

- **Primitive shocks:** a set of $K_\varepsilon \times 1$ random vectors $\{\boldsymbol{\varepsilon}_i\}_{i=1}^N$, which are independent of each other and of any exogenous variables.
- **Exogenous variables:** a set of $K_{\tilde{x}} \times 1$ vectors $\{\tilde{\boldsymbol{x}}_i\}_{i=1}^N$, treated as determined *outside* the model.
- **Structural parameters:** a $K_\theta \times 1$ vector $\boldsymbol{\theta}$, whose elements represent meaningful quantities within the model.
- **Endogenous variables:** a set of $K_w \times 1$ vectors $\{\boldsymbol{w}_i\}_{i=1}^N$, uniquely determined by the model in terms of primitive shocks, exogenous variables, and structural parameters.

Econometrician: the *matchmaker* that "ships" a statistical model (e.g. conditional mean regression) with (a class of) economic models.

- Stronger link $\Rightarrow$ more reliable economics conclusion.
- Weaker link $\Rightarrow$ less reliable economic conclusion.
- No link $\Rightarrow$ $E[Y \mid \tilde{\boldsymbol{x}}]$ is a purely statistical quantity (e.g. best predictor).

3.1 Economic Models

A useful micro- or macro-economic model, indexed by m, characterises an equilibrium through a system of equations:

$$\mathcal{F}_m\Big(\{\boldsymbol{w}_i\}, \{\tilde{\boldsymbol{x}}_i\}, \{\boldsymbol{\varepsilon}_i\}, \boldsymbol{\theta}\Big) = \mathbf{0},$$

which guarantees a unique solution:

$$\boldsymbol{w}_i = g_{mi}\Big(\{\tilde{\boldsymbol{x}}_i\}, \{\boldsymbol{\varepsilon}_i\}, \boldsymbol{\theta}\Big), \qquad i = 1, \ldots, n,$$

although it is typically not possible to obtain $g_{mi}(\cdot)$ explicitly.

In general, $\boldsymbol{w}_i, \tilde{\boldsymbol{x}}_i, \boldsymbol{\varepsilon}_i$, and $\boldsymbol{\theta}$ are specific to a given model m (and so are $K_w, K_{\tilde{x}}, K_\varepsilon, K_\theta$).

If we consider a class of economic models $m = 1, \ldots, M$, it is often possible to identify common components:

- $\boldsymbol{w}_i = (\boldsymbol{w}_{i,0}', \boldsymbol{w}_{i,m}')'$, where $\boldsymbol{w}_{i,0}$ are common to all models, and $\boldsymbol{w}_{i,m}$ are specific to model m.
- $\tilde{\boldsymbol{x}}_i = (\tilde{\boldsymbol{x}}_{i,0}', \tilde{\boldsymbol{x}}_{i,m}')'$, where $\tilde{\boldsymbol{x}}_{i,0}$ are common to all models, and $\tilde{\boldsymbol{x}}_{i,m}$ are specific to model m.
- $\boldsymbol{\varepsilon}_i = (\boldsymbol{\varepsilon}_{i,0}', \boldsymbol{\varepsilon}_{i,m}')'$, where $\boldsymbol{\varepsilon}_{i,0}$ have the same meaning across all models, and $\boldsymbol{\varepsilon}_{i,m}$ are specific to model m.
- $\boldsymbol{\theta} = (\boldsymbol{\theta}_0', \boldsymbol{\theta}_m')'$, where $\boldsymbol{\theta}_0$ are parameters with common meaning across all models, and $\boldsymbol{\theta}_m$ are specific to model m.

Example 3.1.1. Suppose w_{i1} (the first element of $\boldsymbol{w}_i$) and $\tilde{x}_{i1}$ (the first element of $\tilde{\boldsymbol{x}}_i$) are common to all models. We may define $\boldsymbol{\theta}$ such that

$$\frac{dw_{i1}}{d\tilde{x}_{i1}} = \theta_1, \qquad \theta_1 \in \boldsymbol{\theta}_0.$$

This makes sense because $\frac{dw_{i1}}{d\tilde{x}_{i1}}$ quantifies a relationship between variables common to all models in the class.

3.2 The Matchmaking: LR and Economic Theory

We have data, assumed generated by random variables Y_i and $\boldsymbol{x}_i$, $i = 1, \ldots, n$. We consider inference on $E[Y_i \mid \boldsymbol{x}_i]$ using the LR model

$$Y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + U_i$$

and Assumptions A1–A5 plus either A6(i) or A6(ii).

We can match the LR to:

- **a specific model m:**
  - Obtain inference on model-specific effects,
  - Test model-specific implications,
  - Evaluate how well the economic model fits the data (relative to others).

- **a class of models** $m = 1, \ldots, M$**:**
  - Obtain robust inference on effects with common meaning,
  - Develop "stylised facts" to help improve theoretical models.

Example 3.2.1 (Automotive supply and demand)**.** Suppose we have data on quantities sold, prices, and characteristics of automobiles (non-price characteristics include size, power, fuel consumption, etc.). We want inference on demand elasticities (own- and cross- prices and characteristics). These elasticities quantify substitution patterns and are useful for policy.

If we formulate a LR, key practical questions are:

1. How do we assign data to Y_i and $\boldsymbol{x}_i$?
2. How do we interpret and verify assumptions on U_i?
3. How do we interpret $\boldsymbol{\beta}$?

3.3 BLP (1995, Econometrica): Differentiated Products

BLP develops a supply-demand model for markets with differentiated products (e.g. automobiles). There are H households and N products. Each household h has income y_h and preferences for product characteristics

$$\boldsymbol{\phi}_h = (\phi_{h1}, \ldots, \phi_{hK})'.$$

Utility of household h for product i is

$$u_{ih} = h(y_h - p_i) \exp\left\{\tilde{\boldsymbol{x}}_i' \boldsymbol{\phi}_h + \xi_i + \zeta_{ih}\right\},$$

where

- p_i is the price of product i;
- $\tilde{\boldsymbol{x}}_i$ are K characteristics of product i for which households have heterogeneous preferences;
- ξ_i is the aggregate effect of product characteristics for which households have homogeneous preferences;
- ζ_{ih} is the aggregate effect of idiosyncratic tastes, i.i.d. across households and products;
- $h(\cdot)$ is a strictly positive function.

BLP make the following general assumptions:

- ζ_{ih} follows the extreme value distribution.
- $(y_h, \boldsymbol{\phi}_h)$ are distributed across households independent of $\{\tilde{\boldsymbol{x}}_i\}$, with density $f_{y,\phi}$, $E[\boldsymbol{\phi}_h \mid y_h] \equiv \boldsymbol{\mu}_\phi$ and $\mathrm{V}(\boldsymbol{\phi}_h \mid y_h) \equiv \boldsymbol{V}_\phi$.
- Each household chooses the one product that yields maximum utility u_{ih} from the set of N products.

BLP also consider specific models of utility and price-setting:

- $m = 1$: $h(y_h - p_i) = (y_h - p_i)^{\alpha_1}$ and firms are perfectly competitive.

- $m = 2$: $h(y_h - p_i) = \exp\{-\alpha_2 p_i\}$, $\boldsymbol{V}_\phi = \mathbf{0}$, and firms are perfectly competitive.
- $m = 3$: $h(y_h - p_i) = (y_h - p_i)^{\alpha_1}$ and firms are monopolistic.
- $m = 4$: $h(y_h - p_i) = \exp\{-\alpha_2 p_i\}$, $\boldsymbol{V}_\phi = \mathbf{0}$, and firms are monopolistic.

In the BLP setting, the relative market shares are given by:

$$\ln(s_i/s_0) = \mathcal{D}_{mi}\Big(\{\ln p_j\}, \{\tilde{\boldsymbol{x}}_j\}, \{\xi_j\}, (\boldsymbol{\phi}', \alpha_m)'\Big), \qquad i = 1, \ldots, N,$$

where $s_0 = 1 - \sum_{i=1}^N s_i$ and $\alpha_3 = \alpha_1$, $\alpha_4 = \alpha_2$.

Prices are determined by the type of competition assumed:

- **Perfect competition:** prices equal marginal costs:

$$\ln p_i = \tilde{\boldsymbol{x}}_i'\boldsymbol{\gamma} + \omega_i, \qquad i = 1, \ldots, N.$$

- **Monopolistic competition:** prices have endogenous mark-ups:

$$\ln p_i = \mathcal{S}_{mi}\Big(\{\ln s_j\}, \{\tilde{\boldsymbol{x}}_j\}, \{\xi_j\}, \{\omega_j\}, (\boldsymbol{\phi}', \boldsymbol{\gamma}', \alpha_m)'\Big), \qquad i = 1, \ldots, N.$$

Set

$$\boldsymbol{w}_i = (\ln(s_i/s_0), \ln p_i)', \quad \boldsymbol{\varepsilon}_i = (\xi_i, \omega_i)', \quad \boldsymbol{\theta} = (\boldsymbol{\phi}', \boldsymbol{\gamma}', \alpha_m)', \quad \boldsymbol{\theta}_0 = (\boldsymbol{\phi}', \boldsymbol{\gamma}')'.$$

For $m = 1$ and $m = 2$, equilibrium is characterised by

$$\mathcal{F}_m(\cdot) \equiv \begin{pmatrix} \ln(s_1/s_0) - \mathcal{D}_{m1}(\{\ln p_j\}, \{\tilde{\boldsymbol{x}}_j\}, \{\xi_j\}, (\boldsymbol{\phi}', \alpha_m)') \\ \vdots \\ \ln(s_N/s_0) - \mathcal{D}_{mN}(\{\ln p_j\}, \{\tilde{\boldsymbol{x}}_j\}, \{\xi_j\}, (\boldsymbol{\phi}', \alpha_m)') \\ \ln p_1 - \tilde{\boldsymbol{x}}_1'\boldsymbol{\gamma} - \omega_1 \\ \vdots \\ \ln p_N - \tilde{\boldsymbol{x}}_N'\boldsymbol{\gamma} - \omega_N \end{pmatrix} = \mathbf{0}.$$

For $m = 3$ and $m = 4$, equilibrium is characterised by

$$\mathcal{F}_m(\cdot) \equiv \begin{pmatrix} \ln(s_1/s_0) - \mathcal{D}_{m1}(\{\ln p_j\}, \{\tilde{\boldsymbol{x}}_j\}, \{\xi_j\}, (\boldsymbol{\phi}', \alpha_m)') \\ \vdots \\ \ln(s_N/s_0) - \mathcal{D}_{mN}(\{\ln p_j\}, \{\tilde{\boldsymbol{x}}_j\}, \{\xi_j\}, (\boldsymbol{\phi}', \alpha_m)') \\ \ln p_1 - \mathcal{S}_{m1}(\{\ln s_j\}, \{\tilde{\boldsymbol{x}}_j\}, \{\xi_j\}, \{\omega_j\}, (\boldsymbol{\phi}', \boldsymbol{\gamma}', \alpha_m)') \\ \vdots \\ \ln p_N - \mathcal{S}_{mN}(\{\ln s_j\}, \{\tilde{\boldsymbol{x}}_j\}, \{\xi_j\}, \{\omega_j\}, (\boldsymbol{\phi}', \boldsymbol{\gamma}', \alpha_m)') \end{pmatrix} = \mathbf{0}.$$

Since we are mainly interested in **demand elasticities**, focus on the demand equations and consider a linear approximation to $\mathcal{D}_m(\cdot)$.

If $\mathcal{D}_{mi}(\cdot)$ is differentiable w.r.t. $\ln p_i$, $\tilde{\boldsymbol{x}}_i$, and ξ_i, a first-order Taylor expansion around

$$p_i \approx p^*, \qquad \tilde{\boldsymbol{x}}_i \approx \mathbf{0}, \qquad \xi_i \approx 0$$

yields

$$\ln(s_i/s_0) \approx \delta_0 + \delta_1 \ln p_i + \delta_2 \sum_{j \neq i} \ln p_j + \tilde{\boldsymbol{x}}_i'\boldsymbol{\delta}_3 + \sum_{j \neq i} \tilde{\boldsymbol{x}}_j'\boldsymbol{\delta}_4 + U_i,$$

where

$$\delta_1 = \frac{\partial \ln(s_i/s_0)}{\partial \ln p_i}, \quad \delta_2 = \frac{\partial \ln(s_i/s_0)}{\partial \ln p_j}, \quad \boldsymbol{\delta}_3 = \frac{\partial \ln(s_i/s_0)}{\partial \tilde{\boldsymbol{x}}_i}, \quad \boldsymbol{\delta}_4 = \frac{\partial \ln(s_i/s_0)}{\partial \tilde{\boldsymbol{x}}_j},$$

evaluated at $p_i = p^*$, $\tilde{\boldsymbol{x}}_i = \mathbf{0}$, $\xi_i = 0$, and

$$U_i = \xi_i + \rho\bar{\xi}, \qquad \bar{\xi} = \frac{1}{N}\sum_{i=1}^{N} \xi_i.$$

The linear approximation is more accurate when p_i, $\tilde{\boldsymbol{x}}_i$, and ξ_i are less dispersed and close to p^*, $\mathbf{0}$, 0.

3.4 The Match Between LR and Linearised BLP

We can establish a match as follows:

- $n = N$, $k = 1 + 2(1 + K_{\tilde{x}})$.
- $Y_i = \ln(s_i/s_0)$,

$$\boldsymbol{x}_i = \Big(1,\ \ln p_i,\ \sum_{j\neq i} \ln p_j,\ \tilde{\boldsymbol{x}}_i',\ \sum_{j\neq i} \tilde{\boldsymbol{x}}_j'\Big)'.$$

- $\boldsymbol{\beta} = \boldsymbol{\delta} \equiv (\delta_0, \delta_1, \delta_2, \boldsymbol{\delta}_3', \boldsymbol{\delta}_4')'$.
- $U_i = \xi_i + \rho\bar{\xi}$.

Note that $\boldsymbol{\delta}$ contains constant demand elasticities, evaluated at $p_i \approx p^*$, $\tilde{\boldsymbol{x}}_i \approx \mathbf{0}$, $\xi_i \approx 0$. We match these constant elasticities to moments of $(Y_i, \boldsymbol{x}_i)$: $E[\boldsymbol{x}_i Y_i]$ and $E[\boldsymbol{x}_i \boldsymbol{x}_i']$.

- One advantage: these constant elasticities are robust to specifications of $f_{y,\phi}$.
- One disadvantage: we cannot measure how elasticities vary with prices, market shares, and characteristics.

If the LR satisfies $E[U_i \mid \boldsymbol{x}_i] = 0$ (A1), then we can *only* match it to a BLP model where

$$E[\xi \mid \{p_i\}, \{\tilde{\boldsymbol{x}}_i\}] = 0.$$

Suppose

$$E[\xi_i \mid \{\tilde{\boldsymbol{x}}_i\}] = 0, \qquad E[\omega_i \mid \{\tilde{\boldsymbol{x}}_i\}] = 0, \qquad E[\xi_i\omega_i \mid \{\tilde{\boldsymbol{x}}_i\}] = 0.$$

Then for $m = 1$ and $m = 2$, p_i is essentially exogenous. In this case, an LR satisfying $E[U_i \mid \boldsymbol{x}_i] = 0$ can be matched to $m = 1$ or $m = 2$, but **not** $m = 3$ or $m = 4$.

For $m = 1$ or $m = 2$, suppose $\mathrm{C}(\xi_i, \bar{\xi} \mid \{p_i\}, \{\tilde{\boldsymbol{x}}_i\}) = 0$ and

$$\mathrm{V}(\xi_i \mid \{p_i\}, \{\tilde{\boldsymbol{x}}_i\}) = \mathrm{V}(\xi_i) = \sigma_\xi^2.$$

Hence,

$$\mathrm{C}(U_i, U_j \mid X) = \mathrm{C}(U_i, U_j) = \rho\sigma_\xi^2/N.$$

If the LR satisfies $\mathrm{C}(U_i, U_j \mid X) = 0$ (A4), then we can only match it to a BLP model with $\rho = 0$. This is satisfied by $m = 2$ (in which case also $\delta_2 = 0$ and $\boldsymbol{\delta}_4 = \mathbf{0}$) but not $m = 1$.

In summary:

- A LR satisfying A1–A5 and either A6(i) or A6(ii) can only be matched to a differentiated product market model with no interaction between product/consumer characteristics and with exogenous prices.

  BLP have a lot to say about implications on substitution patterns of the utility specification in $m = 2$ (and $m = 4$).

- To derive a statistical model that matches to $m = 1$, we can start with the LR, but relax A4–A5. Implications:
  - The OLS estimator $\hat{\boldsymbol{\beta}}$ is still unbiased but not efficient anymore.
  - A6(i) is not useful in constructing CIs.
  - A6(ii) is not satisfied for any finite N because $\mathrm{C}(U_i, U_j \mid X) = \mathrm{C}(U_i, U_j) = \rho\sigma_\xi^2/N$, but the dependence vanishes as $N \to \infty$.
  - We can replace A6(ii) with an alternative assumption such that the (hypothetical) OLS estimator $\hat{\boldsymbol{\beta}}$ is still asymptotically normal.
  - A (hypothetical) LR with $N \to \infty$ matches to a BLP model with "infinitely many products", each satisfying identical assumptions.

- To derive a statistical model that matches to $m = 3$ or $m = 4$, we need a simultaneous equations model (Week 6).

3.5 Robustifying the LR Match

What about matching to BLP models with

$$\mathrm{V}(\xi_i \mid \{p_i\}, \{\tilde{\boldsymbol{x}}_i\}) = \sigma_{\xi_i}^2 \ ?$$

If ξ_i and ξ_j remain conditionally uncorrelated, such models can be matched to a (hypothetical) LR with heteroskedastic residuals:

$$E(U_i^2 \mid X) = \sigma_{u_i}^2.$$

Extending the LR to heteroskedastic residuals is easy: we may use the same estimator $\hat{\boldsymbol{\beta}}$ to construct CIs, but need to adjust the standard errors.

The adjusted SEs are obtained from a heteroskedasticity-robust estimator of the variance matrix $\mathrm{V}(\hat{\boldsymbol{\beta}} \mid X)$:

$$\widehat{\mathrm{V}}^{HR}(\hat{\boldsymbol{\beta}}) = (\boldsymbol{X}'\boldsymbol{X})^{-1} \left(\sum_{i=1}^{n} \hat{U}_i^2 \boldsymbol{x}_i \boldsymbol{x}_i' \right) (\boldsymbol{X}'\boldsymbol{X})^{-1}.$$

- The robust estimator is widely used as the default variance matrix estimate in most applied work.

- In `R`, the `lm_robust` function in the `estimatr` package implements OLS with heteroskedasticity-robust SEs.

If we want inference on elasticities for products with different market shares, characteristics, and/or prices, the earlier linearisation (w.r.t. $\ln p_i$, $\tilde{\boldsymbol{x}}_i$, and ξ_i simultaneously) is not sufficiently flexible.

Alternative: linearise $\mathcal{D}_{mi}(\cdot)$ w.r.t. ξ_i only. This yields

$$\ln(s_i/s_0) \approx \mu_i\Big(\{p_j\}, \{\tilde{\boldsymbol{x}}_j\}, \boldsymbol{\theta} \Big) + \xi_i + \frac{1}{N} \sum_{j=1}^{N} \rho_i\Big(\{p_j\}, \{\tilde{\boldsymbol{x}}_j\}, \boldsymbol{\theta} \Big) \xi_j.$$

With exogenous prices, it can be matched to a *non-linear* statistical model:

$$Y_i = \mu_i(X, \boldsymbol{\vartheta}) + U_i,$$

where

$$\mathrm{C}(U_i, U_j \mid X) = \sigma_\xi^2 \, \rho_i\Big(\{p_j\}, \{\tilde{\boldsymbol{x}}_j\}, \boldsymbol{\theta} \Big)/N,$$

and

$$\mathrm{V}(U_i \mid X) = \sigma_\xi^2 \left(1 + \frac{2\rho_i(\cdot)}{N} + \sum_{j=1}^{N} \left(\frac{\rho_i(\cdot)}{N} \right)^2 \right) \equiv \sigma_{u_i}^2(X, \boldsymbol{\vartheta}).$$

Note again that $\mathrm{C}(U_i, U_j \mid X) \to 0$ and $\mathrm{V}(U_i \mid X) \to \sigma_\xi^2$ as $N \to \infty$.

3.6 M-Estimation

There are many non-linear statistical models that allow for more flexibility at the expense of more complexity / less parsimony: Non-linear Least Squares, Quantile Regressions, Semi-Parametric models, etc.

Most estimators can be formulated as solutions to an optimisation problem of the form:

$$\hat{\boldsymbol{\vartheta}} = \arg\min_{\boldsymbol{\vartheta}} S_n(\boldsymbol{\vartheta}), \qquad S_n(\boldsymbol{\vartheta}) = \frac{1}{n} \sum_{i=1}^{n} s(Y_i, \boldsymbol{x}_i, \boldsymbol{\vartheta}).$$

These are often referred to as M-Estimators. Note that $S_n(\boldsymbol{\vartheta})$ is an objective function that can be evaluated using available data and a particular value of $\boldsymbol{\vartheta}$, given $s(\cdot)$.

Example 3.6.1. OLS: $s(Y_i, \boldsymbol{x}_i, \boldsymbol{b}) = (Y_i - \boldsymbol{x}_i'\boldsymbol{b})^2$.

For $i = 1, \ldots, N$, $N \geq n$: Assume there exists a unique

$$\boldsymbol{\vartheta}_0 = \arg\min_{\boldsymbol{\vartheta}} E[S_N(\boldsymbol{\vartheta})].$$

Let

$$\tilde{\boldsymbol{\vartheta}} = \arg\min_{\boldsymbol{\vartheta}} S_N(\boldsymbol{\vartheta}).$$

Under assumptions in Hansen (2022, Ch. 22),

$$\sqrt{N}\,(\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{V}_\vartheta).$$

For the LR,

$$\boldsymbol{\beta} = \arg\min_{\boldsymbol{b}} E\Big[(Y_i - \boldsymbol{x}_i'\boldsymbol{b})^2 \mid X\Big]$$

We can adjust the LR, or another statistical model by re-formulating $S_N(\cdot)$ and checking that all technical assumptions are satisfied to find a corresponding M-estimator.

3.7 Summary

To obtain **economic inference** from data, we need to match a **statistical model** to a class of **economic models**.

The LR is a semi-parametric model that is practically simple to implement and has well-known properties. However, the LR model can often only be matched to a restrictive class of economic models, which limits the economic interpretation of the statistical results.

To obtain more robust inference, we may adjust the standard LR model to establish a match to a more robust class of economic models. One simple adjustment is to focus on quantiles instead of the conditional mean, which leads to the QR (and the QR can be further adjusted).

In practice, it helps to view the OLS and QR estimators as special cases of a more general methodology: M-estimation. Because we know the general properties, this helps us deduce practical inference methods for the adjusted statistical models.

4 The Likelihood Function and Maximum Likelihood Estimation

4.1 The Maximum Likelihood Principle

Suppose we have a realisation $\{y_i, x_{i1}, \ldots, x_{ik}\}_{i=1}^n$ of the random variables $\{Y_i, X_{i1}, \ldots, X_{ik}\}_{i=1}^n$ and we are considering a family of distribution models for $(\boldsymbol{y}, \boldsymbol{X})$, denoted by F.

Definition 4.1.1 (Maximum Likelihood Principle (MLP))**.** Choose the distribution model within F that maximises the likelihood of randomly drawing the given realisation.

Formally: choose the distribution whose PDF evaluated at the realisation,

$$f(y_1, \ldots, y_n, x_{11}, \ldots, x_{nk}),$$

is the maximum in F.

If two or more PDFs attain the maximum value, then the distributions are called *observationally equivalent*.

Remark. This is a purely statistical principle. All of the parameters here are just statistical parameters.

Example 4.1.2. Suppose $n = 10$ observations are

$$5, 0, 1, 1, 0, 3, 2, 3, 4, 1.$$

If F contains all possible distributions, the MLP dictates that we should trivially choose

$$\Pr(Y_1 = 5, Y_2 = 0, Y_3 = 1, Y_4 = 1, Y_5 = 0, Y_6 = 3, Y_7 = 2, Y_8 = 3, Y_9 = 4, Y_{10} = 1) = 1.$$

However, F is not very useful for extracting useful information from this data.

Key question: what are we looking for? MLE is goal specific! Suppose the goal is to predict Y_{11}.

Then, a useful family may consist of independent and identical Poisson distributions with unknown mean θ, which we denote by F_θ.

Within F_θ, each candidate distribution is characterised by

$$f(y_1, \ldots, y_n; \theta) = \prod_{i=1}^{n} \frac{e^{-\theta}\theta^{y_i}}{y_i!}, \qquad \text{for any } n = 1, \ldots, 11.$$

As in Lecture 3, we again refer to θ as a (statistical) parameter, which is not a random variable, and in this case, F_θ as a parametric family.

For the given realisation,

$$f(y_1, \ldots, y_{10}; \theta) = \frac{e^{-10\theta}\theta^{\sum_{i=1}^{10} y_i}}{\prod_{i=1}^{10} y_i!} = \frac{e^{-10\theta}\theta^{20}}{207,365}.$$

The MLP now boils down to: choose θ to maximise the parametric PDF $f(y_1, \ldots, y_{10}; \theta)$.

Definition 4.1.3. The parametric PDF is called the likelihood function of θ, denoted $L(\theta; D)$, where $D = \{y_1, \ldots, y_{10}\}$ is the set of all realisations.

Because maximising $L(\theta; D)$ subject to θ is equivalent to maximising any monotonic transformation of $L(\theta; D)$, it is often convenient to work with the log-likelihood $\ln L(\theta; D)$.

In this example, we choose

$$\theta^* = \arg\max_\theta \ln L(\theta; D) = \arg\max_\theta(-10\theta + 20\ln\theta - 207,365) = 2.$$

Hence, we would forecast Y_{11} from a Poisson distribution with PDF

$$f(y_{11}) = \frac{e^{-2}2^{y_{11}}}{y_{11}!}.$$

Note that $f(y_{11})$ characterises all forecast uncertainty: we can derive the expectation, variance, quantiles, etc. It is **Not** a point estimate.

4.2 The MLP in Econometrics

Recall that the equilibrium of an economic model with a single endogenous variable Y_i can be expressed

$$Y_i = g_{mi}(\boldsymbol{x}_i, \{\epsilon_i\}, \boldsymbol{\theta}), \qquad i = 1, \ldots, n,$$

where the vector $\boldsymbol{x}_i$ contains strictly exogenous variables, and $\{\epsilon_i\}$ are primitive shocks, distributed independent of $\{\boldsymbol{x}_i\}$.

The PDF of the joint distribution $(\{\boldsymbol{x}_i\}, \{\epsilon_i\})$, if one exists, is

$$\begin{aligned} f_m(\{\boldsymbol{x}_i\}, \{\epsilon_i\}; \boldsymbol{\theta}) &= f_m(\{\boldsymbol{x}_i\}; \boldsymbol{\theta}) \prod_{i=1}^{n} f_m(\epsilon_i; \boldsymbol{\theta}), \quad \text{(independence)} \\ &= f(\{\boldsymbol{x}_i\}) \prod_{i=1}^{n} f_m(\epsilon_i; \boldsymbol{\theta}). \quad \text{(exogeneity)} \end{aligned}$$

Hence, specifying $f_m(\epsilon_i; \boldsymbol{\theta})$, for $i = 1, \ldots, n$, through

$$f_m(\boldsymbol{X}, \boldsymbol{\varepsilon}; \boldsymbol{\theta}) \overset{\text{through } g_{mi}}{\longrightarrow} f_m(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\theta}),$$

implies the PDF

$$f_m(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\theta}) = f_m(\boldsymbol{X}; \boldsymbol{\theta})\, f_m(\boldsymbol{y} \mid \boldsymbol{X}; \boldsymbol{\theta}) = f(\boldsymbol{X}) f_m(\boldsymbol{y} \mid \boldsymbol{X}; \boldsymbol{\theta}).$$

Since ϵ_i are primitive shocks in model m, the PDFs $f_m(\epsilon_i; \boldsymbol{\theta})$ and $f_m(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\theta})$ are specific to a given economic model.

Note that we have written the PDFs $f_m(\epsilon_i; \boldsymbol{\theta})$ and $f_m(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\theta})$ as functions of random variables, which are defined within a particular economic model.

To obtain a likelihood, we must assume that we have a realisation

$$D = \{y_1, \ldots, y_n, x_{11}, \ldots, x_{nk}\}$$

from RVs that match precisely to the random vectors $\boldsymbol{y}$ and $\boldsymbol{X}$ defined in the economic model. Given the match, we may view the PDF $f_m(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\theta})$ evaluated at D as the likelihood $L(\boldsymbol{\theta}; D)$ for the structural parameters in model m.

Suppose we consider a family $F_{\boldsymbol{\vartheta}}$ of statistical models for

$$(Y_1, \ldots, Y_n, X_{11}, \ldots, X_{nk})$$

that match precisely to the economic model m.

- There exists some $\boldsymbol{\vartheta}_0 = \boldsymbol{\theta}$
- MLP: choose the distribution with PDF $f_m(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}^*)$.
- $\boldsymbol{\vartheta}^*$ depends on $f_m(\epsilon_i; \boldsymbol{\theta})$ and $g_{mi}(\cdot)$, but not $f(\boldsymbol{X})$.

Because $f_m(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\theta}) = f(\boldsymbol{X}) f_m(\boldsymbol{y} \mid \boldsymbol{X}; \boldsymbol{\theta})$ and $f(\boldsymbol{X})$ does not depend on $\boldsymbol{\vartheta}$,

$$\boldsymbol{\vartheta}^* = \arg\max_{\boldsymbol{\vartheta}} f_m(y_1, \ldots, y_n, x_{11}, \ldots, x_{nk}; \boldsymbol{\vartheta}) = \arg\max_{\boldsymbol{\vartheta}} f_m(y_1, \ldots, y_n \mid x_{11}, \ldots, x_{nk}; \boldsymbol{\vartheta}).$$

What if we consider matching statistical models such as the linear regression (LR) to a class of economic models?

Matching the LR to BLP models:

$$Y_i = \boldsymbol{x}_i' \boldsymbol{\beta} + U_i, \qquad i = 1, \ldots, n,$$

where

- $\boldsymbol{\beta}$ (some moments) is a function of $\boldsymbol{\theta}$ (parameter with economics meaning);
- $\boldsymbol{x}_i$ is function of $\{p_i\}$ and $\{\tilde{\boldsymbol{x}}_i\}$;
- $U_i = \xi_i + \rho \bar{\xi}$.

Recall that $\{\xi_i\}$ are (primitive) household utility shocks: the distribution of $\{\xi_i\}$ has PDF

$$f(\{\xi_i\}; \boldsymbol{\theta}) = \prod_{i=1}^{n} f(\xi_i; \boldsymbol{\theta}).$$

Because $\{\xi_i\}$, $\{\tilde{\boldsymbol{x}}_i\}$ are independent,

$$f(\{\tilde{\boldsymbol{x}}_i\}, \{\xi_i\}; \boldsymbol{\theta}) = f(\{\tilde{\boldsymbol{x}}_i\}) \prod_{i=1}^{n} f(\xi_i; \boldsymbol{\theta}).$$

If firms are perfectly competitive, $\{p_i\}$ and $\{\xi_i\}$ are independent, such that

$$f(\{p_i\}, \{\tilde{\boldsymbol{x}}_i\}, \{\xi_i\}; \boldsymbol{\theta}) = f(\{p_i\}, \{\tilde{\boldsymbol{x}}_i\}) \prod_{i=1}^{n} f(\xi_i; \boldsymbol{\theta}) = f(\boldsymbol{X}) \prod_{i=1}^{n} f(\xi_i; \boldsymbol{\theta}),$$

and therefore, using conditional probability,

$$f(\{\xi_i\} \mid \boldsymbol{X}; \boldsymbol{\theta}) = \frac{f(\boldsymbol{X}, \{\xi_i\}; \boldsymbol{\theta})}{f(\boldsymbol{X})} = \frac{f(\boldsymbol{X}) \prod_{i=1}^{n} f(\xi_i; \boldsymbol{\theta})}{f(\boldsymbol{X})} = \prod_{i=1}^{n} f(\xi_i; \boldsymbol{\theta}).$$

We use $f(\{\xi_i\} \mid \boldsymbol{X}; \boldsymbol{\theta}) = \prod_{i=1}^{n} f(\xi_i; \boldsymbol{\theta})$ and $U_i = \xi_i + \rho\bar{\xi}$ to derive the PDF for $(\boldsymbol{u} \mid \boldsymbol{X})$, and consequently, $(\boldsymbol{y} \mid \boldsymbol{X})$.

BLP, like most economic models, does not explicitly detail $f(\xi_i; \boldsymbol{\theta})$: any additional specification of $f(\xi_i; \boldsymbol{\theta})$ restricts the class of BLP models.

Key Question: in what sense is this important? How does one specification of $f(\xi_i; \boldsymbol{\theta})$ vs another specification affect the model implications?

Start with $\xi_i \sim \mathcal{N}(0, \sigma^2_{\xi_i})$: symmetric at zero, mesokurtic, defined by $\sigma^2_{\xi_i}$. If any property of the Normal distribution is restrictive, how and why?

- Perhaps not useful to model ξ_i as a primitive shock?
- If yes, then model it in terms of more primitive components!

If a match to BLP models with different $f(\xi_i; \boldsymbol{\theta})$ specifications is desired, then derive and analyse all implied likelihoods!

Example 4.2.1 (Normal shocks and $\rho = 0$)**.** Simplest example: $\xi_i \sim \mathcal{N}(0, \sigma^2_\xi)$ and $\rho = 0$. Then, $(U_i \mid \boldsymbol{X}) \sim \mathcal{N}(0, \sigma^2_\xi)$ and

$$f(U_i \mid \boldsymbol{x}_i; \sigma^2_\xi) = \frac{1}{\sqrt{2\pi\sigma^2_\xi}} \exp\left\{-\frac{U_i^2}{2\sigma^2_\xi}\right\},$$

which implies

$$f(Y_i \mid \boldsymbol{x}_i; \sigma^2_\xi) = \frac{1}{\sqrt{2\pi\sigma^2_\xi}} \exp\left\{-\frac{(Y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^2}{2\sigma^2_\xi}\right\}.$$

For a given realisation $D = \{y_1, \ldots, y_n, x_{11}, \ldots, x_{nk}\}$, the likelihood is

$$L(\boldsymbol{\beta}, \sigma^2_\xi; D) = \frac{f(x_{11}, \ldots, x_{nk})}{(2\pi\sigma^2_\xi)^{n/2}} \exp\left\{-\frac{1}{2\sigma^2_\xi} \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{k} x_{ij}\beta_j\right)^2\right\},$$

and the log-likelihood is

$$\ln L(\boldsymbol{\beta}, \sigma^2_\xi; D) = \ln f(x_{11}, \ldots, x_{nk}) - \frac{n}{2}\ln(2\pi\sigma^2_\xi) - \frac{1}{2\sigma^2_\xi} \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{k} x_{ij}\beta_j\right)^2.$$

Maximising $\ln L(\boldsymbol{\beta}, \sigma_\xi^2; D)$ yields

$$\boldsymbol{\beta}^* = \left(\sum_{i=1}^n \begin{pmatrix} x_{i1}^2 & \cdots & x_{i1}x_{ik} \\ \vdots & \ddots & \vdots \\ x_{ik}x_{i1} & \cdots & x_{ik}^2 \end{pmatrix} \right)^{-1} \sum_{i=1}^n \begin{pmatrix} x_{i1} \\ \vdots \\ x_{ik} \end{pmatrix} y_i, \qquad \sigma_\xi^{*2} = \frac{1}{n} \sum_{i=1}^n \left(y_i - \sum_{j=1}^k x_{ij}\beta_j^* \right)^2.$$

Compare these to:

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}, \qquad \hat{\sigma}_y^2 = \frac{1}{n}(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})'(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}).$$

$\boldsymbol{\beta}^*, \sigma_\xi^{*2}$ are equivalent to a realisation of $\hat{\boldsymbol{\beta}}, \hat{\sigma}_y^2$.

4.3 Maximum Likelihood Estimator

Why is the MLP useful? It underpins the statistical method: maximum likelihood estimation.

Definition 4.3.1. Suppose we consider the PDF $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta})$ as a function of random variables, and set

$$\hat{\boldsymbol{\vartheta}} = \arg\max_{\boldsymbol{\vartheta}} f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}).$$

$\hat{\boldsymbol{\vartheta}}$ is a random variable since it is a function of $\boldsymbol{y}$ and $\boldsymbol{X}$; we call it the Maximum Likelihood Estimator (MLE).

$$\boldsymbol{\vartheta}^* = \arg\max_{\boldsymbol{\vartheta}} L(\boldsymbol{\vartheta}; D)$$

is one realisation of the MLE, obtained by evaluating $\hat{\boldsymbol{\vartheta}}$ at D, which is one realisation of $\boldsymbol{y}$ and $\boldsymbol{X}$.

The key concept underlying maximum likelihood estimation is the following assumption.

Assumption 4.3.2. Within $F_{\boldsymbol{\vartheta}}$, there exists an "ideal" distribution with PDF $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)$, such that for every $\boldsymbol{\vartheta}$,

$$\mathbb{E}[\ln f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta})] = \int_{(\boldsymbol{y}, \boldsymbol{X})} \ln f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta})\, f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)\, d(\boldsymbol{y}, \boldsymbol{X}) < \infty.$$

Remark. The PDF $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)$ is "ideal" in the sense that it in can be matched to a class of economic models: e.g., $\boldsymbol{\vartheta}_0$ matches to the structural parameters $\boldsymbol{\theta}$ in the BLP models.

Example 4.3.3. $\boldsymbol{\vartheta}_0$ is a function of $\boldsymbol{\vartheta}$ — e.g., $\boldsymbol{\beta}, \sigma_y^2$ in the LR are functions of structural BLP parameters.

The PDF $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)$ satisfies a version of the MLP extended to the expected log-likelihood:

$$\boldsymbol{\vartheta}_0 = \arg\max_{\boldsymbol{\vartheta}} \mathbb{E}[\ln f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta})],$$

because for every $\boldsymbol{\vartheta} \neq \boldsymbol{\vartheta}_0$, Jensen's Inequality yields:

$$\begin{aligned}\mathbb{E}[\ln f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta})] - \mathbb{E}[\ln f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)] &= \mathbb{E}\left[\ln\left(\frac{f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta})}{f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)}\right)\right] \\ &\leq \ln\left(\mathbb{E}\left[\frac{f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta})}{f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)}\right]\right) = 0.\end{aligned}$$

Remark. Many textbooks refer to $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)$ as the PDF of a true distribution. In economics, the concept of a true distribution is not necessary, and often not useful.

4.3.1 Specification Uncertainty

Specification uncertainty is inevitable. Suppose we consider M families $F_{1,\boldsymbol{\vartheta}_1}, \ldots, F_{M,\boldsymbol{\vartheta}_M}$ as reasonable candidates to model $(\boldsymbol{y}, \boldsymbol{X})$. Setting

$$\boldsymbol{\vartheta}_0 = (\boldsymbol{\vartheta}_1', \ldots, \boldsymbol{\vartheta}_M')',$$

an "ideal" PDF may be formulated:

$$f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0) = \omega_1 f_1(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_1) + \cdots + \omega_M f_M(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_M),$$

with weights $\omega_1, \ldots, \omega_M$ non-negative and $\sum_{m=1}^M \omega_m = 1$.

Note that if $f_m(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_m) = f(\boldsymbol{X}) f_m(\boldsymbol{y} \mid \boldsymbol{X}; \boldsymbol{\vartheta}_m)$, then

$$f(\boldsymbol{y} \mid \boldsymbol{X}; \boldsymbol{\vartheta}_0) = \omega_1 f_1(\boldsymbol{y} \mid \boldsymbol{X}; \boldsymbol{\vartheta}_1) + \cdots + \omega_M f_M(\boldsymbol{y} \mid \boldsymbol{X}; \boldsymbol{\vartheta}_M),$$

$$\mathbb{E}[g(\boldsymbol{y}) \mid \boldsymbol{X}] = \omega_1 \mathbb{E}_1[g(\boldsymbol{y}) \mid \boldsymbol{X}] + \cdots + \omega_M \mathbb{E}_M[g(\boldsymbol{y}) \mid \boldsymbol{X}].$$

How do we set the weights? Decision Theory:

$$\{\hat{\omega}_m\} = \arg\max_{\omega_1, \ldots, \omega_M} \sum_{m=1}^M \omega_m \ln\left(\omega_m f_m(\boldsymbol{y} \mid \boldsymbol{X}; \hat{\boldsymbol{\vartheta}}_m)\right),$$

where

$$\hat{\boldsymbol{\vartheta}}_m = \arg\max_{\boldsymbol{\vartheta}_m} f_m(\boldsymbol{y} \mid \boldsymbol{X}; \boldsymbol{\vartheta}_m).$$

Solution:

$$\hat{\omega}_m = \frac{f_m(\boldsymbol{y} \mid \boldsymbol{X}; \hat{\boldsymbol{\vartheta}}_m)}{\sum_{m=1}^M f_m(\boldsymbol{y} \mid \boldsymbol{X}; \hat{\boldsymbol{\vartheta}}_m)}.$$

Intuition: If $L_1(\boldsymbol{\vartheta}_1^*; D) > L_2(\boldsymbol{\vartheta}_2^*; D)$, then $F_{1,\boldsymbol{\vartheta}_1}$ is attributed more relative weight (relative to $F_{2,\boldsymbol{\vartheta}_2}$) in carrying out inference: $\{L_m(\boldsymbol{\vartheta}_m^*; D)\}$ quantify specification uncertainty.

Remark. (i) If you try many families and in the end their likelihood values are similar and the resulting intervals are also similar, that suggests specification uncertainty may not be very important, because your inference is very robust.

(ii) But if the likelihood values are very similar while the intervals, or the interpretations implied by the intervals, are completely different, then specification uncertainty is a real and important issue.

Alternatives models can be compared using log-likelihoods or information criteria (IC). IC combine the log-likelihood with a function of k_m, which is the dimension of $\boldsymbol{\vartheta}_m$. General formula:

$$\mathrm{IC}_m = -2 \ln L_m(\boldsymbol{\vartheta}_m^*; D) + c(k_m, n).$$

The term $c(k_m, n)$ is increasing in k_m and often interpreted as a penalty for including k_m unknowns in the model. IC_m is often interpreted as a tradeoff between flexibility (larger $L_m(\boldsymbol{\vartheta}_m^*; D)$) and parsimony (smaller k_m).

Popular information criteria:

- AIC: $c(k_m, n) = 2k_m$;
- BIC: $c(k_m, n) = k_m \ln n$;
- HQC: $c(k_m, n) = 2k_m \ln(\ln n)$.

Remark. (i) If several models have very similar likelihood or IC values, and they also produce similar intervals and conclusions, then your conclusion is fairly robust to specification, because whichever model you use gives almost the same inference.

(ii) But if different models produce very different inference, then specification uncertainty is genuinely important and cannot be ignored.

4.3.2 Properties of the MLE

If $\ln f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}) = \sum_{i=1}^n \ln f(y_i, \boldsymbol{x}_i; \boldsymbol{\vartheta})$, then the MLE is also an M-Estimator with $s(y_i, \boldsymbol{x}_i; \boldsymbol{\vartheta}) = -\ln f(y_i, \boldsymbol{x}_i; \boldsymbol{\vartheta})$:

$$\hat{\boldsymbol{\vartheta}} = \arg\min_{\boldsymbol{\vartheta}} \frac{1}{n} \sum_{i=1}^n \Big(-\ln f(y_i, \boldsymbol{x}_i; \boldsymbol{\vartheta}) \Big).$$

Recall from Lecture 3 that for $i = 1, \ldots, N$, $N \geq n$, the corresponding $\tilde{\boldsymbol{\vartheta}}$ satisfies

$$\sqrt{N}(\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{V}_\vartheta).$$

Setting

$$\boldsymbol{G}(\boldsymbol{\vartheta}) = \mathbb{E}\left[\frac{\partial s}{\partial \boldsymbol{\vartheta}} \frac{\partial s}{\partial \boldsymbol{\vartheta}'}\right], \qquad \boldsymbol{H}(\boldsymbol{\vartheta}) = \mathbb{E}\left[\frac{\partial^2 s}{\partial \boldsymbol{\vartheta} \partial \boldsymbol{\vartheta}'}\right],$$

$\boldsymbol{V}_\vartheta$ is given by

$$\boldsymbol{V}_\vartheta = \boldsymbol{H}(\boldsymbol{\vartheta}_0)^{-1} \boldsymbol{G}(\boldsymbol{\vartheta}_0) \boldsymbol{H}(\boldsymbol{\vartheta}_0)^{-1}.$$

When $s(y_i, \boldsymbol{x}_i; \boldsymbol{\vartheta}) = -\ln f(y_i, \boldsymbol{x}_i; \boldsymbol{\vartheta})$, we have the information matrix equality:

$$\boldsymbol{G}(\boldsymbol{\vartheta}_0) = \boldsymbol{H}(\boldsymbol{\vartheta}_0).$$

Therefore, the MLE satisfies

$$\sqrt{N}(\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0) \xrightarrow{d} \mathcal{N}\Big(\mathbf{0}, \boldsymbol{H}(\boldsymbol{\vartheta}_0)^{-1}\Big).$$

- $N\boldsymbol{H}(\boldsymbol{\vartheta}_0)$ is called the Fisher Information Matrix.
- $\frac{1}{N}\boldsymbol{H}(\boldsymbol{\vartheta}_0)^{-1}$ is the asymptotic variance matrix of the MLE.
- $\frac{1}{N}\boldsymbol{H}(\boldsymbol{\vartheta}_0)^{-1}$ is equivalent to the Cramér-Rao Lower Bound: the MLE has the "smallest" possible asymptotic variance of all consistent estimators.
- This is a generalisation of the Gauss-Markov Theorem from Lecture 2.

4.3.3 Interpretation

The MLE is used to approximate an "ideal" PDF $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)$, which is matched to (a class of) economic models.

- For inference from the matched economic models, the MLE provides the most efficient method for extracting useful information from the available data.

What if we aim to obtain inference from economic models that do not match precisely to a specific PDF from a parametric family $F_{\boldsymbol{\vartheta}}$? Question re-phrased: How do we deal with specification uncertainty?

4.3.4 Alternative Approaches to Specification Uncertainty

- **Option 1:** Consider M families $\{F_{m,\boldsymbol{\vartheta}_m}\}$ that match to alternative economic models and set the "ideal" PDF to
$$f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0) = \sum_{m=1}^{M} \omega_m f_m(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_m).$$
  - Use MLE to approximate each $f_m(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_m)$, $m = 1, \ldots, M$.
  - Use $L_m(\boldsymbol{\vartheta}_m^*; D)$ to heuristically assess the relative weights $\{\hat{\omega}_m\}$.
  - Obtain inference from
$$f(\boldsymbol{y}, \boldsymbol{X}; \hat{\boldsymbol{\vartheta}}) = \sum_{m=1}^{M} \hat{\omega}_m f_m(\boldsymbol{y}, \boldsymbol{X}; \hat{\boldsymbol{\vartheta}}_m)$$
    or qualitatively interpret the results from $\{f_m(\boldsymbol{y}, \boldsymbol{X}; \hat{\boldsymbol{\vartheta}}_m)\}$ based on $\{\hat{\omega}_m\}$.
- **Option 2:** Consider M families $\{F_{m,\boldsymbol{\vartheta}_m}\}$ that match to alternative economic models and set the "ideal" PDF to
$$f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0) = \sum_{m=1}^{M} \omega_m f_m(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_m).$$
  - Use Bayesian Methods for exact inference on $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)$ (Lecture 5).
- **Option 3:** Treat the "ideal" PDF $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)$ as unspecified, but $\boldsymbol{\vartheta}_0$ matches to some structural $\boldsymbol{\theta}$ in a class of economic models.
  - Specify a function $s(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta})$ that satisfies
$$\boldsymbol{\vartheta}_0 = \arg\min_{\boldsymbol{\vartheta}} \mathbb{E}[s(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta})].$$
  - Use another PDF $\tilde{f}(\cdot)$ and set $s(\cdot) = -\ln \tilde{f}(\cdot)$: Quasi-MLE with asymptotic variance
$$\frac{1}{N} \boldsymbol{H}(\boldsymbol{\vartheta}_0)^{-1} \boldsymbol{G}(\boldsymbol{\vartheta}_0) \boldsymbol{H}(\boldsymbol{\vartheta}_0)^{-1}.$$
  - Use the Generalised Method of Moments (Lecture 6).

5 The Bayesian Framework

5.1 Econometric Inference and Sources of Information

Recall: For a distribution $(\boldsymbol{y}, \boldsymbol{X})$, the parametric PDF

$$f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta})$$

characterises a family of statistical models $\mathcal{F}_{\boldsymbol{\vartheta}}$ for extracting useful information from the data. The functional form of $f(\cdot)$ is defined by the family $\mathcal{F}_{\boldsymbol{\vartheta}}$, and the parameter vector $\boldsymbol{\vartheta}$ quantifies the unknown part of $\mathcal{F}_{\boldsymbol{\vartheta}}$.

To match the statistical models to economic models, assume there exists an "ideal" distribution in $\mathcal{F}_{\boldsymbol{\vartheta}}$ defined by $\boldsymbol{\vartheta}_0$.

Example 5.1.1 (Conditionally normal model)**.**

$$f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0) = f(\boldsymbol{X})\, \phi\Big(\boldsymbol{y} \mid \boldsymbol{X}\boldsymbol{\beta},\, \sigma_y^2 \boldsymbol{I}_n\Big)\,,$$

where $\phi(\boldsymbol{y} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the PDF of the multivariate normal distribution: $\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Here,

$$\boldsymbol{\vartheta}_0 = \Big(\boldsymbol{\beta}', \sigma_y^2\Big).$$

Lecture 2 identities:

$$\boldsymbol{\beta} = \mathbb{E}[\boldsymbol{x}_i \boldsymbol{x}_i']^{-1} \mathbb{E}[\boldsymbol{x}_i Y_i], \qquad \sigma_y^2 = \mathbb{E}[Y_i^2] - \mathbb{E}[\boldsymbol{x}_i Y_i]' \, \mathbb{E}[\boldsymbol{x}_i \boldsymbol{x}_i']^{-1} \mathbb{E}[\boldsymbol{x}_i Y_i].$$

Essential statistical question: What can we learn from the data about the "ideal" distribution $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)$?

Essential econometric question: What can we learn from the data about the economic models matched to $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)$?

Learning from an estimator: use $\hat{\boldsymbol{\vartheta}}$ (e.g. MLE, OLS, etc.) to construct confidence intervals (CIs) for elements of $\boldsymbol{\vartheta}_0$.

Recall: except for OLS with A6(i), CIs constructed from $\hat{\boldsymbol{\vartheta}}$ have no established properties with respect to $\boldsymbol{\vartheta}_0$.

CIs constructed from $\hat{\boldsymbol{\vartheta}}$ approximate CIs constructed from $\tilde{\boldsymbol{\vartheta}}$: the hypothetical estimator formulated using additional RVs

$$\{Y_{n+i}, \boldsymbol{x}_{n+i}\}_{i=1}^{\infty}$$

that satisfy additional assumptions.

CIs from $\tilde{\boldsymbol{\vartheta}}$ quantify uncertainty regarding statistical models in $\mathcal{F}_{\boldsymbol{\vartheta}}$ from which we interpret information in the data regarding economic phenomena.

Property 5.1.2 (Two sources of information)**.** *Econometric inference always combines two sources of information:*

1. the econometrician provides information about the distribution $(\boldsymbol{y}, \boldsymbol{X})$*;*

2. the data provides information about the distribution $(\boldsymbol{y}, \boldsymbol{X})$.

MLE:

1. Econometrician provides information regarding the distribution $(\boldsymbol{y}, \boldsymbol{X})$, up to unknown $\boldsymbol{\vartheta}_0$.
2. Econometrician provides information regarding the distribution

$$(\{Y_{n+i}, \boldsymbol{x}_{n+i}\}_{i=1}^{\infty} \mid \boldsymbol{y}, \boldsymbol{X}),$$

   up to unknown $\boldsymbol{\vartheta}_0$.
3. Data provides information regarding the unknown $\boldsymbol{\vartheta}_0$.

Less information from the econometrician means more flexibility for extracting information from data; more information from the econometrician means more parsimony in extracting information from data.

Efficiency = optimal balance between flexibility and parsimony.

5.2 Bayesian Framework

The Bayesian approach replaces the estimator with an alternative link between data and statistical models.

Assume there exists a random vector $\mathring{\boldsymbol{\vartheta}}$ such that the distribution $(\mathring{\boldsymbol{\vartheta}}, \boldsymbol{y}, \boldsymbol{X})$ has a PDF $f(\mathring{\boldsymbol{\vartheta}}, \boldsymbol{y}, \boldsymbol{X})$. The random vector $\mathring{\boldsymbol{\vartheta}}$ captures all available information about $\boldsymbol{\vartheta}_0$.

Definition 5.2.1 (Prior distribution)**.** The marginal distribution with density

$$f(\mathring{\boldsymbol{\vartheta}}) = \int f(\mathring{\boldsymbol{\vartheta}}, \boldsymbol{y}, \boldsymbol{X})\, d(\boldsymbol{y}, \boldsymbol{X})$$

is called the *prior distribution* and summarises all information about $\boldsymbol{\vartheta}_0$ available from sources outside the data.

Definition 5.2.2 (Posterior distribution)**.** The conditional distribution $(\mathring{\boldsymbol{\vartheta}} \mid \boldsymbol{y}, \boldsymbol{X})$ with density $f(\mathring{\boldsymbol{\vartheta}} \mid \boldsymbol{y}, \boldsymbol{X})$, evaluated at the realisation D and denoted $(\mathring{\boldsymbol{\vartheta}} \mid D)$, is called the *posterior distribution* and summarises all information about $\boldsymbol{\vartheta}_0$ available from all sources, including the data.

The conditional density

$$f(\boldsymbol{y}, \boldsymbol{X} \mid \mathring{\boldsymbol{\vartheta}}) = \frac{f(\mathring{\boldsymbol{\vartheta}}, \boldsymbol{y}, \boldsymbol{X})}{f(\mathring{\boldsymbol{\vartheta}})}$$

is conceptually equivalent to the density $f(\boldsymbol{y}, \boldsymbol{X} \mid \boldsymbol{\vartheta}_0)$, with $\boldsymbol{\vartheta}_0$ replaced by $\mathring{\boldsymbol{\vartheta}}$.

In practice, the most common approach is to specify:

1. the prior density $f(\mathring{\boldsymbol{\vartheta}})$ using information from economic theory, previous empirical work, etc.;
2. the conditional density $f(\boldsymbol{y}, \boldsymbol{X} \mid \mathring{\boldsymbol{\vartheta}})$ derived from $f(\boldsymbol{y}, \boldsymbol{X} \mid \boldsymbol{\vartheta}_0)$.

Definition 5.2.3 (Bayesian likelihood and marginal likelihood)**.** The density $f(\boldsymbol{y}, \boldsymbol{X} \mid \mathring{\boldsymbol{\vartheta}})$ evaluated at the realisation D is the *Bayesian likelihood* $f(D \mid \mathring{\boldsymbol{\vartheta}})$.

The marginal density

$$f(\boldsymbol{y}, \boldsymbol{X}) = \int f(\mathring{\boldsymbol{\vartheta}}, \boldsymbol{y}, \boldsymbol{X})\, d\mathring{\boldsymbol{\vartheta}},$$

evaluated at the realisation D, is the *marginal likelihood* $f(D)$.

Theorem 5.2.4 (Bayes' theorem)**.** *Given the distribution* $(\mathring{\boldsymbol{\vartheta}}, \boldsymbol{y}, \boldsymbol{X})$ *with prior density* $f(\mathring{\boldsymbol{\vartheta}})$ *and likelihood* $f(D \mid \mathring{\boldsymbol{\vartheta}})$*, we have*

$$\Pr\Big(0 < f(D) < \infty\Big) = 1,$$

and the posterior density is

$$f(\mathring{\boldsymbol{\vartheta}} \mid D) = \frac{f(\mathring{\boldsymbol{\vartheta}})\, f(D \mid \mathring{\boldsymbol{\vartheta}})}{f(D)}.$$

This "result" is almost trivial!

The posterior density $f(\mathring{\boldsymbol{\vartheta}} \mid D)$ is a complete characterisation of uncertainty regarding $\boldsymbol{\vartheta}_0$. We can compute any posterior moment $\mathbb{E}[g(\mathring{\boldsymbol{\vartheta}}) \mid D]$, quantile q_τ such that $\Pr(g(\mathring{\boldsymbol{\vartheta}}) \leq q_\tau \mid D) = \tau$, probability interval $[l_\alpha, u_\alpha]$ such that $\Pr(l_\alpha < g(\mathring{\boldsymbol{\vartheta}}) \leq u_\alpha \mid D) = \alpha$, etc.

Example 5.2.5. Given all available information, $\mathring{\vartheta}_j$ is between $l_{0.95}$ and $u_{0.95}$ with 95% probability.

The distribution $(\mathring{\boldsymbol{\vartheta}} \mid D)$ may be viewed as updating the prior with observed D; this updating is coherent:

$$f(\mathring{\boldsymbol{\vartheta}} \mid Y_1 = y_1, Y_2 = y_2) = \frac{f(\mathring{\boldsymbol{\vartheta}})\, f(y_1, y_2 \mid \mathring{\boldsymbol{\vartheta}})}{f(y_1, y_2)} = \frac{f(\mathring{\boldsymbol{\vartheta}} \mid Y_1 = y_1)\, f(y_2 \mid Y_1 = y_1, \mathring{\boldsymbol{\vartheta}})}{f(y_2 \mid Y_1 = y_1)}.$$

MLE:

1. Econometrician provides information regarding the distribution $(\boldsymbol{y}, \boldsymbol{X})$, up to unknown $\boldsymbol{\vartheta}_0$.
2. Econometrician provides information regarding the distribution

   $$(\{Y_{n+i}, \boldsymbol{x}_{n+i}\}_{i=1}^{\infty} \mid \boldsymbol{y}, \boldsymbol{X}),$$

   up to unknown $\boldsymbol{\vartheta}_0$.
3. Data provides information regarding the unknown $\boldsymbol{\vartheta}_0$.

Posterior:

1. Econometrician provides information regarding the distribution $(\boldsymbol{y}, \boldsymbol{X})$, up to unknown $\boldsymbol{\vartheta}_0$.
2. Econometrician provides information regarding the unknown $\boldsymbol{\vartheta}_0$ from prior sources.
3. Data provides information regarding the unknown $\boldsymbol{\vartheta}_0$.

5.3 Bayesian Normal Linear Regression

Consider the conditionally normal linear regression (LR):

$$Y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + U_i, \qquad (\boldsymbol{U} \mid \boldsymbol{X}) \sim \mathcal{N}(\boldsymbol{0}, \sigma_u^2 \boldsymbol{I}_n).$$

Recall that conditional normality implies A1, A4 and A5. Assumptions A2 and A3 are also not necessary in the Bayesian framework.

The Bayesian likelihood is obtained by evaluating the PDF:

$$f(\boldsymbol{y} \mid \boldsymbol{X}, \mathring{\boldsymbol{\beta}}, \mathring{\sigma}_u^2) = (2\pi\mathring{\sigma}_u^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\mathring{\sigma}_u^2}(\boldsymbol{y} - \boldsymbol{X}\mathring{\boldsymbol{\beta}})'(\boldsymbol{y} - \boldsymbol{X}\mathring{\boldsymbol{\beta}})\right\},$$

at the realisation D.

Assume the prior density is

$$f(\mathring{\boldsymbol{\beta}}, \mathring{\sigma}_u^2) = f(\mathring{\sigma}_u^2)\, f(\mathring{\boldsymbol{\beta}} \mid \mathring{\sigma}_u^2),$$

where:

$$f(\mathring{\sigma}_u^2) = \frac{(s^2)^\nu}{\Gamma(\nu)}(\mathring{\sigma}_u^2)^{-\nu-1} \exp\left\{-\frac{s^2}{\mathring{\sigma}_u^2}\right\} \qquad \text{(Inverse-Gamma)},$$

$$f(\mathring{\boldsymbol{\beta}} \mid \mathring{\sigma}_u^2) = (2\pi\mathring{\sigma}_u^2)^{-\frac{k}{2}} |\boldsymbol{V}|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2\mathring{\sigma}_u^2}(\mathring{\boldsymbol{\beta}} - \boldsymbol{b})'\boldsymbol{V}^{-1}(\mathring{\boldsymbol{\beta}} - \boldsymbol{b})\right\} \qquad \text{(Normal)}.$$

The specification implies the marginal prior distribution for $\mathring{\beta}_j$, $j = 1, \ldots, k$ is

$$\sqrt{\frac{\nu}{s^2 V_{jj}}}\,(\mathring{\beta}_j - b_j) \sim t_{2\nu}.$$

$\nu, s^2, \boldsymbol{b}, \boldsymbol{V}$ are prior tuning parameters. They are related to prior moments:

$$\mathbb{E}[\mathring{\sigma}_u^2] = \frac{s^2}{\nu - 1}, \quad \mathrm{V}(\mathring{\sigma}_u^2) = \frac{\mathbb{E}[\mathring{\sigma}_u^2]}{(\nu-2)^2}, \quad \mathbb{E}[\mathring{\beta}_j] = b_j, \quad \mathrm{V}(\mathring{\beta}_j) = \mathbb{E}[\mathring{\sigma}_u^2]\, V_{jj}.$$

The posterior density is:

$$f(\mathring{\sigma}_u^2 \mid \boldsymbol{y}, \boldsymbol{X}) = \frac{(\bar{s}^2)^{\bar{\nu}}}{\Gamma(\bar{\nu})}(\mathring{\sigma}_u^2)^{-\bar{\nu}-1} \exp\left\{-\frac{\bar{s}^2}{\mathring{\sigma}_u^2}\right\} \qquad \text{(Inverse-Gamma)},$$

$$f(\mathring{\boldsymbol{\beta}} \mid \mathring{\sigma}_u^2, \boldsymbol{y}, \boldsymbol{X}) = (2\pi\mathring{\sigma}_u^2)^{-\frac{k}{2}} |\bar{\boldsymbol{V}}|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2\mathring{\sigma}_u^2}(\mathring{\boldsymbol{\beta}} - \bar{\boldsymbol{b}})'\bar{\boldsymbol{V}}^{-1}(\mathring{\boldsymbol{\beta}} - \bar{\boldsymbol{b}})\right\} \qquad \text{(Normal)},$$

where

$$\bar{\nu} = \nu + \frac{n}{2}, \qquad \bar{s}^2 = s^2 + \frac{1}{2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b})'(\boldsymbol{I}_n + \boldsymbol{X}\boldsymbol{V}\boldsymbol{X}')^{-1}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}),$$

$$\bar{\boldsymbol{V}} = \left(\boldsymbol{V}^{-1} + \boldsymbol{X}'\boldsymbol{X}\right)^{-1}, \qquad \bar{\boldsymbol{b}} = \bar{\boldsymbol{V}}\left(\boldsymbol{V}^{-1}\boldsymbol{b} + \boldsymbol{X}'\boldsymbol{y}\right).$$

The marginal likelihood is

$$f(\boldsymbol{y} \mid \boldsymbol{X}) = (2\pi)^{-\frac{n}{2}} \frac{\Gamma(\bar{\nu})\, |\bar{\boldsymbol{V}}|^{\frac{1}{2}}\, (\bar{s}^2)^{-\bar{\nu}}}{\Gamma(\nu)\, |\boldsymbol{V}|^{\frac{1}{2}}\, (s^2)^{-\nu}}.$$

5.4 Monte Carlo Experiment

Perform an experiment by generating (pseudo-)random numbers on a computer.

Set "true" data generating process (DGP) to be a conditionally normal LR with

$$k = 3, \qquad \boldsymbol{\beta} = (0, -1, 1)', \qquad \sigma_u^2 = 0.5.$$

Set the prior tuning parameters:

$$\nu = 5, \qquad s^2 = 1, \qquad \boldsymbol{b} = \boldsymbol{0}, \qquad \boldsymbol{V} = \boldsymbol{I}_3.$$

Compare the prior, posterior, likelihood and MLE-derived 95% confidence intervals for β_3 from fictitious data sets with

$$n = 5, \quad n = 30, \quad n = 100, \quad n = 1000.$$

The slides provide four example plots:

- Example: Regression with $n = 5$.
- Example: Regression with $n = 30$.
- Example: Regression with $n = 100$.
- Example: Regression with $n = 1000$.

Each plot overlays (for β_3): a vertical line at the true β_3, the prior density, the posterior density, the likelihood, and the OLS and MLE 95% CIs.

6 Parameter Identification and Instrumental Variable Estimation

6.1 Identification

Recall:

- A *statistical model* is a distribution with PDF $f(\boldsymbol{y}, \boldsymbol{X})$ such that the data $\boldsymbol{D}$ is one realisation of $\boldsymbol{y}$ and $\boldsymbol{X}$.
- For a parametric family $\mathcal{F}_\vartheta$, the PDF is $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta})$, where f is fixed and differences in distribution are attributed entirely to different values of $\boldsymbol{\vartheta}$.

Definition 6.1.1. If two or more PDFs evaluated at $\boldsymbol{D}$ attain the maximum, the distributions are called *observationally equivalent*.

Definition 6.1.2. All possible values of $\boldsymbol{\vartheta}$ are called a *parameter space*, denoted Θ; e.g., $\Theta = \mathbb{R}^k$.

Also recall that we match an "ideal" distribution with PDF $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0)$ to economic equilibrium models, and

$$\boldsymbol{\vartheta}_0 = \arg\max_{\boldsymbol{\vartheta}} \ \mathbb{E}[\ln f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta})] \,.$$

The ideal distribution is observationally equivalent to another distribution if $L(\boldsymbol{\vartheta}_0; \boldsymbol{D}) = L(\boldsymbol{\vartheta}; \boldsymbol{D})$ for some $\boldsymbol{\vartheta} \in \Theta$.

Definition 6.1.3 (Identification of a parameter vector)**.** The parameter vector $\boldsymbol{\vartheta}_0$ is *identified* if there is no other observationally equivalent parameter in Θ. This is equivalent to:

$$f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}_0) = f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\vartheta}) \quad \Leftrightarrow \quad \boldsymbol{\vartheta}_0 = \boldsymbol{\vartheta}.$$

All estimators (OLS, MLE, etc.) require that $\boldsymbol{\vartheta}_0$ is identified. However, a posterior distribution does not require full identification.

Example 6.1.4 (OLS identification via a rank condition)**.** Consider

$$Y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + U_i, \qquad \mathbb{E}[U_i \mid \boldsymbol{x}_i] = 0, \qquad \mathrm{V}(U_i \mid \boldsymbol{X}) = \sigma_u^2.$$

The parameter space is $\mathbb{R}^k \times \mathbb{R}_+$.

Since $\mathbb{E}[U_i \mid \boldsymbol{x}_i] = 0 \Rightarrow \mathbb{E}[\boldsymbol{x}_i U_i] = 0$, $\boldsymbol{\beta}$ is identified if

$$\mathbb{E}[\boldsymbol{x}_i\,(Y_i - \boldsymbol{x}_i'\boldsymbol{\beta}_0)] = \mathbb{E}[\boldsymbol{x}_i\,(Y_i - \boldsymbol{x}_i'\boldsymbol{\beta}_1)] \;\Rightarrow\; \boldsymbol{\beta}_0 = \boldsymbol{\beta}_1.$$

Rearranging:

$$\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']\,(\boldsymbol{\beta}_0 - \boldsymbol{\beta}_1) = \mathbf{0}.$$

So $\boldsymbol{\beta}_0 = \boldsymbol{\beta}_1$ if $\operatorname{rank}\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i'] = k$, since then

$$\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']\,(\boldsymbol{\beta}_0 - \boldsymbol{\beta}_1) = \mathbf{0} \;\Rightarrow\; \boldsymbol{\beta}_0 = \boldsymbol{\beta}_1.$$

Example 6.1.5 (Failure of identification under perfect collinearity)**.** Consider

$$Y_i = \alpha_0 + \alpha_1 X_i + \alpha_2(0.5X_i) + U_i, \qquad \mathbb{E}[U_i \mid \boldsymbol{x}_i] = 0, \qquad \mathrm{V}(U_i \mid \boldsymbol{X}) = \sigma_u^2,$$

with $\boldsymbol{x}_i = (1,\ X_i,\ 0.5X_i)'$. Then

$$\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i'] = \begin{pmatrix} 1 & \mathbb{E}[X_i] & 0.5\mathbb{E}[X_i] \\ \mathbb{E}[X_i] & \mathbb{E}[X_i^2] & 0.5\mathbb{E}[X_i^2] \\ 0.5\mathbb{E}[X_i] & 0.5\mathbb{E}[X_i^2] & 0.25\mathbb{E}[X_i^2] \end{pmatrix}.$$

Since the third row is $0.5\times$ the second row, $\operatorname{rank}\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i'] = 2$. Hence $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \alpha_2)'$ is not identified.

However, set

$$\beta_0 = \alpha_0, \qquad \beta_1 = \alpha_1 + 0.5\alpha_2.$$

Then $\boldsymbol{\beta} = (\beta_0, \beta_1)'$ is identified.

In general, if $\boldsymbol{\vartheta}_0$ is not identified, there exists a dimension-reducing transformation of $\boldsymbol{\vartheta}$ that is.

6.2 Structural Identification

In the BLP example, we matched a linear regression to the demand equations:

$$\ln(s_i/s_0) = D_{mi}\Big(\{\ln p_i\}, \{\tilde{\boldsymbol{x}}_i\}, \{\xi_i\}, (\boldsymbol{\phi}', \alpha_m)'\Big), \qquad i = 1, \ldots, N,$$

by applying a first-order approximation. The regression coefficients $\boldsymbol{\beta}$ were matched exactly to structural parameters $\boldsymbol{\delta}$, interpreted as elasticities. In this case, $\boldsymbol{\beta}$ identified $\Rightarrow$ $\boldsymbol{\delta}$ identified: we can compute $\boldsymbol{\delta}$ precisely from $\boldsymbol{\beta}$.

However, in BLP models with monopolistic competition, there is also a set of supply equations:

$$\ln p_i = S_{mi}\Big(\{\ln s_i\}, \{\tilde{\boldsymbol{x}}_i\}, \{\xi_i\}, \{\omega_i\}, (\boldsymbol{\phi}', \boldsymbol{\gamma}', \alpha_m)'\Big), \qquad i = 1, \ldots, N.$$

To match a statistical model to BLPs with monopolistic competition, we can apply a first-order approximation of the supply equation and expand the linear regression to a system of linear regressions:

$$\begin{aligned} Y_{i1} + \beta_{12} Y_{i2} + \boldsymbol{x}_i' \boldsymbol{\gamma}_1 &= U_{i1}, \\ Y_{i2} + \beta_{21} Y_{i1} + \boldsymbol{x}_i' \boldsymbol{\gamma}_2 &= U_{i2}, \end{aligned}$$

where $\mathbb{E}[U_{i1} \mid \boldsymbol{X}] = \mathbb{E}[U_{i2} \mid \boldsymbol{X}] = 0$.

The system of linear regressions is called a *linear structural equations model (SEM)*. The term *structural* comes from the assumption that β_{12}, β_{21}, $\boldsymbol{\gamma}_1$ and $\boldsymbol{\gamma}_2$ can be matched precisely to parameters in the first-order approximation of the BLP equilibrium conditions.

Structural identification: two important questions.

1. Are linear SEM parameters identified from the data?
2. Are BLP parameters identified from linear SEM parameters?

The G-equation SEM is written as

$$\boldsymbol{y}_i' \boldsymbol{B} + \boldsymbol{x}_i' \boldsymbol{\Gamma} = \boldsymbol{u}_i',$$

where $\boldsymbol{y}_i, \boldsymbol{B}, \boldsymbol{x}_i, \boldsymbol{\Gamma}, \boldsymbol{u}_i$ have dimensions

$$(G \times 1),\ (G \times G),\ (k \times 1),\ (k \times G),\ (G \times 1),$$

respectively.

Other notation (same model):

$$\boldsymbol{B}\boldsymbol{y}_i + \boldsymbol{\Gamma}\boldsymbol{x}_i = \boldsymbol{u}_i, \qquad (G \times G)(G \times 1) + (G \times k)(k \times 1) = (G \times 1),$$

and for the g-th equation,

$$\boldsymbol{y}_i' \boldsymbol{\beta}_g + \boldsymbol{x}_i' \boldsymbol{\Gamma}_g = U_{ig}, \qquad (1 \times G)(G \times 1) + (1 \times k)(k \times 1) = (1 \times 1),$$

where $\boldsymbol{\beta}_g$ and $\boldsymbol{\Gamma}_g$ denote the g-th columns of $\boldsymbol{B}$ and $\boldsymbol{\Gamma}$.

Set

$$\boldsymbol{Y}_{(n \times G)} = \begin{pmatrix} \boldsymbol{y}_1' \\ \vdots \\ \boldsymbol{y}_n' \end{pmatrix}, \qquad \boldsymbol{X}_{(n \times k)} = \begin{pmatrix} \boldsymbol{x}_1' \\ \vdots \\ \boldsymbol{x}_n' \end{pmatrix}, \qquad \boldsymbol{U}_{(n \times G)} = \begin{pmatrix} \boldsymbol{u}_1' \\ \vdots \\ \boldsymbol{u}_n' \end{pmatrix}.$$

The entire system may be written $\boldsymbol{Y}\boldsymbol{B} + \boldsymbol{X}\boldsymbol{\Gamma} = \boldsymbol{U}$ and the statistical model has PDF $f(\boldsymbol{Y}, \boldsymbol{X}; \vartheta_0)$.

Assumption 6.2.1. • SEM1: $\boldsymbol{B}$ is nonsingular and has rank G.

- SEM2: $\operatorname{rank} \mathbb{E}[\boldsymbol{x}_i \boldsymbol{x}_i'] = k$.
- SEM3: $\mathbb{E}[\boldsymbol{u}_i \mid \boldsymbol{x}_i] = \boldsymbol{0}$ and $\mathbb{E}[\boldsymbol{u}_i \boldsymbol{u}_i' \mid \boldsymbol{X}] = \boldsymbol{\Sigma}$ is a symmetric positive definite matrix.

- SEM14: $\mathbb{E}[\boldsymbol{u}_i\boldsymbol{u}_j' \mid \boldsymbol{X}] = \mathbf{0}$ for all $i \neq j$.

Assumption SEM1 implies we can solve for all endogenous variables in terms of all exogenous variables:

$$\boldsymbol{y}_i' + \boldsymbol{x}_i'\boldsymbol{\Gamma}\boldsymbol{B}^{-1} = \boldsymbol{u}_i'\boldsymbol{B}^{-1}.$$

Equivalently,

$$\boldsymbol{y}_i' = \boldsymbol{x}_i'\boldsymbol{\Pi} + \boldsymbol{v}_i', \qquad \boldsymbol{\Pi} = -\boldsymbol{\Gamma}\boldsymbol{B}^{-1}, \qquad \boldsymbol{v}_i' = \boldsymbol{u}_i'\boldsymbol{B}^{-1}.$$

This is called the *reduced form* of the linear SEM. $\mathbb{E}[\boldsymbol{v}_i \mid \boldsymbol{x}_i] = \mathbf{0}$ and

$$\mathbb{E}[\boldsymbol{v}_i\boldsymbol{v}_i' \mid \boldsymbol{X}] = (\boldsymbol{B}^{-1})'\boldsymbol{\Sigma}\boldsymbol{B}^{-1} \equiv \boldsymbol{\Omega}.$$

$\boldsymbol{\Pi}$ and $\boldsymbol{\Omega}$ are the *reduced form parameters.*

Under assumptions SEM2–SEM4, the reduced form parameters are identified; e.g.,

$$\boldsymbol{\Pi} = \mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}\mathbb{E}[\boldsymbol{x}_i\boldsymbol{y}_i'].$$

The structural form parameters are identified if there is a unique $(\boldsymbol{B}, \boldsymbol{\Gamma}, \boldsymbol{\Sigma})$ such that

$$\boldsymbol{\Pi} = -\boldsymbol{\Gamma}\boldsymbol{B}^{-1} \quad \text{and} \quad \boldsymbol{\Omega} = (\boldsymbol{B}^{-1})'\boldsymbol{\Sigma}\boldsymbol{B}^{-1}.$$

Number of parameters must match.

- Reduced form: exactly $Gk + \frac{1}{2}G(G+1)$.
- Structural form: up to $G^2 + Gk + \frac{1}{2}G(G+1)$.

Identification of structural form parameters requires restrictions. Interpretation: insufficient information in the data about all structural form parameters.

We introduce information from an external source about some parameters to allow the data to provide information about other parameters. External source is economic theory: match $\boldsymbol{B}$ and $\boldsymbol{\Sigma}$ to structural parameters in economic models to elicit restrictions.

6.3 Introduction to Instrumental Variables

Suppose we are only interested in the structural form parameters of one equation (e.g., demand elasticities):

$$Y_{i1} + \beta_{12}Y_{i2} + \boldsymbol{z}_{i1}'\boldsymbol{\gamma}_1 = U_{i1},$$

where we match Y_{i1} to $\ln(s_i/s_0)$ and Y_{i2} to $\ln p_i$.

This regression cannot be implemented with OLS if the supply equation is:

$$Y_{i2} + \beta_{21}Y_{i1} + \boldsymbol{z}_{i2}'\boldsymbol{\gamma}_2 = U_{i2},$$

with $\mathbb{E}[U_{i1}U_{i2} \mid \boldsymbol{z}_{i1}, \boldsymbol{z}_{i2}] = 0$ and $\beta_{21} \neq 0$.

The reduced form of the supply equation is

$$Y_{i2} = \boldsymbol{z}_i'\boldsymbol{\pi}_2 + V_{i2}, \qquad V_{i2} = \frac{U_{i2} - \beta_{21}U_{i1}}{1 - \beta_{12}\beta_{21}}.$$

Hence,

$$\mathbb{E}[U_{i1}V_{i2} \mid \boldsymbol{z}_i] \neq 0 \quad \text{and} \quad \mathbb{E}[U_{i1}Y_{i2} \mid \boldsymbol{z}_i] \neq 0.$$

Option 1: Estimate the linear SEM in reduced form, impose restrictions on $(\boldsymbol{B}, \boldsymbol{\Gamma}, \boldsymbol{\Sigma})$ and derive the structural form. This will produce inference on both the demand and supply elasticities.

Option 2: Suppose $\boldsymbol{z}_i$ is $k_z \times 1$ and set $\boldsymbol{x}_i = (\boldsymbol{z}_{1i}', Y_{i2})'$, which is $k_x \times 1$. Assume $\operatorname{rank} \mathbb{E}[\boldsymbol{x}_i \boldsymbol{z}_i'] = k_x$, then use the estimator:

$$\hat{\boldsymbol{\beta}}_{\text{2SLS}} = \left(\boldsymbol{X}'\boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}\boldsymbol{Z}'\boldsymbol{X}\right)^{-1}\left(\boldsymbol{X}'\boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}\boldsymbol{Z}'\boldsymbol{y}\right).$$

Moreover, $\operatorname{rank} \mathbb{E}[\boldsymbol{x}_i \boldsymbol{z}_i'] = k_x$ implies $k_z \geq k_x$; if $k_z = k_x$ then

$$\hat{\boldsymbol{\beta}}_{\text{2SLS}} \equiv \hat{\boldsymbol{\beta}}_{\text{IV}} = (\boldsymbol{Z}'\boldsymbol{X})^{-1}\boldsymbol{Z}'\boldsymbol{y}.$$

$\hat{\boldsymbol{\beta}}_{\text{2SLS}}$ is called the *two-stage least squares* estimator and $\hat{\boldsymbol{\beta}}_{\text{IV}}$ is called the *instrumental variables* estimator.

$\boldsymbol{z}_i$ is a vector of instrumental variables (IV) for Y_{i2}. It contains all exogenous variables in the linear SEM. Key features:

- $\boldsymbol{z}_i$ is uncorrelated with U_{i1} [exogenous IVs]
- $\boldsymbol{z}_i$ is correlated with Y_{i2} (unless $\boldsymbol{\gamma}_2 = \boldsymbol{0}$). [relevant IVs]

$k_z \geq k_x$ implies $\boldsymbol{z}_{i2}$ contains at least one exogenous variable that is not in $\boldsymbol{z}_{1i}$. This corresponds to an *exclusion restriction* on $\boldsymbol{\Gamma}$ in the linear SEM.

It is the key restriction that allows for partial identification: β_{21} and $\boldsymbol{\gamma}_1$ are identified, even if β_{21} and $\boldsymbol{\gamma}_2$ might not be.

6.4 The Generalized Method of Moments (GMM)

In many economic models, the equilibrium characterising system of equations

$$F_m\Big(\{\boldsymbol{w}_i\}, \{\tilde{\boldsymbol{x}}_i\}, \{\varepsilon_i\}, \boldsymbol{\theta}\Big) = \boldsymbol{0},$$

can be uniquely solved for ε_i:

$$\varepsilon_i = h_{mi}\Big(\{\boldsymbol{w}_i\}, \{\tilde{\boldsymbol{x}}_i\}, \boldsymbol{\theta}\Big), \qquad i = 1, \ldots, n.$$

Then, independence of $\{\varepsilon_i\}$ and $\{\tilde{\boldsymbol{x}}_i\}$ implies that for $\boldsymbol{z}_i = \zeta(\tilde{\boldsymbol{x}}_i)$,

$$\mathbb{E}\Big[h_{mi}\Big(\{\boldsymbol{w}_i\}, \{\tilde{\boldsymbol{x}}_i\}, \boldsymbol{\theta}\Big)\boldsymbol{z}_i'\Big] = \boldsymbol{0}, \qquad i = 1, \ldots, n.$$

Extending to $i = 1, \ldots, N$ and applying the WLLN yields:

$$\frac{1}{N}\sum_{i=1}^{N} h_{mi}\Big(\{\boldsymbol{w}_i\}, \{\tilde{\boldsymbol{x}}_i\}, \boldsymbol{\theta}\Big)\boldsymbol{z}_i' \xrightarrow{p} \mathbb{E}\Big[h_{mi}\Big(\{\boldsymbol{w}_i\}, \{\tilde{\boldsymbol{x}}_i\}, \boldsymbol{\theta}\Big)\boldsymbol{z}_i'\Big] = \boldsymbol{0}.$$

This motivates constructing a loss function as follows. Let

$$\mu_{ij}(\boldsymbol{\theta}) = h_{mi}\Big(\{\boldsymbol{w}_i\}, \{\tilde{\boldsymbol{x}}_i\}, \boldsymbol{\theta}\Big)z_{ij},$$

and set

$$\boldsymbol{M}_i(\boldsymbol{\theta}) = \left(\mu_{i1}(\boldsymbol{\theta})', \ldots, \mu_{ik_z}(\boldsymbol{\theta})'\right)'.$$

Note that $\boldsymbol{M}_i(\boldsymbol{\theta})$ is a $k_\varepsilon k_z \times 1$ vector whose elements depend on $\{\boldsymbol{w}_i\}$, $\{\tilde{\boldsymbol{x}}_i\}$ and $\boldsymbol{\theta}$.

Next, set the loss function to be:

$$S_n(\boldsymbol{\theta}) = \left(\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{M}_i(\boldsymbol{\theta})\right)' \boldsymbol{W}_n \left(\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{M}_i(\boldsymbol{\theta})\right),$$

where $\boldsymbol{W}_n$ is a full rank, square, symmetric weighting matrix.

In the spirit of M-estimation, the GMM estimator is derived as:

$$\hat{\boldsymbol{\theta}}_{\text{GMM}} = \arg\min_{\boldsymbol{\theta}} S_n(\boldsymbol{\theta}).$$

The properties of $\hat{\boldsymbol{\theta}}_{\text{GMM}}$ are analogous to the M-estimator.

In this formulation, GMM is an approach to estimate the structural parameters $\boldsymbol{\theta}$ directly. This is the most common use of GMM in econometrics; it is equivalent to specifying a distribution with $f(\boldsymbol{Y}, \boldsymbol{X}; \vartheta_0)$ where:

- $\boldsymbol{\vartheta}_0 = \boldsymbol{\theta}$;
- a specific economic theory informs certain moments;
- the PDF form f is not fully specified.

Example 6.4.1. Let

$$h_{mi1}(Y_{i1}, Y_{i2}, \boldsymbol{z}_{i1}, \beta_{21}, \boldsymbol{\gamma}_1) = Y_{i1} + \beta_{21} Y_{i2} + \boldsymbol{z}_i'\boldsymbol{\gamma}_1 = Y_{i1} - \boldsymbol{x}_i'\boldsymbol{\beta},$$

and $\boldsymbol{z}_i = (\boldsymbol{z}_{i1}', \boldsymbol{z}_{i2}')'$ as in the 2SLS/IV example. Then,

$$S_n(\boldsymbol{\theta}) = \left(\frac{1}{n}\sum_{i=1}^{n}(Y_{i1} - \boldsymbol{x}_i'\boldsymbol{\beta})\right)' \boldsymbol{W}_n \left(\frac{1}{n}\sum_{i=1}^{n}(Y_{i1} - \boldsymbol{x}_i'\boldsymbol{\beta})\right),$$

and

$$\hat{\boldsymbol{\beta}}_{\text{GMM}} = (\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{y}).$$

If $\boldsymbol{W}_n = \sigma^2(\boldsymbol{Z}'\boldsymbol{Z})^{-1}$, then $\hat{\boldsymbol{\beta}}_{\text{GMM}} = \hat{\boldsymbol{\beta}}_{\text{2SLS}}$.

7 Inference from Data I: Hypothesis Testing

7.1 Hypotheses

Hypothesis tests attempt to assess whether there is evidence contrary to a proposed restriction.

Let $\boldsymbol{\psi} = r(\boldsymbol{\vartheta})$ be a $q \times 1$ vector representing some transformation of a $k \times 1$ vector of model parameters $\boldsymbol{\vartheta}$.

Example 7.1.1 (Linear regression examples of $r(\boldsymbol{\vartheta})$)**.** Consider the linear regression model

$$Y_i = \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + U_i,$$

and set, for example:

- $\boldsymbol{\psi} = r(\beta_1, \beta_2, \beta_3, \sigma_u^2) = \beta_1$;
- $\boldsymbol{\psi} = r(\beta_1, \beta_2, \beta_3, \sigma_u^2) = \beta_2 - \beta_1$;
- $\boldsymbol{\psi} = r(\beta_1, \beta_2, \beta_3, \sigma_u^2) = (\beta_2 - \beta_1,\ \beta_3 - \beta_2)'$, etc.

A *point hypothesis* concerning $\boldsymbol{\psi}$ is a proposed restriction such as $\boldsymbol{\psi} = \boldsymbol{\psi}_0$ where $\boldsymbol{\psi}_0$ is a hypothesized value chosen by the researcher.

- The *null hypothesis* is $H_0 : \boldsymbol{\psi} = \boldsymbol{\psi}_0$.
- The *alternative hypothesis* is $H_1 : \boldsymbol{\psi} \neq \boldsymbol{\psi}_0$.

The parameter space Θ is the union of parameter values that satisfy the restriction in H_0 and those that do not (equivalently those in H_1):

$$\Theta = \{\boldsymbol{\vartheta} \in \Theta : r(\boldsymbol{\vartheta}) = \boldsymbol{\psi}_0\} \ \cup \ \{\boldsymbol{\vartheta} \in \Theta : r(\boldsymbol{\vartheta}) \neq \boldsymbol{\psi}_0\}.$$

In hypothesis testing, every parameter value in Θ either satisfies H_0 or H_1. Equivalently, H_0 and H_1 split the family of distributions $\mathcal{F}_{\boldsymbol{\vartheta}}$ into distributions satisfying H_0 and distributions satisfying H_1.

The goal is to assess whether or not the restriction in H_0 is compatible with the observed data:

- Is the realisation $\boldsymbol{D}$ incompatible with the hypothesis, in the sense that $\boldsymbol{D}$ would be highly unlikely if the hypothesis were true?
- Equivalently: is $\boldsymbol{D}$ too extreme to be generated by the distributions in $\mathcal{F}_{\boldsymbol{\vartheta}}$ that satisfy H_0?

7.2 Decision Rule

A hypothesis test may either *Reject* H_0 or *Fail to reject* H_0. The decision is based on a realisation of random variables, so it is a mapping from the sample space to the decision set.

It is convenient to express this mapping as a real-valued function of random variables called a test statistic:

$$T(\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\psi}_0),$$

e.g., t-statistic, F-statistic, etc. Evaluating at the realisation $\boldsymbol{D}$ yields

$$\tau = T(\boldsymbol{D}, \boldsymbol{\psi}_0).$$

Definition 7.2.1 (Decision rule via critical value)**.** Given a critical value c:

- Reject H_0 if $\tau > c$.
- Fail to reject H_0 if $\tau \leq c$.

7.3 Designing Test Statistics

Designing a hypothesis test typically boils down to designing an appropriate test statistic.

If the test statistic has a known distribution, then c is selected such that the distributions in $\mathcal{F}_{\boldsymbol{\vartheta}}$ satisfying H_0 yield

$$\Pr_0(T(\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\psi}_0) > c) = \alpha,$$

where α is a pre-determined value called the *significance level*.

To compute c given α, we must know the CDF of $T(\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\psi}_0)$:

- This CDF is derived from distributions in $\mathcal{F}_{\boldsymbol{\vartheta}}$ satisfying H_0.
- This CDF cannot depend on any unknown quantities.
- When the CDF is fully known, $T(\cdot)$ is called *pivotal*.

7.4 Example: LR and OLS

Test statistics in practice are typically simple transformations of estimators.

Recall the OLS estimator $\hat{\boldsymbol{\beta}}$, $\hat{\sigma}_y^2$, and define

$$\hat{t}(\psi_0) = \sqrt{\frac{n-k}{n}}\,\frac{\hat{\beta}_j - \psi_0}{\widehat{\mathrm{se}}(\hat{\beta}_j)}.$$

Then, the test statistic for $H_0 : \beta_j = \psi_0$ is

$$T(\boldsymbol{Y}, \boldsymbol{X}, \psi_0) = \left|\hat{t}(\psi_0)\right|.$$

If A6(i) holds, then distributions in $\mathcal{F}_{\boldsymbol{\vartheta}}$ satisfying H_0 have PDFs

$$f(\boldsymbol{y}, \boldsymbol{X}; \beta_1, \ldots, \beta_{j-1}, \psi_0, \beta_{j+1}, \ldots, \beta_k, \sigma_y^2) = \prod_{i=1}^{n} \varphi\left(Y_i - \sum_{l \neq j} \beta_l X_{il} - \psi_0 X_{ij},\ \sigma_y^2\right),$$

and

$$\hat{t}(\psi_0) \sim t_{n-k}.$$

Hence,

$$\Pr_0(T(\boldsymbol{Y}, \boldsymbol{X}, \psi_0) > c) = 2\Big(1 - F_t(c; n-k)\Big) = \alpha,$$

where $F_t(\cdot)$ is the CDF of the Student-t distribution, so the critical value is

$$c = F_t^{-1}(1 - \alpha/2).$$

Given a realisation of $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}_y^2$, we can compute $\tau = T(\boldsymbol{D}, \psi_0)$ for any desired value of ψ_0.

Suppose $\psi_0 = 0$.

- If $\tau > c$, then $\boldsymbol{D}$ is an extreme draw from a distribution with PDF

  $$f(\boldsymbol{y}, \boldsymbol{X}; \beta_1, \ldots, \beta_{j-1}, 0, \beta_{j+1}, \ldots, \beta_k, \sigma_y^2),$$

  which is not compatible with $\boldsymbol{D}$. Decision: reject the distributions in $\mathcal{F}_{\boldsymbol{\vartheta}}$ satisfying H_0 as unreasonable probability models, thereby reducing the set of candidate models in $\mathcal{F}_{\boldsymbol{\vartheta}}$.
- If $\tau \leq c$, then $\boldsymbol{D}$ is a reasonable draw from such a distribution. Decision: do nothing (fail to reject).

7.5 The Significance Level

As $\alpha \to 1$, almost every $\boldsymbol{D}$ will be a reasonable draw; as $\alpha \to 0$, almost every $\boldsymbol{D}$ will be an extreme draw.

- α is sometimes called the *size* of the test.
- For every reject decision based on some α, there exists another $\tilde{\alpha} < \alpha$ that would yield a fail to reject.
- For every fail to reject decision based on some α, there exists another $\tilde{\alpha} > \alpha$ that would yield a reject.

The significance level is also known as the probability of type I error.

Correct interpretation: if hypothetical repeated samples were drawn independently from distributions in $\mathcal{F}_{\boldsymbol{\vartheta}}$ satisfying H_0, say $\boldsymbol{D}^{(1)}, \ldots, \boldsymbol{D}^{(D)}$, and the same test was carried out for each $\boldsymbol{D}^{(d)}$, then we would obtain R rejections, with

$$\frac{R}{D} \to \alpha \quad \text{as } D \to \infty.$$

Each rejection is an error because it would conclude that $\boldsymbol{D}^{(d)}$ is not compatible with the distribution it was sampled from.

Type I error is often interpreted as: "reject H_0 when it is true."

Confusion:

- H_0 restricts a family of distributions (i.e., probability models), which is fictitious by nature. Therefore, H_0 cannot be literally "true" or "untrue" without specifying a context.
- Type I error is often written as $\Pr(\text{reject } H_0 \mid H_0)$. Confusion: H_0 involves parameters, which are not random variables, so a probability conditional on H_0 is not defined in standard probability theory. Interpreting this requires a specific context.

7.6 Power of the Test

Let

$$\Pr_1(T(\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\psi}_0) > c) = \pi(\boldsymbol{\psi})$$

denote the probability obtained from a distribution in $\mathcal{F}_{\boldsymbol{\vartheta}}$ satisfying H_1, i.e., with $r(\boldsymbol{\vartheta}) = \boldsymbol{\psi} \neq \boldsymbol{\psi}_0$.

Then, for all $\boldsymbol{\psi} \in \{\boldsymbol{\psi} = r(\boldsymbol{\vartheta}),\ \boldsymbol{\vartheta} \in \Theta : \boldsymbol{\psi} \neq \boldsymbol{\psi}_0\}$, the function $\pi(\boldsymbol{\psi})$ is called the *power function* of the test.

The power function measures the test's ability to distinguish distributions in $\mathcal{F}_{\boldsymbol{\vartheta}}$: those satisfying H_0 vs those satisfying H_1.

A well-designed test statistic exhibits increasing power for $\boldsymbol{\psi}$ "farther away" from $\boldsymbol{\psi}_0$; e.g., larger $\|\boldsymbol{\psi} - \boldsymbol{\psi}_0\|$ yields larger $\pi(\boldsymbol{\psi})$.

Power is also known as the probability of avoiding type II error.

Correct interpretation: if repeated samples were drawn independently from a distribution in $\mathcal{F}_{\boldsymbol{\vartheta}}$ satisfying H_1 (with $\boldsymbol{\psi} \neq \boldsymbol{\psi}_0$), say $\boldsymbol{D}^{(1)}, \ldots, \boldsymbol{D}^{(D)}$, and the same test was carried out for each $\boldsymbol{D}^{(d)}$, then:

- we would obtain R rejections and $D - R$ failures to reject;
- $R/D \to \pi(\boldsymbol{\psi})$ as $D \to \infty$;
- each failure to reject is an error because it fails to distinguish distributions satisfying H_0 from the one that $\boldsymbol{D}^{(d)}$ was sampled from.

Useful intuition: the choice of c involves a trade-off between size and power.

- Given a test statistic, increasing the critical value decreases the size, but also decreases the power.

7.7 Approximate Hypothesis Tests

An exact distribution of an estimator can only be derived in special cases (e.g., conditionally normal LR). The same is true for test statistics: $\Pr_0(T(\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\psi}_0) > c)$ is rarely known.

Most hypothesis tests rely on the approximation:

$$\lim_{N\to\infty} \Pr_0\Big(T(\tilde{\boldsymbol{Y}}, \tilde{\boldsymbol{X}}, \boldsymbol{\psi}_0) > c\Big) = \alpha.$$

Equivalently,

$$T(\tilde{\boldsymbol{Y}}, \tilde{\boldsymbol{X}}, \boldsymbol{\psi}_0) \xrightarrow{d} Z,$$

where the distribution of Z has CDF F_Z, and the critical value is computed from

$$1 - F_Z(c) = \alpha.$$

When a test statistic is designed such that F_Z is known (when $\Pr_0(T(\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\psi}_0) > c)$ is not), it is called an *asymptotic test*.

Replacing A6(i) with A6(ii) yields:

$$\tilde{t}(\psi_0) = \frac{\tilde{\beta}_j - \psi_0}{\tilde{\text{se}}(\tilde{\beta}_j)}, \qquad T(\tilde{\boldsymbol{Y}}, \tilde{\boldsymbol{X}}, \psi_0) = \left|\tilde{t}(\psi_0)\right|.$$

For distributions in $\mathcal{F}_{\boldsymbol{\vartheta}}$ satisfying H_0,

$$\tilde{t}(\psi_0) \xrightarrow{d} \mathcal{N}(0,1), \qquad T(\tilde{\boldsymbol{Y}}, \tilde{\boldsymbol{X}}, \psi_0) \xrightarrow{d} Z,$$

with

$$F_Z(c) = 1 - 2\Phi(c) \quad \Rightarrow \quad c = \Phi^{-1}(1 - \alpha/2),$$

where Φ is the standard normal CDF.

The power function of this test is, for any $\beta_j \neq 0$,

$$\pi(\beta_j) = \Pr_1\left(\left|\tilde{t}(\beta_j) + \frac{\beta_j - \psi_0}{\tilde{\text{se}}(\tilde{\beta}_j)}\right| > c\right).$$

For any distribution in $\mathcal{F}_{\boldsymbol{\vartheta}}$ with $\beta_j \neq \psi_0$, we have $\tilde{t}(\beta_j) \xrightarrow{d} \mathcal{N}(0,1)$ and $\tilde{\text{se}}(\tilde{\beta}_j) \xrightarrow{p} 0$, so that

$$\lim_{N\to\infty} \pi(\beta_j) \to 1.$$

In any asymptotic test:

- $T(\tilde{\boldsymbol{Y}}, \tilde{\boldsymbol{X}}, \boldsymbol{\psi}_0)$ is *asymptotically pivotal* iff F_Z is fully known;
- c is the *asymptotic critical value*;
- α is the *asymptotic size*;
- the test is *consistent* iff $\lim_{N\to\infty} \pi(\boldsymbol{\psi}) \to 1$.

7.8 P-values

The binary outcome (Reject H_0 / Fail to reject H_0) does not fully summarize the evidence in the data. Either outcome ignores the magnitude of the realised test statistic τ, which suggests a degree of evidence against H_0.

An alternative to the binary outcome based on a pre-determined α is the p-value:

$$\text{p-value} = \Pr_0(T(\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\psi}_0) > \tau)\,.$$

- The p-value is a decreasing function of τ.
- The p-value is the smallest significance level at which H_0 cannot be rejected.
- Since $\tau > c \Leftrightarrow$ p-value $< \alpha$, reject H_0 at significance level α iff p-value $< \alpha$.

The p-value is often misinterpreted, misrepresented and used to make unjustified claims. Common misconceptions:

- The p-value does not indicate the size or importance of an economically meaningful effect.
- In particular, a smaller p-value does not mean a relationship is "more significant".
- The p-value cannot be used to rank different models or compare families of distributions.
- The p-value is not the probability that "H_0 is true" or "H_1 is false".

However, the p-value may be viewed as a measure of the strength of information against H_0. It is good empirical practice to compute and report the p-value, which contains more useful information than the binary decision or appended (*), (**), (***).

7.9 Statistical Significance

Testing requires a pre-determined significance level α. The most common practice is to set $\alpha = 5\%$.

- This is a by-product of traditional tables of critical values and statistical software.
- There is no scientific justification for $\alpha = 5\%$.

When a test rejects H_0 at the 5% significance level it is common to say that the realised test statistic is *statistically significant*.

Confusion: this is often extended to parameters $\boldsymbol{\vartheta}$ with $H_0 : \vartheta_j = 0$:

- If H_0 is rejected, conclude ϑ_j is statistically significant.
- Interpretation requires specific context (often missing).
- It is often incorrectly interpreted as implying a "substantial" or "important" economic effect.

When approximating a test with an asymptotic distribution of the test statistic, we aim for a consistent test. But if $\tilde{\boldsymbol{\vartheta}} \xrightarrow{p} \boldsymbol{\vartheta}_0$, then for every fixed α, a consistent test is guaranteed to reject as $N \to \infty$ any H_0 except $H_0 : \boldsymbol{\psi} = r(\boldsymbol{\vartheta}_0)$.

Consequently, the statement "ϑ_j is statistically significant" does not distinguish between $\vartheta_{0j} = 10^{-100}$ and $\vartheta_{0j} = 10^{100}$. If ϑ_{0j} is matched to a structural θ that is interpreted as an economic relationship, then $\theta = 10^{-100}$ vs $\theta = 10^{100}$ are economically very different, but equally statistically significant.

7.10 Hypothesis Tests vs Confidence Intervals

Recall that in the LR satisfying A6(i), the test statistic for $H_0 : \beta_j = \psi_0$ is $T(\boldsymbol{Y}, \boldsymbol{X}, \psi_0) = \left|\hat{t}(\psi_0)\right|$.

Recall also that the $(1-\alpha) \times 100\%$ confidence interval (CI) is

$$\hat{\beta}_j \pm c\sqrt{\frac{n}{n-k}}\,\widehat{\mathrm{se}}(\hat{\beta}_j), \qquad F_t(c) = 1 - \alpha/2.$$

ψ_0 is below the lower bound of the CI if and only if

$$\sqrt{\frac{n-k}{n}}\,\frac{\hat{\beta}_j - \psi_0}{\widehat{\mathrm{se}}(\hat{\beta}_j)} > c,$$

and ψ_0 is above the upper bound of the CI if and only if

$$\sqrt{\frac{n-k}{n}}\,\frac{\psi_0 - \hat{\beta}_j}{\widehat{\mathrm{se}}(\hat{\beta}_j)} > c.$$

Consequently, ψ_0 is outside of the CI if and only if $\left|\hat{t}(\psi_0)\right| > c$.

Equivalently, a hypothesis test with $H_0 : \beta_j = \psi_0$ and test statistic $T(\boldsymbol{Y}, \boldsymbol{X}, \psi_0)$ will reject H_0 at significance level α for any ψ_0 outside the $(1-\alpha) \times 100\%$ CI for β_j.

This example extends to all hypothesis tests with scalar H_0 (i.e., $q = 1$): the CI for $\boldsymbol{\psi} = r(\boldsymbol{\vartheta})$ contains information on all $\boldsymbol{\psi}_0$ that would result in rejecting H_0 at significance level α.

- The CI supersedes scalar hypothesis tests, including all significance tests.

7.11 Joint Hypothesis Tests

Hypothesis tests only add inferential value when H_0 involves $q > 1$ restrictions.

In principle, we can also construct confidence sets for the $q \times 1$ vector $\boldsymbol{\psi} = r(\boldsymbol{\vartheta})$:

- Confidence sets for $q = 2$ are generalised ellipsoids.
- Confidence sets for $q > 2$ can be derived mathematically, but are difficult to visualise.
- Confidence sets for $q > 1$ adhere to the same principle: any $\boldsymbol{\psi}_0$ outside the set is rejected as the H_0, and a H_0 with any $\boldsymbol{\psi}_0$ inside the set cannot be rejected.
- Confidence sets for $q > 1$ are extremely rare in practice.

Thus, joint hypothesis tests are used to test $H_0 : \boldsymbol{\psi} = \boldsymbol{\psi}_0$ (equivalently, check whether $\boldsymbol{\psi}_0$ is inside the set).

8 Inference from Data II: Posterior Analysis

8.1 Posterior Inference

Bayesian posterior inference is rooted in simulation. Every draw

$$\boldsymbol{\vartheta}^{\circ(d)} \sim \Big(\boldsymbol{\vartheta}^{\circ} \mid D\Big)$$

yields a possible model

$$f(Y, X; \boldsymbol{\vartheta}_0) \;\leftrightarrow\; f\Big(Y, X; \boldsymbol{\vartheta}^{\circ(d)}\Big).$$

If $\boldsymbol{\vartheta}_0$ is matched to the structural $\boldsymbol{\theta}$ of an economic model, then from each $\boldsymbol{\vartheta}^{\circ(d)}$ we obtain:

- a possible calibration $\boldsymbol{\theta}^{(d)}$ of the economic model;
- possible theory-derived effects;
- possible counterfactuals;
- possible policy interventions;
- possible predictions.

8.2 Probit Example

Recall from Tutorial 4 the analysis of private insurance purchase using the Health and Retirement Study (HRS) data.

- Observe insurance purchase as a binary outcome along with individual characteristics.
- Use Probit regression to obtain inference on average marginal effects and predicted choice probabilities.

We matched the Probit regression to the demand for private insurance utility model:

$$Y_i^* = \mu_u + \tilde{\boldsymbol{x}}_{o,i}'\boldsymbol{\theta}_o - \sigma_u U_i,$$

such that

$$\Pr(Y_i = 1 \mid \boldsymbol{x}_i) = \Pr(Y_i^* \geq 0 \mid \boldsymbol{x}_i) = \Phi(\boldsymbol{x}_i'\boldsymbol{\beta}).$$

A Bayesian probit regression may be constructed as:

$$\Pr(Y_i = 1 \mid \boldsymbol{x}_i, \boldsymbol{\beta}^{\circ}) = \Phi(\boldsymbol{x}_i'\boldsymbol{\beta}^{\circ}), \qquad \boldsymbol{\beta}^{\circ} \sim \mathcal{N}(\boldsymbol{b}, \boldsymbol{V}).$$

For every posterior draw $\boldsymbol{\beta}^{\circ(d)}$, we obtain a calibrated utility model:

$$\frac{Y_i^*}{\sigma_u} = \boldsymbol{x}_i'\boldsymbol{\beta}^{\circ(d)} - U_i.$$

We can use posterior draws to simulate:

- (normalised) utilities Y_i^*/σ_u for observed individuals, $i = 1, \ldots, n$;
- predicted (normalised) utilities for unobserved individuals, $i > n$;

- predicted choice probabilities $\Pr(Y_i^* \geq 0 \mid \boldsymbol{x}_i)$ for unobserved individuals, $i > n$.

Set $\boldsymbol{b} = \boldsymbol{0}$, $\boldsymbol{V} = 10\boldsymbol{I}_k$ and use Gibbs sampling to simulate.

	Mean	Median	Std. Dev.	2.5%	97.5%
retire	0.0419	0.0421	0.0182	0.0058	0.0775
age	-0.0032	-0.0032	0.0024	-0.0081	0.0015
hstatusg	0.0704	0.0703	0.0196	0.0316	0.1083
hhincome	0.0004	0.0004	0.0001	0.0002	0.0007
educyear	0.0250	0.0250	0.0030	0.0193	0.0311
married	0.1294	0.1297	0.0198	0.0905	0.1684
hisp	-0.1675	-0.1668	0.0392	-0.2451	-0.0933

	Estimate	Std. Error	2.5%	97.5%
retire	0.0419	0.0181	0.0064	0.0775
age	-0.0031	0.0024	-0.0079	0.0016
hstatusg	0.0699	0.0194	0.0318	0.1080
hhincome	0.0004	0.0002	0.0001	0.0007
educyear	0.0251	0.0029	0.0194	0.0308
married	0.1267	0.0189	0.0896	0.1638
hisp	-0.1564	0.0328	-0.2207	-0.0922

i	retire	age	hstatusg	pincome	educyear	married	hisp
149	0	67	0	0.047	3	0	1
292	0	65	0	0.091	12	0	0
259	1	65	0	0.081	16	0	0

i	Mean	Median	Std. Dev.	2.5%	97.5%
149	-1.9788	-1.9423	0.9385	-3.8921	-0.2875
292	-1.1642	-1.0678	0.7554	-2.8621	-0.0697
259	-0.9589	-0.8512	0.6764	-2.5195	-0.0444

8.3 Bayesian Approach to Specification Uncertainty

Recall that a Bayesian model is fully characterised by the joint PDF:

$$f(\boldsymbol{\vartheta}^\circ, Y, X).$$

All specification uncertainty amounts to different specifications of $f(\boldsymbol{\vartheta}^\circ, Y, X)$:

- **Prior uncertainty:** alternative specifications of $f(\boldsymbol{\vartheta}^\circ)$.
- **Likelihood uncertainty:** alternative specifications of $f(Y, X \mid \boldsymbol{\vartheta}^\circ)$.
- **Joint uncertainty:** uncertainty in $f(\boldsymbol{\vartheta}^\circ, Y, X)$ due to prior, likelihood, or both.

In the Bayesian framework, specification uncertainty is treated with random variables.

Consider a set of models $f_m(\boldsymbol{\vartheta}^\circ, Y, X)$, for $m = 1, \ldots, M$. Define the discrete RV m°, with support in $\{1, \ldots, M\}$.

- $f(\boldsymbol{\vartheta}^\circ, Y, X \mid m^\circ = m) = f_m(\boldsymbol{\vartheta}^\circ, Y, X)$;
- $f(\boldsymbol{\vartheta}^\circ, Y, X) = \sum_{m=1}^{M} \Pr(m^\circ = m)\, f(\boldsymbol{\vartheta}^\circ, Y, X \mid m^\circ = m)$.

$\Pr(m^\circ = 1), \ldots, \Pr(m^\circ = M)$ are **prior relative model weights**.

- $\Pr(m^\circ = m) \geq 0$ and $\sum_{m=1}^{M} \Pr(m^\circ = m) = 1$.
- **Intuition:** a prior distribution over candidate models, summarising information about models $m = 1, \ldots, M$ available from sources outside the data.

The RV m° is used to implement **Bayesian Model Averaging**.

8.4 Bayesian Model Averaging

The Bayesian model can be redefined so that m° is also a parameter:

$$f(\boldsymbol{\vartheta}^\circ, m^\circ, Y, X) = f(m^\circ)\, f(\boldsymbol{\vartheta}^\circ, Y, X \mid m^\circ) \equiv \Pr(m^\circ = m)\, f(\boldsymbol{\vartheta}^\circ, Y, X \mid m^\circ = m).$$

- The **full posterior** is

$$f(\boldsymbol{\vartheta}^\circ, m^\circ \mid D) = \frac{f(\boldsymbol{\vartheta}^\circ, m^\circ, D)}{\displaystyle\int \sum_{m=1}^{M} \Pr(m^\circ = m)\, f(\boldsymbol{\vartheta}^\circ, D \mid m^\circ = m)\, d\boldsymbol{\vartheta}^\circ}.$$

- The **model-m posterior** is

$$f(\boldsymbol{\vartheta}^\circ \mid m^\circ = m, D) = \frac{f_m(\boldsymbol{\vartheta}^\circ, D)}{\displaystyle\int f_m(\boldsymbol{\vartheta}^\circ, D)\, d\boldsymbol{\vartheta}^\circ}.$$

- The **model-averaged posterior** is

$$f(\boldsymbol{\vartheta}^\circ \mid D) = \sum_{m=1}^{M} \Pr(m^\circ = m \mid D)\, f(\boldsymbol{\vartheta}^\circ \mid m^\circ = m, D).$$

$\Pr(m^\circ = 1 \mid D), \ldots, \Pr(m^\circ = M \mid D)$ are **posterior model weights**:

$$\Pr(m^\circ = m \mid D) = \frac{\Pr(m^\circ = m) \int f_m(\boldsymbol{\vartheta}^\circ, D)\, d\boldsymbol{\vartheta}^\circ}{\sum_{m=1}^{M} \Pr(m^\circ = m) \int f_m(\boldsymbol{\vartheta}^\circ, D)\, d\boldsymbol{\vartheta}^\circ} = \frac{\Pr(m^\circ = m)\, f_m(D)}{f(D)}.$$

Here $f_m(D)$ is the model-m marginal likelihood and $f(D)$ is the model-averaged marginal likelihood.

- **Intuition:** a posterior distribution over candidate models, summarising information about $m = 1, \ldots, M$ from all sources, including the data.
- Posterior weights summarise all specification uncertainty.

If $\Pr(m^\circ = m) = 1/M$, then

$$\Pr(m^\circ = m \mid D) = \frac{f_m(D)}{\sum_{m=1}^{M} f_m(D)}.$$

The **Bayes factor** is

$$BF_{l,m} = \frac{f_l(D)}{f_m(D)}.$$

- The evidence in the data in favour of model l relative to m.
- If $\Pr(m^\circ = m) = 1/M$, then $BF_{l,m} = \Pr(m^\circ = l \mid D) \big/ \Pr(m^\circ = m \mid D)$.

8.5 Bayesian Hypothesis Testing

Competing hypotheses can always be formulated in terms of model comparison. For the private insurance example:

- H_0: (all else equal) education reduces utility from private health insurance;
- H_1: (all else equal) education increases utility from private health insurance.

In terms of the economic model:

$$H_0 : \ \theta_{o,5} \leq 0 \quad \text{vs.} \quad H_1 : \ \theta_{o,5} > 0.$$

In terms of the Probit regression, this translates to:

$$H_0 : \ \beta_6 \leq 0 \quad \text{vs.} \quad H_1 : \ \beta_6 > 0.$$

Recall (Lecture 8) that $H_0 : \beta_6 \leq 0$ vs. $H_1 : \beta_6 > 0$ cannot be compared directly using an MLE-based approach. Instead, we re-formulate:

1. Assume $\beta_6 \geq 0$ (i.e. restrict the parameter space Θ).
2. Test $H_0 : \beta_6 = 0$ vs. $H_1 : \beta_6 > 0$.

The test rejects H_0 if it is sufficiently incompatible with D; otherwise we fail to reject.

In a Bayesian Probit regression, we formulate H_0 and H_1 as two models which differ in terms of priors.

Let $1_S(x)$ be an indicator function: yields 1 if $x \in S$, 0 otherwise.

- **Model 1:** $f_1(\beta_6^\circ) = 2\phi(\beta_6^\circ; 0, 10)\, 1_{(-\infty,0]}(\beta_6^\circ)$.
- **Model 2:** $f_2(\beta_6^\circ) = 2\phi(\beta_6^\circ; 0, 10)\, 1_{(0,\infty)}(\beta_6^\circ)$.

(Here $\phi(\cdot; 0, 10)$ denotes the $\mathcal{N}(0, 10)$ density.)

The relative evidence in favour of H_0 is $\Pr(m^\circ = 1 \mid D)$; in favour of H_1 is $\Pr(m^\circ = 2 \mid D)$.

Because the half-normal prior densities $f_1(\beta_6^\circ)$ and $f_2(\beta_6^\circ)$ are *nested* within the normal prior density, we do not need to do anything special to compare the two hypotheses. In particular, if $\Pr(m^\circ = 1) = \Pr(m^\circ = 2) = 0.5$, then

$$\Pr(m^\circ = 1 \mid D) = \Pr(\beta_6^\circ \leq 0 \mid D), \qquad \Pr(m^\circ = 2 \mid D) = \Pr(\beta_6^\circ > 0 \mid D).$$

In the HRS example, we estimate $\Pr(\beta_6^\circ > 0 \mid D) = 1$ and conclude evidence overwhelmingly supports H_1. In fact, we estimate $\Pr(\beta_6^\circ > 0.039 \mid D) = 1$.

8.6 Bayesian Tests of Restrictions

The same approach applies to testing exact restrictions such as $H_0 : \beta_6 = 0$ vs. $H_1 : \beta_6 \neq 0$.

- **Model 1:** prior CDF $F_1(\beta_6^\circ) = 1_{[0,\infty)}(\beta_6^\circ) \ \Rightarrow \ \Pr(\beta_6^\circ = 0) = 1$.
- **Model 2:** prior CDF $F_2(\beta_6^\circ) = \Phi\Big(\beta_6^\circ / \sqrt{10}\Big) \ \Rightarrow \ \beta_6^\circ \sim \mathcal{N}(0, 10)$.

The model posterior $\Pr(m^\circ = m \mid D)$, $m = 1, 2$, again summarises all evidence regarding H_0 and H_1.

In this case, the normal prior does not nest Model 1:

$$\beta_6^\circ \sim \mathcal{N}(0, 10) \quad \Rightarrow \quad \Pr(\beta_6^\circ = 0) = 0,$$

which contradicts $F_1(\beta_6^\circ)$.

A prior that does nest Model 1 and Model 2 has CDF:

$$F(\beta_6^\circ) = \Pr(m^\circ = 1)\, 1_{[0,\infty)}(\beta_6^\circ) + \Pr(m^\circ = 2)\, \Phi\Big(\beta_6^\circ / \sqrt{10}\Big).$$

In practice, three approaches to testing exact restrictions are:

1. Use the "mixture prior" to derive a nesting posterior.
2. Derive model-specific posteriors individually and compute marginal likelihoods or Bayes factors.
3. Derive the Model 2 posterior and compute Bayes factors.

Computing Bayes factors is simplest in Bayesian regressions with natural conjugate priors (Lecture 5 / Tutorial 5). Consider $H_0 : \beta_j = 0$, $H_1 : \beta_j \neq 0$:

- Model 1: regression with X_{ij} excluded;
- Model 2: unrestricted.

In both cases, $f_m(\boldsymbol{y} \mid \boldsymbol{X})$ has a simple formula; hence

$$BF_{1,2} = \frac{f_1(\boldsymbol{y} \mid \boldsymbol{X})}{f_2(\boldsymbol{y} \mid \boldsymbol{X})}$$

is easy to compute without simulation.

In models with nonconjugate priors, such as the Probit regression, $BF_{1,2}$ can still be computed from the posterior of Model 2 using the Savage–Dickey density ratio (SSDR):

$$BF_{1,2} = \frac{f_{\beta_6^\circ \mid \boldsymbol{y}, \boldsymbol{X}}(0 \mid D)}{f_{\beta_6^\circ}(0)},$$

where $f_{\beta_6^\circ \mid \boldsymbol{y}, \boldsymbol{X}}$ is the marginal posterior and $f_{\beta_6^\circ}$ the prior in Model 2.

Not surprisingly, we find $BF_{1,2} \approx 0 \Rightarrow \Pr(m^\circ = 1 \mid D) \approx 0$; however, restrictions close to the posterior mode are strongly supported.

Restriction	0	0.04	0.06	0.07	0.08	0.10
$BF_{1,2}$	0.00	0.51	169.70	380.89	204.99	0.50
$\Pr(m^\circ = 1 \mid D)$	0.000	0.337	0.994	0.997	0.995	0.334

It is easy to tell which restrictions will be supported—and to what extent—from the simple histogram of posterior draws (e.g. of β_6°): restrictions near the bulk of the histogram (near the posterior mode) tend to be strongly supported; restrictions far in the tails tend to be weakly supported.

8.7 Types of Testable Hypotheses

The Bayesian approach using model posterior distributions is very flexible.

Hypotheses involving parameters, which lead to models with alternative priors, may be:

- **multiple:** e.g. $H_0 : \beta_6 = 0$, $H_1 : \beta_6 = 0.04$, $H_2 : \beta_6 = 0.06$, $H_3 : \beta_6 = 0.07$, $H_4 : \beta_6 = 0.08$, $H_5 : \beta_6 = 0.10$, $H_6 : \beta_6 \in \mathbb{R}$;
- **overlapping:** e.g. $H_0 : \beta_6 \leq 0$, $H_1 : \beta_6 > 0$, $H_2 \in \mathbb{R}$.

Hypotheses may involve features of the likelihood: e.g. $H_0 : U_i$ is normal vs. $H_1 : U_i$ is logistic.

Every hypothesis that can be translated to a specification $f_m(\boldsymbol{\vartheta}^\circ, Y, X)$ may be evaluated using $\Pr(m^\circ = m \mid D)$ relative to the other $M - 1$ hypotheses.

9 Non-parametric and Flexible Parametric Models

9.1 Flexible Distribution Models

Recall that an "ideal" distribution with PDF $f(Y, X; \vartheta_0)$, which is presumed to match to a class of economic models, is an element of a parametric family $\mathcal{F}_\vartheta$, with k_ϑ parameters.

- A key feature of $\mathcal{F}_\vartheta$ is that the dimension k_ϑ is finite.
- The k_ϑ-dimensional parameter vector ϑ_0 is matched to structural parameters θ in economic models.

A parametric PDF can be made more flexible by setting

$$f(Y, X; \vartheta_0) = \sum_{m=1}^{M} \omega_m f_m(Y, X; \vartheta_m),$$

with $f_m(Y, X; \vartheta_m)$ matched to specific economic models.

- More components $M \Rightarrow$ more flexibility, but less parsimony.
- Weights ω_m can be set using either information criteria or Bayesian model averaging.

9.2 Non-parametric Distribution Models

The extreme: set $M \to \infty$ to obtain infinite flexibility, but with zero parsimony; obviously not practical.

The alternative: represent $f(Y, X; \vartheta_0)$ with a series expansion

$$f(Y, X; \vartheta_0) = \sum_{m=1}^{\infty} \omega_m \phi_m(Y, X; \vartheta_m),$$

where $\{\phi_m(\cdot)\}$ are basis functions.

- Basis functions have no connection to economic models.
- ϑ_0 is infinite-dimensional and no match to structural θ.
- This is a completely top-down approach.

9.3 Basis Functions

Definition 9.3.1 (Basis (informal))**.** A *basis* is a set of objects that span a certain mathematical space, so that every object in the space can be written as a series expansion (a linear combination of basis objects).

Example 9.3.2. Two-dimensional vectors ($\mathbb{R}^2$): a basis is

$$\{(1,0)', (0,1)'\}$$

because every two-dimensional vector $\boldsymbol{x}$ can be written:

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Example 9.3.3. Space of infinitely differentiable (at zero) functions: a basis is

$$\left\{1, x, x^2, \ldots\right\}$$

because every element $f(x)$ can be written:

$$f(x) = \sum_{m=1}^{\infty} \omega_m \phi_m(x),$$

where $\phi_m(x) = x^{m-1}$ and

$$\omega_m = \frac{\left.\dfrac{d^{m-1} f}{dx^{m-1}}\right|_{x=0}}{(m-1)!}.$$

The space of all PDFs: all functions $f(Y, X)$ that satisfy

1. $f(Y, X) \geq 0$ and
2. $\int f(Y, X)\, d(Y, X) = 1$.

Many known options exist to construct a basis for this space.

- Polynomials: Legendre polynomials, Bernstein polynomials, etc.
- Splines: piecewise linear splines, B-splines, etc.
- Densities: Beta densities (Bernstein polynomials), normal densities, Gamma densities, etc.

All bases above can be represented as known functions $\{\phi_m(\cdot)\}$ with (possibly) unknown elements $\{\vartheta_m\}$.

Example 9.3.4. $\phi_m(Y, X; \boldsymbol{\mu}_m, \boldsymbol{V}_m)$ is the multivariate normal density with mean $\boldsymbol{\mu}_m$ and covariance $\boldsymbol{V}_m$.

9.4 Series Expansion of a PDF

Consequently, let ϑ_0 contain all unknown quantities $\{\vartheta_m\}$ along with all unknown weights $\{\omega_m\}$.

Every PDF can be represented with a series expansion up to unknown quantities ϑ_0 as

$$f(Y, X; \vartheta_0) = \sum_{m=1}^{\infty} \omega_m \phi_m(Y, X; \vartheta_m),$$

with an appropriate choice of basis $\{\phi_m(\cdot)\}$.

However, any given basis restricts $f(Y, X; \vartheta_0)$ be an element of the space spanned by $\{\phi_m(\cdot)\}$.

Example 9.4.1. If $\phi_m(Y) = Y^{m-1}$ then

$$f(Y) = \sum_{m=1}^{\infty} \frac{\left.\dfrac{d^{m-1} f}{dY^{m-1}}\right|_{Y=0}}{(m-1)!} Y^{m-1}$$

is a PDF that is infinitely differentiable at zero.

9.5 Inference from Non-parametric PDFs

The series expansion of a PDF is a convenient representation. How do we match it to any economic theory for inference?

Since this is purely a top-down approach, the focus is on prediction.

Most common implementation in practice:

1. Assume $(y_i, \boldsymbol{x}_i)$ are iid for all $i = 1, \ldots, n+1$:

$$f(\tilde{Y}, \tilde{X}; \vartheta_0) = \prod_{i=1}^{n+1} f(y_i, \boldsymbol{x}_i; \vartheta_0).$$

2. Specify: $f(y_i, \boldsymbol{x}_i; \vartheta_0) = \sum_{m=1}^{\infty} \omega_m \phi_m(y_i, \boldsymbol{x}_i; \vartheta_m)$.
3. Obtain inference on:

$$\mathbb{E}[y_{n+1} \mid \boldsymbol{x}_{n+1}] = \frac{\sum_{m=1}^{\infty} \omega_m \int y_{n+1}\, \phi_m(y_{n+1}, \boldsymbol{x}_{n+1}; \vartheta_m)\, dy_{n+1}}{\sum_{m=1}^{\infty} \omega_m \int \phi_m(y_{n+1}, \boldsymbol{x}_{n+1}; \vartheta_m)\, dy_{n+1}}.$$

9.6 Example: Normal Mixtures

Multivariate normal densities,

$$\phi_m(\boldsymbol{w}; \boldsymbol{\mu}_m, \boldsymbol{V}_m) = (2\pi)^{-\frac{k_w}{2}} |\boldsymbol{V}_m|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}(\boldsymbol{w} - \boldsymbol{\mu}_m)' \boldsymbol{V}_m^{-1} (\boldsymbol{w} - \boldsymbol{\mu}_m)\right\},$$

form a basis for the space of absolutely continuous PDFs.

Therefore, if the PDF of $(y_i, \boldsymbol{x}_i)$ is absolutely continuous, then

$$f(y_i, \boldsymbol{x}_i; \{\omega_m\}, \{\boldsymbol{\mu}_m\}, \{\boldsymbol{V}_m\}) = \sum_{m=1}^{\infty} \omega_m \phi_m(\boldsymbol{w}_i; \boldsymbol{\mu}_m, \boldsymbol{V}_m),$$

where

$$\boldsymbol{w}_i = \begin{pmatrix} y_i \\ \boldsymbol{x}_i \end{pmatrix}, \qquad \boldsymbol{\mu}_m = \begin{pmatrix} \mu_{y,m} \\ \boldsymbol{\mu}_{x,m} \end{pmatrix}, \qquad \boldsymbol{V}_m = \begin{pmatrix} V_{y,m} & \boldsymbol{V}_{yx,m} \\ \boldsymbol{V}'_{yx,m} & \boldsymbol{V}_{x,m} \end{pmatrix}.$$

We obtain $\mathbb{E}[y_{n+1} \mid \boldsymbol{x}_{n+1}] = (1, \boldsymbol{x}'_{n+1})\boldsymbol{\beta}(\boldsymbol{x}_{n+1})$, where

$$\boldsymbol{\beta}(\boldsymbol{x}_{n+1}) = \sum_{m=1}^{\infty} \eta_m(\boldsymbol{x}_{n+1}) \begin{pmatrix} \mu_{y,m} - \boldsymbol{V}_{yx,m}\boldsymbol{V}_{x,m}^{-1}\boldsymbol{\mu}_{x,m} \\ \boldsymbol{V}_{yx,m}\boldsymbol{V}_{x,m}^{-1}\boldsymbol{x}_{n+1} \end{pmatrix},$$

$$\eta_m(\boldsymbol{x}_{n+1}) = \frac{\omega_m \phi_m(\boldsymbol{x}_{n+1}; \boldsymbol{\mu}_{x,m}, \boldsymbol{V}_{x,m})}{\sum_{m=1}^{\infty} \omega_m \phi_m(\boldsymbol{x}_{n+1}; \boldsymbol{\mu}_{x,m}, \boldsymbol{V}_{x,m})}.$$

Clearly, $\mathbb{E}[y_{n+1} \mid \boldsymbol{x}_{n+1}]$ is not linear because $\boldsymbol{\beta}(\boldsymbol{x}_{n+1})$ depends on $\boldsymbol{x}_{n+1}$ (except for special cases). However, to compute $\boldsymbol{\beta}(\boldsymbol{x}_{n+1})$ we need $\{\omega_m\}$, $\{\boldsymbol{\mu}_m\}$ and $\{\boldsymbol{V}_m\}$.

9.7 Non-parametric Models in Practice

A model with infinitely many unknown quantities cannot be directly implemented. Nonzero parsimony needs to be introduced in practice. There are three standard approaches.

1. Use a finite mixture ($M < \infty$) approximation.
2. Use a kernel density to impose restrictions across $\{\vartheta_m\}$.
3. Use Bayesian methods with a Dirichlet process prior to impose restrictions across $\{\vartheta_m\}$.

A finite mixture approximation can be implemented using either MLE or Bayesian methods.

9.8 Kernel Density Estimation

A popular approach to inference on the conditional mean of a non-parametric model uses the kernel density estimator (KDE). The KDE is closely related to the histogram estimator of a univariate PDF; setting $-\infty < l_y < \min\{Y_i\}$, $\max\{Y_i\} < u_y < \infty$,

$$\hat{f}(y_{n+1}) = \frac{1}{nh} \sum_{i=1}^{n} \sum_{b=1}^{B} \mathbf{1}_{[l_y+h(b-1),\, l_y+hb)}(Y_i)\ \mathbf{1}_{[l_y+h(b-1),\, l_y+hb)}(y_{n+1}),$$

where B is number of bins and $h = \dfrac{u_y - l_y}{B}$ is the bin interval.

A kernel density is a "smoothing" function $K(z)$; it typically satisfies:

- $K(z) = K(-z)$ for all z; (Symmetric)
- $K(z) \geq 0$ for all z; (Non-Negative)
- $\int K(z)\, dz = 1$; (Normalised)
- $\int zK(z)\, dz = 0$; (Centered)
- $\int z^m K(z)\, dz < \infty$ for all $m = 2, 3, \ldots$ (Finite Moments)

For a distribution with density $f(\tilde{Y}) = \prod_{i=1}^{n+1} f(Y_i)$, assume the univariate density $f(Y_{n+1})$ is infinitely differentiable at $f(y_{n+1})$. There exists a series expansion $f(y_{n+1}) = \sum_{m=1}^{\infty} \omega_m \phi_m(y_{n+1})$,

where setting $Z = (y_{n+1} - Y_i)/h$ for some $h > 0$,

$$\omega_1 = h^{-1}\mathbb{E}[K(Z)] = h^{-1} \int K\Big(\frac{y_{n+1} - y_i}{h}\Big) f(y_i)\, dy_i,$$

$$\omega_2 = \mathbb{E}[Z] = \int zK(z)\, dz = 0,$$

$$\omega_m = h^{m-1}\mathbb{E}[Z^{m-1}] = h^{m-1} \int z^{m-1}K(z)\, dz, \qquad m = 3, 4, \ldots,$$

and

$$\phi_1(y_{n+1}) = 1, \qquad \phi_m(y_{n+1}) = \frac{(-1)^m}{(m-1)!} \frac{d^{m-1}f}{dY_i^{m-1}}(y_{n+1}), \qquad m = 2, 3, \ldots.$$

Note that $\lim_{h\to 0} \omega_m = 0$ for all $m \geq 3$, and

$$\hat{\omega}_1 = \frac{1}{nh} \sum_{i=1}^{n} K\Big(\frac{y_{n+1} - Y_i}{h}\Big)$$

is an unbiased estimator of ω_1 for any kernel density $K(z)$. Consequently, the KDE is given by:

$$\hat{f}(y_{n+1}) \equiv \hat{\omega}_1 = \frac{1}{nh} \sum_{i=1}^{n} K\Big(\frac{y_{n+1} - Y_i}{h}\Big).$$

The KDE is a biased estimator:

$$\mathbb{E}[\hat{f}(y_{n+1})] = f(y_{n+1}) - \sum_{m=3}^{\infty} \omega_m \phi_m(y_{n+1}),$$

but the bias vanishes as $h \to 0$.

9.9 Properties of the KDE

The variance of the KDE is:

$$\mathrm{V}(\hat{f}(y_{n+1})) = \frac{1}{nh} f(y_{n+1}) \int K(z)^2\, dz + \frac{1}{n} \sum_{l=0}^{\infty} h^l g_l(y_{n+1}),$$

where $g_l(y_{n+1}) < \infty$ for all $l = 0, 1, \ldots$; hence, $\mathrm{V}(\hat{f}(\cdot)) \to \infty$ as $h \to 0$.
Let $\tilde{f}(y_{N+1})$ the corresponding hypothetical KDE for $N > n$, such that

$$\mathbb{E}[\tilde{f}(y_{N+1})] = f(y_{N+1}) - \sum_{m=3}^{\infty} \omega_m \phi_m(y_{N+1}),$$

$$\mathrm{V}(\tilde{f}(y_{N+1})) = \frac{1}{Nh} f(y_{N+1}) \int K(z)^2\, dz + \frac{1}{N} \sum_{l=0}^{\infty} h^l g_l(y_{N+1}).$$

The asymptotic bias and asymptotic variance both vanish as $N \to \infty$, provided that $h \to 0$ and $Nh \to 0$.

9.10 KDE in Practice

To implement the KDE in practice, we choose

- a kernel density $K(z)$, and
- the bandwidth (smoothing parameter) h.

Some examples of kernels:

$$\text{Uniform Kernel: } K(z) = \frac{1}{2}\,\mathbf{1}_{(-1,1)}(z);$$
$$\text{Triangular Kernel: } K(z) = (1-|z|)\,\mathbf{1}_{(-1,1)}(z);$$
$$\text{Epanechnikov Kernel: } K(z) = \frac{3}{4}(1-z^2)\,\mathbf{1}_{(-1,1)}(z);$$
$$\text{Gaussian Kernel: } K(z) = \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{z^2}{2}\right\}.$$

Different kernels weigh differently the observations $y_1, \ldots, y_n$.

- The uniform kernel puts equal weights within the interval $[y_{n+1} - h,\, y_{n+1} + h]$.
- The triangular kernel puts weight (inversely) proportional to the distance from y_{n+1} in the interval $[y_{n+1} - h,\, y_{n+1} + h]$.
- The Epanechnikov kernel is bell-shape on the interval $[y_{n+1} - h,\, y_{n+1} + h]$.
- The Gaussian kernel is bell-shape on $\mathbb{R}$.

It turns out that the choice of h is much more important than the choice of $K(z)$!

- h too large $\rightarrow$ estimated density too smooth ("flat line");
- h too small $\rightarrow$ estimated density too noisy ("jagged").

9.11 Multivariate KDE

The KDE methodology may be extended to multivariate distributions by defining multivariate kernels.

Curse of dimensionality: implementing KDE with densities involving more than four variables is not practical.

A non-parametric bivariate density $f(y_{n+1}, \boldsymbol{x}_{n+1})$ can be constructed using KDE and a sample $D = \{y_i, \boldsymbol{x}_i\}$ to estimate a nonlinear conditional mean $\mathbb{E}[Y_{n+1} \mid X_{n+1} = \boldsymbol{x}_{n+1}]$.

- This is known as the non-parametric regression.
- The resulting conditional mean estimator is called the Nadaraya–Watson estimator.

10 Time-series Models

10.1 Review of Time-Series Models

Time-series models are specialised statistical models, designed for data consisting of ordered observations.

In economic settings, time-series data (TS data) is collected for a group of entities at multiple points in time. The most common examples are in:

- **Macroeconomics**—e.g., real GDP, inflation and unemployment indicators collected quarterly from 1950Q1 to 2025Q2.
- **Finance**—e.g., closing share prices of S&P500 stocks collected weekly from 2010W1 to 2025W26.

Time-series models facilitate using data to forecast a variety of important indicators and guide critical decisions:

- medium-term inflation forecasts for monetary policy;
- long-term GDP growth forecasts for development strategies;
- short-term orders forecasts for supply chain management;

Time-series models facilitate using data to answer questions about **dynamic relationships**, which are central to many policy decisions:

- How does inflation respond to a monetary policy shock within the first three months?
- What is the long run increase in consumption following a decrease in the tax rate?
- How quickly do sales respond to higher marketing expenditures?
- How does news about future productivity affect business cycles?

Definition 10.1.1 (Stochastic process)**.** A stochastic process is a collection of random vectors that are ordered, presumably (for convenience) in time.

- Each random vector $\boldsymbol{y}_t = (Y_{1t}, \ldots, Y_{n_y t})'$ is $n_y \times 1$.
- The process is denoted by $\{\boldsymbol{y}_t : t \in \mathbb{Z}\}$ or simply $\{\boldsymbol{y}_t\}$.
- The lag operator L is often useful:

$$\boldsymbol{y}_{t-h} \equiv L^h \boldsymbol{y}_t, \qquad \boldsymbol{y}_{t+h} \equiv L^{-h} \boldsymbol{y}_t.$$

- All the random vectors are related according to probabilities.
- TS data is one realisation of a sub-process: $\{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_T\}$.

Forecasts and dynamic relationships extrapolate patterns from TS data.

- The patterns are modeled as auto-covariances: $\mathrm{C}(\boldsymbol{y}_t, \boldsymbol{y}_{t+h})$.
- Covariance-stationary:

$$\mathrm{C}(\boldsymbol{y}_t, \boldsymbol{y}_{t+h}) \equiv \boldsymbol{\Gamma}(h) \quad \text{is independent of } t.$$

- $\boldsymbol{\Gamma}(h)$ plotted against $h \geq 0$ is the auto-covariance function (ACF).
- If $\mathbb{E}[\boldsymbol{y}_t] = \mathbf{0}$, $\boldsymbol{\Gamma}(0)$ is p.d., and $\boldsymbol{\Gamma}(h) = \mathbf{0}$ for all $h \geq 1$, then $\{\boldsymbol{y}_t\}$ is a white-noise process, with no extractable patterns.

Theorem 10.1.2 (Wold Decomposition Theorem)**.** *Let $\{\boldsymbol{y}_t\}$ denote a stochastic process, which is:*

1. *mean-zero: $\mathbb{E}[\boldsymbol{y}_t] = \mathbf{0}$ for all t;*
2. *covariance-stationary: $\mathbb{E}[\boldsymbol{y}_t \boldsymbol{y}_{t+h}'] \equiv \boldsymbol{\Gamma}(h)$ for all t;*
3. *purely non-deterministic—intuitively, no process $\{\boldsymbol{v}_t\}$ exists such that $\boldsymbol{y}_t - \boldsymbol{v}_t$ is perfectly predictable.*

Then,

$$\boldsymbol{y}_t = \sum_{l=0}^{\infty} \boldsymbol{\Phi}_l \boldsymbol{u}_{t-l},$$

where:

- *$\{\boldsymbol{u}_t\}$ is r_u-dimensional white-noise, $r_u \leq n_y$, and $\mathbb{E}[\boldsymbol{u}_t \boldsymbol{u}_t'] = \boldsymbol{\Sigma}_u$;*
- *$\boldsymbol{\Phi}_0 = \boldsymbol{I}_{r_u}$ and $\sum_{l=1}^{\infty} \boldsymbol{\Phi}_l \boldsymbol{\Sigma}_u \boldsymbol{\Phi}_l' < \infty$.*

The ACF can be expressed as:

$$\boldsymbol{\Gamma}(h) = \sum_{l=0}^{\infty} \boldsymbol{\Phi}_l \boldsymbol{\Sigma}_u \boldsymbol{\Phi}_{l+h}', \qquad h = 1, 2, \ldots.$$

10.2 Univariate ARMA Models

The simplest case is the univariate process where $n_y = 1$:

$$Y_t = \sum_{l=0}^{\infty} \Phi_l U_{t-l} \quad \text{with} \quad \mathbb{E}[U_t^2] = \sigma_u^2.$$

Using the lag operator, this is equivalent to

$$Y_t = \Phi(L) U_t, \qquad \Phi(L) = 1 + \sum_{l=1}^{\infty} \Phi_l L^l, \quad \sum_{l=1}^{\infty} \Phi_l^2 < \infty.$$

It is useful to view $\Phi(L)$ as a function: replace the operator L with the variable $z \in \mathbb{C}$ to obtain $\Phi(z)$. The function $\Phi(z) = 1 + \Phi_1 z + \Phi_2 z^2 + \cdots$ can be approximated by a rational function:

$$\Phi(z) \approx \frac{b(z)}{a(z)},$$

where

- $a(z) = 1 - a_1 z - \cdots - a_p z^p$ is a polynomial of degree p and $a(z) \neq 0$ for all $|z| \leq 1$;
- $b(z) = 1 + b_1 z + \cdots + b_q z^q$ is a polynomial of degree q and $b(z) \neq 0$ for all $|z| \leq 1$.

Now replace z again with L:

$$a(L) Y_t \approx b(L) U_t,$$

or equivalently, the model

$$Y_t = a_1 Y_{t-1} + \cdots + a_p Y_{t-p} + U_t + b_1 U_{t-1} + \cdots + b_q U_{t-q}.$$

- This is the ARMA(p, q) model of the mean-zero, covariance-stationary process $\{Y_t\}$.
- It is a useful building block to construct forecasting models for economic processes.

The parameters $a_1, \ldots, a_p, b_1, \ldots, b_q, \sigma_u^2$ are related to auto-covariances through the pseudo-regression:

$$Y_t = \boldsymbol{z}_t' \boldsymbol{\delta} + U_t,$$

where

$$\boldsymbol{z}_t = (Y_{t-1}, \ldots, Y_{t-p}, U_{t-1}, \ldots, U_{t-q})', \qquad \boldsymbol{\delta} = (a_1, \ldots, a_p, b_1, \ldots, b_q)',$$

and

$$\operatorname{rank} \mathbb{E}[\boldsymbol{z}_t \boldsymbol{z}_t'] = p + q.$$

Example 10.2.1 (Forecasting GDP)**.** Suppose we wish to forecast GDP using $x_1, \ldots, x_T$.

- Model the GDP process $\{X_t\}$ as:

$$X_t = \gamma_0 + \gamma_1 t + \sum_{s=-\infty}^{t} Y_t.$$

- Equivalent representations:
  - **GDP growth:** $a(L)\Delta X_t = d + b(L)U_t$, where $\Delta = 1 - L$.
  - **Initial condition:** $a(L)(X_t - X_0) = c + dt + b(L)\sum_{s=1}^{t} U_s$.

Note that $\mathbb{E}[X_t] = \gamma_0 + \gamma_1 t$ and $\mathrm{V}(X_t)$ is not finite.

- This model imposes that $\{X_t\}$ is not stationary and completely unpredictable in the long run:

$$\mathrm{V}[X_{T+h} \mid X_T, \ldots] \to \infty \quad \text{as } h \to \infty.$$

- However, GDP growth $\{\Delta X_t\}$ is stationary $\Rightarrow$ long-run predictable:

$$\lim_{h\to\infty} \mathbb{E}[\Delta X_{T+h} \mid X_T, \ldots] = \gamma_1, \qquad \lim_{h\to\infty} \mathrm{V}[\Delta X_{T+h} \mid X_T, \ldots] = \sigma_u^2 \sum_{l=0}^{\infty} \Phi_l^2 < \infty.$$

10.3 Multivariate ARMA Models

In the general case with $n_y \geq 1$ and $r_u \leq n_y$:

- the Wold Theorem yields: $\boldsymbol{y}_t = \boldsymbol{\Phi}(L)\boldsymbol{u}_t$;
- $\boldsymbol{\Phi}(L)$ is an $n_y \times r_u$ matrix and each entry is $\Phi_{ij}(L)$.

Apply the univariate approach to each $\Phi_{ij}(L)$ and use a rational function approximation:

$$\Phi_{ij}(z) \approx \frac{b_{ij}(z)}{a_{ij}(z)}.$$

This implies there exists a "matrix version" of the polynomial fraction:

$$\boldsymbol{\Phi}(z) = \boldsymbol{A}(z)^{-1}\boldsymbol{B}(z),$$

where

- $\boldsymbol{A}(z)$ is $n_y \times n_y$ with polynomial entries $A_{ij}(z)$ and $|\boldsymbol{A}(z)| \neq 0$ for all $|z| \leq 1$;
- $\boldsymbol{B}(z)$ is $n_y \times r_u$ with polynomial entries $B_{ij}(z)$ and if $r = n$, $|\boldsymbol{B}(z)| \neq 0$ for all $|z| \leq 1$.

Setting $p = \max \deg A_{ij}(z)$ and $q = \max \deg B_{ij}(z)$,

$$\boldsymbol{A}(L)\boldsymbol{y}_t = \boldsymbol{B}(L)\boldsymbol{u}_t$$

is a VARMA(p, q) model of the mean-zero, covariance-stationary $\{\boldsymbol{y}_t\}$.

The parameters of a VARMA(p, q) are the elements of matrices $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_p$, $\boldsymbol{B}_1, \ldots, \boldsymbol{B}_q$ and $\boldsymbol{\Sigma}_u$, where

$$\boldsymbol{A}(L) = \boldsymbol{I}_{n_y} - \boldsymbol{A}_1 L - \cdots - \boldsymbol{A}_p L^p, \qquad \boldsymbol{B}(L) = \boldsymbol{I}_{r_u} + \boldsymbol{B}_1 L + \cdots + \boldsymbol{B}_q L^q.$$

Similar to the univariate ARMA, the VARMA(p, q) may be represented as a pseudo-SEM:

$$\boldsymbol{y}_t' = \boldsymbol{z}_t' \boldsymbol{\Delta} + \boldsymbol{u}_t,$$

where

$$\boldsymbol{z}_t = (\boldsymbol{y}_{t-1}', \ldots, \boldsymbol{y}_{t-p}', \boldsymbol{u}_{t-1}', \ldots, \boldsymbol{u}_{t-q}')', \qquad \boldsymbol{\Delta} = (\boldsymbol{A}_1, \ldots, \boldsymbol{A}_p, \boldsymbol{B}_1, \ldots, \boldsymbol{B}_q)'.$$

However, unlike in univariate ARMAs or SEMs discussed in Lecture 6,

$$\operatorname{rank} \mathbb{E}[\boldsymbol{z}_t \boldsymbol{z}_t'] \leq n(p + q).$$

- $\mathbb{E}[\boldsymbol{z}_t \boldsymbol{z}_t']$ may be singular (recall SEM2 in Lecture 6).
- VARMA(p, q) coefficients may be unidentified.

10.4 Vector Autoregressive Models

To apply the SEM methodology, the vast majority of empirical work has adopted the following transformation.

Assume $n_y = r_u$.

- The Wold Theorem yields $|\boldsymbol{B}(z)| \neq 0$ for all $|z| \leq 1$.
- There exists an $n_y \times n_y$ matrix

$$\boldsymbol{\Pi}(z) = \boldsymbol{B}(z)^{-1}\boldsymbol{A}(z) = \boldsymbol{I}_n - \boldsymbol{\Pi}_1 L - \cdots - \boldsymbol{\Pi}_l L^l - \cdots,$$

  with elements $\Pi_{ij}(z) = \sum_{l=0}^{\infty} \Pi_{ij,l} L^l$ and $\sum_{l=0}^{\infty} |\Pi_{ij,l}| < \infty$.

Because the series $\sum_{l=0}^{\infty} |\Pi_{ij,l}|$ converges, it follows that

$$\lim_{l \to \infty} \Pi_{ij,l} = 0;$$

that is, higher-order terms $\boldsymbol{\Pi}_l$ are negligible.

This motivates using a truncated

$$\boldsymbol{\Pi}(L) = \boldsymbol{I}_{n_y} - \boldsymbol{\Pi}_1 L - \cdots - \boldsymbol{\Pi}_p L^p$$

as a reasonable approximation:

$$\boldsymbol{y}_t' = \boldsymbol{x}_t'\boldsymbol{\Pi} + \boldsymbol{u}_t,$$

where

$$\boldsymbol{x}_t' = (\boldsymbol{y}_{t-1}', \ldots, \boldsymbol{y}_{t-p}')', \qquad \boldsymbol{\Pi} = (\boldsymbol{\Pi}_1, \ldots, \boldsymbol{\Pi}_p)'.$$

This is known as the VAR(p) model of $\{\boldsymbol{y}_t\}$.

- $\mathbb{E}[\boldsymbol{x}_t\boldsymbol{x}_t'] = np$ for any $p \geq 0$, so SEM2 is satisfied.
- Note, however, that p is generally not equivalent to $\max\deg \boldsymbol{A}(z)$ in a corresponding VARMA representation.
- In fact, there always exists a VARMA model for $\{\boldsymbol{y}_t\}$ which is more flexible and more parsimonious than the VAR(p).
- But the VARMA model cannot be implemented with OLS!

10.5 Matching Statistical Models to Economic Theory

To obtain inference on dynamic relationships, we need to match a time-series model such as the VARMA derived from the Wold Theorem to a class of economic models.

In macroeconomics, modern theory develops dynamic, stochastic general equilibrium (DSGE) models.

Example 10.5.1 (The Real Business Cycle (RBC) Model.). • Exogenous variable: A_t (technology);

- Endogenous variables: Y_t (output), C_t (consumption), I_t (investment), N_t (labour), K_t (capital), R_t (interest rate);
- Primitive shock: ϵ_t (technological innovation);
- Structural parameters:

$$\boldsymbol{\theta} = (a, \alpha, \beta, \delta, \eta, \rho, \sigma_\epsilon)'.$$

The RBC equilibrium:

$$\begin{aligned}
Y_t &= C_t + I_t, \\
Y_t &= A_t K_{t-1}^{\alpha} N_t^{1-\alpha}, \\
K_t &= I_t + (1-\delta)K_{t-1}, \\
R_t &= \alpha \frac{Y_t}{K_{t-1}} + 1 - \delta, \\
C_t^{-\eta} &= \beta\, \mathbb{E}_t\left[C_{t+1}^{-\eta} R_{t+1}\right], \\
\frac{Y_t}{N_t} &= \frac{a}{1-\alpha} C_t^{\eta}, \\
\ln A_t &= (1-\rho)\ln A^* + \rho \ln A_{t-1} + \sigma_\epsilon \epsilon_t.
\end{aligned}$$

$\mathbb{E}_t[\cdot]$ denotes the conditional expectation, which conditions on information at time t (realisations of all random variables up to t).

A^* is the technology steady-state: $\mathbb{E}[A_t] = A^*$ iff $\rho < 1$.

- This implies that the six endogenous variables also have steady-states: $Y^*, C^*, I^*, N^*, K^*, R^*$.

10.6 General Properties of DSGEs and State-space Form

The basic elements of most available DSGEs are:

- n_w states, collectively denoted by $\boldsymbol{w}_t$. These include exogenous and endogenous variables; may also include measurement error processes.
- r_ϵ structural shocks, denoted by $\boldsymbol{\epsilon}_t$. These consist of economically interpretable shocks and may also include innovations to measurement errors.
- k_θ structural parameters, denoted by $\boldsymbol{\theta}$.

The structural shocks $\boldsymbol{\epsilon}_t$ are:

- mean-zero (i.e., $\mathbb{E}[\boldsymbol{\epsilon}_t] = \mathbf{0}$);
- uncorrelated in the cross-section (i.e., $\mathrm{V}(\boldsymbol{\epsilon}_t) = \boldsymbol{I}_{r_\epsilon}$).

A first-order (log-linearised) approximation to the equilibrium solution can be expressed as:

$$\boldsymbol{w}_t = \boldsymbol{A}(\boldsymbol{\theta})\boldsymbol{w}_{t-1} + \boldsymbol{B}(\boldsymbol{\theta})\boldsymbol{\epsilon}_t \qquad \text{(State Equation)},$$

where:

- $\boldsymbol{A}(\boldsymbol{\theta})$ is $n_w \times n_w$ and $\boldsymbol{B}(\boldsymbol{\theta})$ is $n_w \times r_\epsilon$;
- $\mathbb{E}[\boldsymbol{w}_t] = \mathbf{0}$; equivalently $\lim_{l\to\infty} \boldsymbol{A}(\boldsymbol{\theta})^l = \mathbf{0}$;
- $\operatorname{rank} \boldsymbol{B}(\boldsymbol{\theta}) = r_\epsilon$ and $\operatorname{rank}(\boldsymbol{A}(\boldsymbol{\theta}), \boldsymbol{B}(\boldsymbol{\theta})) = n_w$;
- n_w, r_ϵ are unique, but $\boldsymbol{w}_t, \boldsymbol{A}(\boldsymbol{\theta}), \boldsymbol{B}(\boldsymbol{\theta})$ are not unique.

It is almost always the case that $n_w \gg r_\epsilon$.

To match a model of $\{\boldsymbol{y}_t\}$ to DSGEs, we assume:

$$\boldsymbol{y}_t = \boldsymbol{C}(\boldsymbol{\theta})\boldsymbol{w}_{t-1} + \boldsymbol{D}(\boldsymbol{\theta})\boldsymbol{\epsilon}_t \qquad \text{(Measurement Equation)},$$

where $\boldsymbol{C}(\boldsymbol{\theta})$ is $n_y \times n_w$ and $\boldsymbol{D}(\boldsymbol{\theta})$ is $n_y \times r_\epsilon$.

Together, the state and measurement equations form a state-space system.

Using the lag operator and re-arranging:

$$\begin{aligned}
\boldsymbol{y}_t &= \Big(\boldsymbol{C}(\boldsymbol{I}_{n_w} - \boldsymbol{A}L)^{-1}\boldsymbol{B}L + \boldsymbol{D}\Big)\boldsymbol{\epsilon}_t \\
&= \underbrace{\boldsymbol{CB} + \boldsymbol{D}}_{\boldsymbol{K}_0}\boldsymbol{\epsilon}_t + \underbrace{\boldsymbol{CAB}}_{\boldsymbol{K}_1} L\boldsymbol{\epsilon}_t + \underbrace{\boldsymbol{CA}^2\boldsymbol{B}}_{\boldsymbol{K}_2} L^2\boldsymbol{\epsilon}_t + \cdots \\
&= \boldsymbol{K}(L)\boldsymbol{\epsilon}_t.
\end{aligned}$$

10.7 Impulse Response Functions

The $n_y \times nr_\epsilon$ matrix $\boldsymbol{K}(z)$ of functions $K_{ij}(z) = \sum_{l=0}^{\infty} K_{ij,l} z^l$ has crucial features:

- Each $K_{ij,l}$ quantifies an impulse response.
- Each $K_{ij}(z)$ is precisely rational—equivalent to a ratio of polynomial functions.

Impulse responses (IR) summarise the dynamic effects of shocks on the economy.

- A realisation of a shock at time t represents new information relative to the past $(t-1, t-2, \ldots)$.
- The new information changes the expected path of $\{\boldsymbol{y}_t\}$ from time t onward $(t, t+1, \ldots)$.
- IR functions (IRFs) quantify the difference in expected paths: after the shock occurs vs. before the shock occurs.

The impulse response of variable i at horizon h to a shock j at time t is:

$$\mathrm{IR}_{ij}(h) = \mathbb{E}_{t-1}[Y_i(t+h) \mid \epsilon_{jt} = 1] - \mathbb{E}_{t-1}[Y_i(t+h)]\,.$$

Since

$$Y_i(t+h) = \sum_{l=0}^{\infty} \sum_{j=1}^{r_\epsilon} K_{ij,l}\, \epsilon_{j(t+h-l)},$$

we can write

$$\begin{aligned}
Y_i(t+h) &= \sum_{l=0}^{h-1} \sum_{j=1}^{r_\epsilon} K_{ij,l}\, \epsilon_{j(t+h-l)} + \sum_{j=1}^{r_\epsilon} K_{ij,h}\, \epsilon_{jt} + \sum_{l=h+1}^{\infty} \sum_{j=1}^{r_\epsilon} K_{ij,l}\, \epsilon_{j(t+h-l)}, \\
\mathbb{E}_{t-1}[Y_i(t+h) \mid \epsilon_{jt} = 1] &= K_{ij,h} + \sum_{l=h+1}^{\infty} \sum_{j=1}^{r_\epsilon} K_{ij,l}\, \epsilon_{j(t+h-l)}, \\
\mathbb{E}_{t-1}[Y_i(t+h)] &= \sum_{l=h+1}^{\infty} \sum_{j=1}^{r_\epsilon} K_{ij,l}\, \epsilon_{j(t+h-l)}.
\end{aligned}$$

Therefore,

$$\mathrm{IR}_{ij}(h) = K_{ij,h}.$$

10.8 The VARMA Representation of a DSGE

Assume $n_y = r_\epsilon$.

Because $\boldsymbol{K}(z)$ is a matrix of rational functions, there exists a "matrix fraction":

$$\boldsymbol{K}(z) = \boldsymbol{\Psi}(z)^{-1}\boldsymbol{\Theta}(z),$$

where

- $\boldsymbol{\Psi}(z)$ is $n_y \times n_y$ with polynomial entries $\Psi_{ij}(z)$ and $|\boldsymbol{\Psi}(z)| \neq 0$ for all $|z| \leq 1$;
- $\boldsymbol{\Theta}(z)$ is $n_y \times n_y$ with polynomial entries $\Theta_{ij}(z)$.

Consequently, the state-space system for a DSGE is equivalent to the VARMA(p, q) representation:

$$\boldsymbol{\Psi}(L)\boldsymbol{y}_t = \boldsymbol{\Theta}(L)\boldsymbol{\epsilon}_t.$$

However, a crucial difference to the VARMA derived from the Wold Theorem is that $|\boldsymbol{\Theta}(z)| = 0$ for some $|z| < 1$ is possible.

- In fact, this will hold in any DSGE where agents anticipate effects—e.g., technology news, policy implementation lags.

10.9 DSGE VARMA vs Wold VARMA and Identification of IRFs

To match a VARMA derived from Wold Theorem, including the VAR approximation, we may set:

- $n_y = r_\epsilon$;
- $\boldsymbol{\Sigma}_u = \boldsymbol{K}_0 \boldsymbol{K}_0'$;
- $\boldsymbol{A}(z) = \boldsymbol{\Psi}(z)$;
- $\boldsymbol{B}(z)\boldsymbol{K}_0\boldsymbol{Q}(z) = \boldsymbol{\Theta}(z)$, where
$$\boldsymbol{Q}(z) = \boldsymbol{I}_{n_y} + \sum_{l=1}^{\infty} \boldsymbol{Q}_l z^l$$
satisfies
$$\boldsymbol{Q}(z)\boldsymbol{Q}(z^{-1})' = \boldsymbol{Q}(z^{-1})'\boldsymbol{Q}(z) = \boldsymbol{I}_{n_y}, \qquad \forall z \in \mathbb{C}.$$

$\boldsymbol{Q}(z)$ is called a para-unitary matrix. To obtain inference from a Wold VARMA in this match, it is necessary to use external information to:

1. derive $\boldsymbol{K}_0$ from $\boldsymbol{\Sigma}_u$;
2. fix $\boldsymbol{Q}(z)$.

If $\boldsymbol{K}_0$ and $\boldsymbol{Q}(z)$ are determined, then

$$\boldsymbol{K}(z) = \boldsymbol{A}(z)^{-1}\boldsymbol{B}(z)\boldsymbol{K}_0\boldsymbol{Q}(z).$$

- This match allows for inference on all IRFs using a Wold VARMA as the statistical model.
- If only IRFs to shock j are required, then only column j of $\boldsymbol{K}_0$ and column j of $\boldsymbol{Q}(z)$ need to be determined with external information.

It is very common practice to set $\boldsymbol{Q}(z) = \boldsymbol{I}_{n_y}$ and use $\boldsymbol{\Pi}(z) \approx \boldsymbol{B}(z)^{-1}\boldsymbol{A}(z)$ to approximate

$$\boldsymbol{K}(z) \approx \boldsymbol{\Pi}(z)^{-1}\boldsymbol{K}_0.$$

- This is the structural vector autoregression (SVAR) methodology.

What type of DSGE models can SVARs be matched to?

- Setting $\boldsymbol{Q}(z) = \boldsymbol{I}_{n_y}$ implies $\boldsymbol{\Theta}(z)$ satisfies $|\boldsymbol{\Theta}(z)| \neq 0$ for all $|z| \leq 1$.
- This implies only DSGEs without policy implementation lags, anticipated technological innovations, etc. can be matched!

11 Panel Models

11.1 Panel Data

Panel data (similar to TS data) refers to structures consisting of observations on entities for multiple time periods.

- Alternatively, the term *longitudinal data* is sometimes used (more common in statistics, etc.).

- Distinguishing feature: presence of multiple observations for each entity, but focus is on the cross-section.

There are several distinct advantages of panel models relative to cross-sectional models:

1. control for unobserved time-invariant endogeneity without instrumental variables;
2. allow for broader forms of heterogeneity;
3. model dynamic relationships and effects (multivariate time-series models).

11.2 Pooled Regression

Some notations:

- S_i is the set of time periods for individual i.
- If the panel is balanced, then $S_i = \{1, \ldots, T\}$.
- If the panel is unbalanced, then $T_i = \dim S_i$.
- Y_{it} is the dependent variable and $\boldsymbol{x}_{it}$ is a $k_x \times 1$ vector of regressors for entity i at time period t.
- $\boldsymbol{y}_i$ is a $T_i \times 1$ vector and $\boldsymbol{X}_i$ is a $T_i \times k_x$ matrix that "stack" RVs indexed by $t \in S_i$, in chronological order, for entity i.
- $\boldsymbol{y} = (\boldsymbol{y}_1', \ldots, \boldsymbol{y}_n')'$ is $\sum_{i=1}^n T_i \times 1$, and $\boldsymbol{X} = (\boldsymbol{X}_1', \ldots, \boldsymbol{X}_n')'$ is $\sum_{i=1}^n T_i \times k$.

Pooled Regression:

$$Y_{it} = \boldsymbol{x}_{it}'\boldsymbol{\beta} + U_{it} \quad \text{and OLS assumptions hold.}$$

- Effectively, ignore panel structure.
- $\mathbb{E}[U_{it} \mid \boldsymbol{X}_i] = 0$ is referred to as *strict mean-independence*; for example, it excludes co-variates such as $X_{ijt} = Y_{i(t-1)}$.

11.3 Random Effects Regression

The pooled regression assumes for every entity i, the residuals $U_{i1}, \ldots, U_{iT}$ are uncorrelated.

One-way error component model:

$$U_{it} = \alpha_i + e_{it}.$$

- α_i is an entity-specific residual effect.
- e_{it} is an idosynchratic error.
- The residuals $U_{i1}, \ldots, U_{iT}$ cluster within entities.

Random Effects Regression: alternative (to OLS) assumptions.

RE1 $\mathbb{E}[e_{it} \mid \boldsymbol{X}_i] = 0$ for all i and t.

RE2 $\mathbb{E}[e_{it}^2 \mid \boldsymbol{X}_i] = \sigma_e^2$ for all i and t.

RE3 $\mathbb{E}[e_{it}e_{js} \mid \boldsymbol{X}_i, \boldsymbol{X}_j] = 0$ for all i, j and $s \neq t$.

RE4 $\mathbb{E}[\alpha_i \mid \boldsymbol{X}_i] = \alpha$ for all i.

RE5 $\mathbb{E}[\alpha_i^2 \mid \boldsymbol{X}_i] = \sigma_\alpha^2$ for all i.

RE6 $\mathbb{E}[\alpha_i \alpha_j \mid \boldsymbol{X}_i, \boldsymbol{X}_j] = 0$ for all $i \neq j$.

RE7 $\mathbb{E}[\alpha_i e_{it} \mid \boldsymbol{X}_i] = 0$ for all i and t.

Assumptions RE1–RE2 imply $\mathbb{E}[\boldsymbol{u}_i \mid \boldsymbol{X}_i] = 0$ and

$$\mathbb{E}[\boldsymbol{u}_i \boldsymbol{u}_i' \mid \boldsymbol{X}_i] = \sigma_e^2 \boldsymbol{\Omega}_i,$$

where

$$\boldsymbol{\Omega}_i = \boldsymbol{I}_{T_i} + \frac{\sigma_\alpha^2}{\sigma_e^2} \mathbf{1}_{T_i} \mathbf{1}_{T_i}',$$

and $\mathbf{1}_{T_i}$ is a $T_i \times 1$ vector of 1s.

- Equivalent equi-correlation: $\sigma_\alpha^2 = \rho \sigma_u^2$ and $\sigma_e^2 = (1-\rho)\sigma_u^2$, then

$$\mathbb{E}[U_{it}^2 \mid \boldsymbol{X}_i] = \sigma_u^2 \quad \text{and} \quad \mathbb{E}[U_{it} U_{is} \mid \boldsymbol{X}_i] = \rho \sigma_u^2 \text{ for all } t \neq s.$$

- $\boldsymbol{\Omega}_i$ depends on i since its dimension depends on T_i.

The RE regression may be implemented with GLS:

$$\widehat{\boldsymbol{\beta}}_{\text{RE}} = \left(\sum_{i=1}^n \boldsymbol{X}_i' \boldsymbol{\Omega}_i^{-1} \boldsymbol{X}_i \right)^{-1} \left(\sum_{i=1}^n \boldsymbol{X}_i' \boldsymbol{\Omega}_i^{-1} \boldsymbol{y}_i \right).$$

- In practice, a feasible GLS estimator is used, with σ_α^2 and σ_e^2 replaced by estimators.

11.4 Fixed Effects Regression

In many applications it is useful to interpret α_i as a time-invariant unobserved missing variable.

- Wage regression: α_i may be the unobserved ability of individual i.
- Investment regression: α_i may be a firm-specific productivity factor.

In such interpretations, α_i is called a fixed effect and the assumption $\mathbb{E}[\alpha_i \mid \boldsymbol{X}_i]$ is very restrictive.

Fixed Effects Regression: relax assumptions RE4–RE7.

Fixed Effects Estimator: transform the regression to eliminate α_i.

Define:

$$\bar{Y}_i = \frac{1}{T_i} \sum_{t \in S_i} Y_{it}, \qquad \dot{Y}_{it} = Y_{it} - \bar{Y}_i,$$

$$\bar{\boldsymbol{x}}_i = \frac{1}{T_i} \sum_{t \in S_i} \boldsymbol{x}_{it}, \qquad \dot{\boldsymbol{x}}_{it} = Y_{it} - \bar{\boldsymbol{x}}_i,$$

$$\bar{U}_i = \frac{1}{T_i} \sum_{t \in S_i} U_{it}, \qquad \dot{U}_{it} = U_{it} - \bar{U}_i,$$

$$\bar{e}_i = \frac{1}{T_i} \sum_{t \in S_i} e_{it}, \qquad \dot{e}_{it} = e_{it} - \bar{e}_i.$$

Note that $\dot{U}_{it} = \dot{e}_{it}$, and RE1–RE3 imply

$$\mathbb{E}[\dot{e}_{it} \mid \boldsymbol{X}_i] = 0, \qquad \mathbb{E}[\dot{e}_{it}^2 \mid \boldsymbol{X}_i] = \frac{T_i - 1}{T_i}\sigma_e^2, \qquad \mathbb{E}[\dot{e}_{it}\dot{e}_{is} \mid \boldsymbol{X}_i] = -\frac{1}{T_i}\sigma_e^2.$$

Since

$$Y_{it} = \boldsymbol{x}_{it}'\boldsymbol{\beta} + \alpha_i + e_{it}, \qquad \bar{Y}_i = \bar{\boldsymbol{x}}_i'\boldsymbol{\beta} + \alpha_i + \bar{e}_i,$$

it implies

$$\dot{Y}_{it} = \dot{\boldsymbol{x}}_{it}'\boldsymbol{\beta} + \dot{e}_{it}. \tag{1}$$

We estimate (1), with the fixed effect eliminated, using GLS.

The GLS estimator in this case is:

$$\widehat{\boldsymbol{\beta}}_{\text{FE}} = \left(\sum_{i=1}^{n} \dot{\boldsymbol{X}}_i'\dot{\boldsymbol{X}}_i\right)^{-1} \left(\sum_{i=1}^{n} \dot{\boldsymbol{X}}_i'\dot{\boldsymbol{y}}_i\right).$$

11.5 The Hierarchical Modelling Interpretation of FEs

Note that $\boldsymbol{x}_{it}$ cannot include RVs that are time-invariant.

Consider the more general specification:

$$Y_{it} = \boldsymbol{x}_{it}'\boldsymbol{\beta}_i + \alpha_i + e_{it},$$

and set:

$$\boldsymbol{X} = \begin{pmatrix} (\mathbf{1}_{T_1}, \boldsymbol{X}_1) \\ \vdots \\ (\mathbf{1}_{T_n}, \boldsymbol{X}_n) \end{pmatrix}, \qquad \boldsymbol{\gamma} = \begin{pmatrix} \alpha_1 \\ \boldsymbol{\beta}_1 \\ \vdots \\ \alpha_n \\ \boldsymbol{\beta}_n \end{pmatrix}.$$

The general FE regression is equivalent to the LR:

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\gamma} + \boldsymbol{e}.$$

- $\boldsymbol{\gamma}$ is $(k_x + 1)n \times 1$; the number of parameters grows with n.
- OLS assumptions A1–A5 are satisfied.
- Asymptotic inference based on A6(ii) requires $T_i \to \infty$.
- Effectively, OLS $\Rightarrow$ no pooling across entities.

To pool information across entities, consider the SEM:

$$(\alpha_i, \boldsymbol{\beta}_i') = (\alpha, \boldsymbol{\beta}') + \boldsymbol{z}_i'\boldsymbol{\Pi} + \boldsymbol{v}_i',$$

$$\mathbb{E}[\boldsymbol{v}_i \mid \boldsymbol{X}_i, \boldsymbol{z}_i] = 0, \qquad \mathrm{V}(\boldsymbol{v}_i \mid \boldsymbol{X}_i, \boldsymbol{z}_i) = \boldsymbol{\Omega} = \begin{pmatrix} \Omega_\alpha & \boldsymbol{\Omega}_{\alpha,\beta} \\ \boldsymbol{\Omega}_{\alpha,\beta}' & \boldsymbol{\Omega}_\beta \end{pmatrix}.$$

In general, $\boldsymbol{z}_i$ is a $k_z \times 1$ vector whose elements are time-invariant, and hence, not included in $\boldsymbol{x}_{it}$.

Let $\boldsymbol{\Pi} = (\boldsymbol{\Pi}_1, \boldsymbol{\Pi}_2)$, where $\boldsymbol{\Pi}_1$ is $k_z \times 1$; the FE regression pools information by setting $\boldsymbol{\Pi}_2 = \mathbf{0}$, $\boldsymbol{\Omega}_{\alpha,\beta} = \mathbf{0}$, $\boldsymbol{\Omega}_\beta = \mathbf{0}$. Equivalently,

- allow $\alpha_i = \alpha + \boldsymbol{z}_i'\boldsymbol{\Pi}_1 + v_{i1}$, but
- restrict $\boldsymbol{\beta}_i = \boldsymbol{\beta}$.

The LR $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\gamma} + \boldsymbol{e}$ is very flexible but not very parsimonious.

The FE regression is not very flexible but very parsimonious.

We can better control the trade-off between flexibility and parsimony by shrinking

$$\boldsymbol{\Pi}_2 \to \boldsymbol{0}, \quad \boldsymbol{\Omega}_{\alpha,\beta} \to \boldsymbol{0}, \quad \boldsymbol{\Omega}_{\beta} \to \boldsymbol{0}.$$

Even better: control the trade-off by shrinking

$$\boldsymbol{\Pi} \to \boldsymbol{0}, \quad \boldsymbol{\Omega} \to \boldsymbol{0}.$$

- This will, in fact, shrink from the LR to the pooled regression (i.e. $\boldsymbol{\Pi} = \boldsymbol{0}$ and $\boldsymbol{\Omega} = \boldsymbol{0}$).
- The shrinking may be implemented using prior distributions.

11.6 The Bayesian Panel Regression

In a Bayesian framework, we may specify:

$$Y_{it} = \boldsymbol{x}_{it}'\mathring{\boldsymbol{\beta}}_i + \mathring{\alpha}_i + e_{it}, \qquad e_{it} \sim N(0, \mathring{\sigma}_i^2),$$

$$(\mathring{\alpha}_i, \mathring{\boldsymbol{\beta}}_i') = (\mathring{\alpha}, \mathring{\boldsymbol{\beta}}') + \boldsymbol{z}_i'\mathring{\boldsymbol{\Pi}} + \boldsymbol{v}_i', \qquad \boldsymbol{v}_i \sim N(0, \mathring{\sigma}_i^2\mathring{\boldsymbol{\Omega}}).$$

Let vec$(\cdot)$ denote matrix vectorisation and $\otimes$ the Kronecker product; setting $\mathring{\boldsymbol{\pi}} = \text{vec}(\mathring{\boldsymbol{\Pi}})$, consider the following prior distributions:

$$\Big((\mathring{\alpha}, \mathring{\boldsymbol{\beta}}')' \mid \mathring{\boldsymbol{\Omega}}\Big) \sim N\Big((a, \boldsymbol{b}')',\ \boldsymbol{V}^{\alpha,\beta}\mathring{\boldsymbol{\Omega}}\Big),$$

$$(\mathring{\boldsymbol{\pi}} \mid \mathring{\boldsymbol{\Omega}}^2) \sim N\Big(\boldsymbol{0},\ \mathring{\boldsymbol{\Omega}} \otimes \boldsymbol{V}^{\pi}\Big),$$

$$(\mathring{\sigma}_i^2 \mid \mathring{\sigma}_e^2) \sim IG(\nu_i + 1, \nu_i\mathring{\sigma}_e^2), \qquad \mathring{\sigma}_e^2 \sim G(\nu_\sigma, s_\sigma^2), \qquad \mathring{\boldsymbol{\Omega}} \sim IW(\nu_\Omega, \boldsymbol{S}_\Omega).$$

$IW(\nu_\Omega, \boldsymbol{S}_\Omega)$ is the Inverse-Wishart distribution with PDF:

$$f(\mathring{\boldsymbol{\Omega}}) = \frac{|\boldsymbol{S}_\Omega|^{\nu_\Omega/2}}{2^{\nu_\Omega/2}\Gamma_{k_x+1}(\nu_\Omega/2)}|\mathring{\boldsymbol{\Omega}}|^{-(\nu_\Omega+k_x/2+1)} \exp\left\{-\frac{1}{2}\text{tr}\Big(\boldsymbol{S}_\Omega\mathring{\boldsymbol{\Omega}}^{-1}\Big)\right\}.$$

The prior tuning parameters $\nu_i, \nu_\Omega, \boldsymbol{S}_\Omega$ and $\boldsymbol{V}^\pi$ control shrinkage.

11.7 Conditional Posterior Densities

The posterior distribution conditional on $\mathring{\alpha}, \mathring{\boldsymbol{\beta}}, \mathring{\boldsymbol{\pi}}, \mathring{\sigma}_e^2$ and $\mathring{\boldsymbol{\Omega}}$ is:

$$\Big(\mathring{\sigma}_i^2 \mid \mathring{\alpha}, \mathring{\boldsymbol{\beta}}, \mathring{\boldsymbol{\pi}}, \mathring{\sigma}_e^2, \mathring{\boldsymbol{\Omega}}, \mathcal{D}\Big) \sim IG\left(\nu_i + 1 + \frac{T_i}{2},\ s_i^2\right),$$

$$\Big((\mathring{\alpha}_i, \mathring{\boldsymbol{\beta}}_i')' \mid \mathring{\sigma}_i^2, \mathring{\alpha}, \mathring{\boldsymbol{\beta}}, \mathring{\boldsymbol{\pi}}, \mathring{\boldsymbol{\Omega}}, \mathcal{D}\Big) \sim N(\boldsymbol{m}_i,\ \mathring{\sigma}_i^2\boldsymbol{V}_i),$$

where setting

$$\tilde{\boldsymbol{m}}_i = \Big((\mathring{\alpha}, \mathring{\boldsymbol{\beta}}') + \boldsymbol{z}_i'\mathring{\boldsymbol{\Pi}}\Big)',$$

$$\boldsymbol{V}_i = \left(\mathring{\boldsymbol{\Omega}}^{-1} + (\mathbf{1}_{T_i}, \boldsymbol{X}_i)'(\mathbf{1}_{T_i}, \boldsymbol{X}_i)\right)^{-1},$$
$$\boldsymbol{m}_i = \boldsymbol{V}_i\left(\mathring{\boldsymbol{\Omega}}^{-1}\tilde{\boldsymbol{m}}_i + (\mathbf{1}_{T_i}, \boldsymbol{X}_i)'\boldsymbol{y}_i\right),$$
$$s_i^2 = \nu_i\mathring{\sigma}_e^2 + \frac{1}{2}\left(\tilde{\boldsymbol{m}}_i'\mathring{\boldsymbol{\Omega}}^{-1}\tilde{\boldsymbol{m}}_i - \boldsymbol{m}_i'\boldsymbol{V}_i^{-1}\boldsymbol{m}_i + \boldsymbol{y}_i'\boldsymbol{y}_i\right).$$

To express the posterior distribution conditional on $\mathring{\alpha}_i, \mathring{\boldsymbol{\beta}}_i$ and $\mathring{\sigma}_i^2$, $i = 1, \ldots, n$, define:

$$\mathring{\boldsymbol{\Gamma}} = \begin{pmatrix} \mathring{\alpha}_1 & \mathring{\boldsymbol{\beta}}_1' \\ \vdots & \vdots \\ \mathring{\alpha}_n & \mathring{\boldsymbol{\beta}}_n' \end{pmatrix}, \qquad \boldsymbol{Z} = \begin{pmatrix} 1 & \boldsymbol{z}_1' \\ \vdots & \vdots \\ 1 & \boldsymbol{z}_n' \end{pmatrix}, \qquad \mathring{\boldsymbol{\Sigma}}_e = \begin{pmatrix} \mathring{\sigma}_1^2 \\ \vdots \\ \mathring{\sigma}_n^2 \end{pmatrix},$$

$$\mathring{\boldsymbol{\Delta}} = \begin{pmatrix} \mathring{\alpha} & \mathring{\boldsymbol{\beta}}' & \mathring{\boldsymbol{\Pi}}_1 & \mathring{\boldsymbol{\Pi}}_2 \end{pmatrix}, \qquad \boldsymbol{M}_\Delta = \begin{pmatrix} a & \boldsymbol{b}' & 0 & 0 \end{pmatrix}, \qquad \boldsymbol{V}_\Delta = \begin{pmatrix} \boldsymbol{V}^{\alpha,\beta} \\ \boldsymbol{V}^{\pi} \end{pmatrix}.$$

Using this notation, we may write:

$$\left(\mathrm{vec}(\mathring{\boldsymbol{\Gamma}}) \mid \mathring{\boldsymbol{\Delta}}, \mathring{\boldsymbol{\Omega}}, \mathring{\boldsymbol{\Sigma}}_e\right) \sim N\Big((\boldsymbol{I}_{k_x+1} \otimes \boldsymbol{Z})\mathrm{vec}(\mathring{\boldsymbol{\Delta}}),\ \mathring{\boldsymbol{\Omega}} \otimes \mathring{\boldsymbol{\Sigma}}_e\Big),$$
$$\left(\mathrm{vec}(\mathring{\boldsymbol{\Delta}}) \mid \mathring{\boldsymbol{\Omega}}\right) \sim N\Big(\mathrm{vec}(\boldsymbol{M}_\Delta),\ \mathring{\boldsymbol{\Omega}} \otimes \boldsymbol{V}_\Delta\Big).$$

The posterior distribution conditional on $\mathring{\alpha}_i, \mathring{\boldsymbol{\beta}}_i$ and $\mathring{\sigma}_i^2$, $i = 1, \ldots, n$, is equivalent to the posterior $(\mathring{\sigma}_e^2, \mathring{\boldsymbol{\Omega}}, \mathring{\boldsymbol{\Delta}} \mid \mathring{\boldsymbol{\Gamma}}, \mathring{\boldsymbol{\Sigma}}_e, \mathcal{D})$:

$$\left(\mathring{\sigma}_e^2 \mid \mathring{\boldsymbol{\Gamma}}, \mathring{\boldsymbol{\Sigma}}_e, \mathcal{D}\right) \sim G\left(\nu_\sigma + \sum_{i=1}^n \nu_i + n,\ s_\sigma^2 + \sum_{i=1}^n \nu_i\mathring{\sigma}_i^2\right),$$
$$\left(\mathring{\boldsymbol{\Omega}} \mid \mathring{\boldsymbol{\Gamma}}, \mathring{\boldsymbol{\Sigma}}_e, \mathcal{D}\right) \sim IW(\nu_\Omega + n, \boldsymbol{S}_\Omega),$$
$$\left(\mathrm{vec}(\mathring{\boldsymbol{\Delta}}) \mid \mathring{\boldsymbol{\Omega}}, \mathring{\boldsymbol{\Gamma}}, \mathring{\boldsymbol{\Sigma}}_e, \mathcal{D}\right) \sim N\Big(\mathrm{vec}(\boldsymbol{M}_\Delta),\ \mathring{\boldsymbol{\Omega}} \otimes \boldsymbol{V}_\Delta\Big),$$

where

$$\boldsymbol{V}_\Delta = \left(\boldsymbol{V}_\Delta^{-1} + \boldsymbol{Z}'\mathring{\boldsymbol{\Sigma}}_e^{-1}\boldsymbol{Z}\right)^{-1}, \qquad \boldsymbol{M}_\Delta = \boldsymbol{V}_\Delta\left(\boldsymbol{V}_\Delta^{-1}\boldsymbol{M}_\Delta + \boldsymbol{Z}'\mathring{\boldsymbol{\Sigma}}_e^{-1}\mathring{\boldsymbol{\Gamma}}\right),$$
$$\boldsymbol{S}_\Omega = \boldsymbol{S}_\Omega + \boldsymbol{M}_\Delta'\boldsymbol{V}_\Delta^{-1}\boldsymbol{M}_\Delta - \boldsymbol{M}_\Delta'\boldsymbol{V}_\Delta^{-1}\boldsymbol{M}_\Delta + \mathring{\boldsymbol{\Gamma}}'\mathring{\boldsymbol{\Sigma}}_e^{-1}\mathring{\boldsymbol{\Gamma}}.$$

12 High-Dimensional Models and Machine Learning

12.1 High-Dimensional Regression

Recall the LR

$$Y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + U_i \quad \text{with} \quad \boldsymbol{\beta} \in \mathbb{R}^k.$$

- For $\operatorname{rank}\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i'] = k$ (A2) to hold, we must have $k < n$.
- Under A6(ii), asymptotic CIs hold for $k < \infty$, $N \to \infty$.
- High dimensional (HD) regression: $k \gg n$—i.e., k can be much larger than n.
- OLS does not work, so we need an alternative estimator.
- To derive asymptotic CIs, we need conditions on relative growth rates: $N > n$ vs $K > k$.

Example 12.1.1. • Series regression with many regressors (tensor product).

- "Saturated" regression model with many binary variables.
- 20 binary variables fully interacted: $2^{20} > 1\text{M}$ regressors.
- Contemporary "big" datasets contain thousands of potential regressors, but many containing low information.

12.2 Vector Norms

High dimensional regression estimators rely on vector norms, which measure the size of the vector.

Definition 12.2.1 (General p-norm)**.** For a vector $\boldsymbol{x} = (x_1, \ldots, x_k)'$, the p-norm is

$$\|\boldsymbol{x}\|_p = \left(\sum_{j=1}^{k} |x_j|^p\right)^{1/p}.$$

- 1-norm: $\|\boldsymbol{x}\|_1 = \sum_{j=1}^{k} |x_j|$.
- 2-norm (Euclidean): $\|\boldsymbol{x}\|_2 = \left(\sum_{j=1}^{k} x_j^2\right)^{1/2}$.
- sup-norm ($p \to \infty$): $\|\boldsymbol{x}\|_\infty = \max_{1 \le j \le k} |x_j|$.
- 0-norm ($p \to 0$): $\|\boldsymbol{x}\|_0 = \sum_{j=1}^{k} \mathbf{1}[x_j \neq 0]$.
- For $p \ge q \ge 1$,
$$\|\boldsymbol{x}\|_q \ge \|\boldsymbol{x}\|_p \ge \|\boldsymbol{x}\|_\infty \qquad \text{(monotonicity)}.$$

12.3 Ridge Estimator

Definition 12.3.1 (Ridge estimator)**.** The ridge estimator is defined as:

$$\widehat{\boldsymbol{\beta}}_{\text{ridge}} = \arg\min_{\boldsymbol{b}} (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}) + \lambda \|\boldsymbol{b}\|_2^2 = (\lambda \boldsymbol{I}_k + \boldsymbol{X}'\boldsymbol{X})^{-1} \boldsymbol{X}'\boldsymbol{y}.$$

- $\lambda > 0$ is the ridge parameter; it must be pre-specified.
- $\widehat{\boldsymbol{\beta}}_{\text{ridge}}$ is well-defined when $k \gg n$ and $\boldsymbol{X}'\boldsymbol{X}$ is singular.

The ridge estimator is a penalized least squares estimator.

- Penalizing large coefficient vectors prevents over-fitting.
- In this sense, λ is as a penalty on the magnitude of the coefficient vector.

The ridge estimator trades some bias for a smaller variance.

- Under certain conditions, this yields overall smaller mean squared error (MSE) than the OLS estimator.

The ridge estimator is not invariant to re-scaling regressors.

- If X_{ij} are re-scaled (e.g. change currency form AUD to USD, etc.), the ridge estimator's properties change.
- It is common to "standardise" the regressors before implementing the ridge estimator.

The ridge estimator can be implemented in R with the `glmnet()` command in the `glmnet` package.

12.4 Ridge Estimator is a Bayesian Estimator

Recall the Bayesian linear regression, with conjugate priors. The mode of the posterior distribution $(\mathring{\boldsymbol{\beta}} \mid \mathcal{D})$ is

$$\boldsymbol{b} = \left(\boldsymbol{V}^{-1} + \boldsymbol{X}'\boldsymbol{X}\right)^{-1}\left(\boldsymbol{V}^{-1}\boldsymbol{b} + \boldsymbol{X}'\boldsymbol{y}\right).$$

Setting $\boldsymbol{b} = \boldsymbol{0}$ and $\boldsymbol{V} = \lambda^{-1}\boldsymbol{I}_k$ yields $\widehat{\boldsymbol{\beta}}_{\text{ridge}} = \boldsymbol{b}$. Therefore, the ridge estimator is simply the posterior mode of a Bayesian regression with a certain prior.

- The full posterior provide a complete characterisation of uncertainty.
- Under regularity conditions, $\Pr(\left\|\mathring{\boldsymbol{\beta}} - \boldsymbol{\beta}\right\|_2 \mid \mathcal{D}) \to 0$ as $K \to 0$ and $N \to 0$.

12.5 K-Fold Cross-Validation (CV)

K-fold CV is a principled method for selecting tuning parameters.

- Consider a set of possible values for the tuning parameters.
- Compute a CV metric for each value of the tuning parameters.
- Set the tuning parameters to the value with the smallest CV metric.

For the ridge estimator, we consider a set of values for λ and compute CV_λ.

1. Randomly sort the observations.
2. Split the sample into folds $k = 1, \ldots, K$ of roughly equal size $n_k \approx n/K$.
3. For fold $k = 1, \ldots, K$,
   (a) Exclude fold k to obtain a sample of $n - n_k$ observations.
   (b) Compute $\widehat{\boldsymbol{\beta}}_{-k}$ with this sample.
   (d) Compute
   $$\text{CV}_{\lambda,k} = n_k^{-1} \sum_{i \in S_k} \left(Y_i - \boldsymbol{x}_i'\widehat{\boldsymbol{\beta}}_{-k}\right)^2,$$
   where S_k indexes all the observations in fold k.
4. Compute
   $$\text{CV}_\lambda = K^{-1} \sum_{k=1}^{K} \text{CV}_{\lambda,k}.$$

Most commonly chosen K is 10 (default for the CV algorithm in R).

12.6 Bayesian Model Averaging

CV has useful theoretical properties. However, a ridge estimator with a particular λ (selected by CV) ignores specification uncertainty.

- It is highly recommended to verify robustness by comparing results for alternative choices of λ.

Alternatively, use Bayesian model averaging (BMA).

- Introduce $\mathring{\lambda}$ with a prior, e.g.

$$\mathring{\sigma}^2 \sim IG(\nu_\sigma, s_\sigma^2), \qquad \mathring{\lambda} \sim G(\nu_\lambda, \delta_\lambda), \qquad (\mathring{\boldsymbol{\beta}} \mid \mathring{\sigma}^2, \mathring{\lambda}) \sim N\left(\mathbf{0}, \frac{\mathring{\sigma}^2}{\mathring{\lambda}} \boldsymbol{I}_k\right).$$

- Use a two-block Gibbs sampler—$(\mathring{\boldsymbol{\beta}}, \mathring{\sigma}^2 \mid \mathring{\lambda}, \mathcal{D})$ and $(\mathring{\lambda} \mid \mathring{\boldsymbol{\beta}}, \mathring{\sigma}^2, \mathcal{D})$—to summarise the posterior $(\mathring{\boldsymbol{\beta}}, \mathring{\sigma}^2, \mathring{\lambda} \mid \mathcal{D})$.

12.7 LASSO Estimator

The least absolute shrinkage and selection operator (LASSO) estimator is defined as:

$$\widehat{\boldsymbol{\beta}}_{\text{LASSO}} = \arg\min_{\boldsymbol{b}} (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}) + \lambda \left\|\boldsymbol{b}\right\|_1.$$

- The LASSO estimator does not have a closed-form formula.
- Solve for $\widehat{\boldsymbol{\beta}}_{\text{LASSO}}$ using numerical optimisation.
- LASSO is also well-defined when $k \gg n$.
- The LASSO parameter λ is called the penalty parameter.
- LASSO provides a mechanism for variable selection.
- $\lambda \to \infty \Rightarrow \widehat{\boldsymbol{\beta}}_{\text{LASSO}} \to \mathbf{0}$; $\lambda \to 0 \Rightarrow \widehat{\boldsymbol{\beta}}_{\text{LASSO}} \to \widehat{\boldsymbol{\beta}}_{\text{OLS}}$.
- For $0 < \lambda < \infty$, LASSO "shinks" some coefficients $\beta_j \to 0$.

In R, use the `glmnet()` command in the `glmnet` package.

- `cv.glmnet()` implements K-fold CV to select λ.

In the Bayesian linear regression, suppose the prior

$$\mathring{\sigma}^2 \sim IG(\nu, s^2), \qquad (\mathring{\boldsymbol{\beta}} \mid \mathring{\sigma}^2) \sim N(\boldsymbol{b}, \mathring{\sigma}^2 \boldsymbol{V}),$$

is replaced with:

$$\Pr(\mathring{\sigma}^2 = 1) = 1, \qquad f(\mathring{\boldsymbol{\beta}} \mid \mathring{\sigma}^2) = \prod_{j=1}^{k} \frac{\lambda}{2} e^{-\lambda|\mathring{\beta}_j|} = \left(\frac{\lambda}{2}\right)^k e^{-\lambda\left\|\mathring{\boldsymbol{\beta}}\right\|_1},$$

where $f(\mathring{\boldsymbol{\beta}} \mid \mathring{\sigma}^2)$ is the PDF of the Laplace distribution.

The mode of the posterior distribution $(\mathring{\boldsymbol{\beta}} \mid \mathcal{D})$ equals $\widehat{\boldsymbol{\beta}}_{\text{LASSO}}$.

- The full posterior provide a complete characterisation of uncertainty.

- Under regularity conditions, $\Pr(\left\|\mathring{\boldsymbol{\beta}} - \boldsymbol{\beta}\right\|_2 \mid \mathcal{D}) \to 0$ as $K \to 0$ and $N \to 0$.

The general Bayesian LASSO used in practice specifies:

$$\mathring{\sigma}^2 \sim IG(\nu_\sigma, s_\sigma^2), \qquad \mathring{\lambda}^2 \sim G(\nu_\lambda, \delta_\lambda),$$

$$f(\mathring{\boldsymbol{\beta}} \mid \mathring{\sigma}^2, \mathring{\lambda}) = \left(\frac{\mathring{\lambda}}{2\sqrt{\mathring{\sigma}^2}}\right)^k \exp\left\{-\frac{\mathring{\lambda}}{\sqrt{\mathring{\sigma}^2}} \left\|\mathring{\boldsymbol{\beta}}\right\|_1\right\}.$$

A Gibbs sampler can be used to simulate draws from the posterior $(\mathring{\boldsymbol{\beta}}, \mathring{\sigma}^2, \mathring{\lambda} \mid \mathcal{D})$—e.g., Park and Casella (2008).

Many variable selection priors that improve upon the Bayesian LASSO have been derived.

Example 12.7.1. Horseshoe prior (Carvalho and Polson, 2010), which is more robust to different sparsity patterns.

12.8 Variable Selection Priors

Example 12.8.1. Posterior mode of a 300×300 correlations matrix using a variable selection prior on 45,150 coefficients.

Kastner (2019).

12.9 Machine Learning

Variable selection priors are designed to facilitate practical simulation algorithms to sample from the full posterior.

- This is stochastic search variable selection (SSVS).
- This is a form of statistical learning: a computer-intensive algorithm to uncover the most probably relationships.

The general set of algorithmic approaches to statistical learning is known as machine learning.

- Originated in computer science, as a branch of artificial intelligence, but without an associated statistical theory.
- Main focus is on point prediction and learning about sparse structures using huge data samples.
- Many association, classification, categorisation, etc. problems can be cast as high-dimensional regressions.

Applying machine learning models and algorithms to economic problems is at the frontier of econometrics.

Example 12.9.1.
- average treatment effects (Tibshirani and Wager, 2019);
- fraud detection (Farbmacher et al., 2020);
- financial asset pricing (Gu et al., 2021);
- market segmentation (Decarolis and Rovigatti, 2021);

- macroeconomic forecasting (Coulombe, 2020) and nowcasting (Chapman and Desai, 2021);
- macroeconomic policies with nonlinear equilibria (Chen et al., 2021; Hinterlang & Täzer, 2021; Zheng et al., 2022).

Key challenges: "big" data structure, interpretability, technical expertise, computational resources.